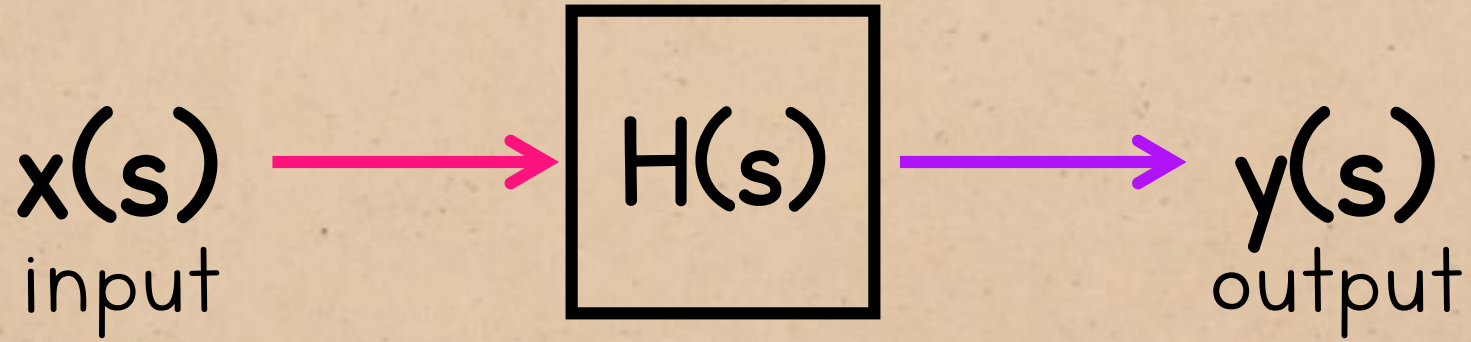




BLOCK DIAGRAMS

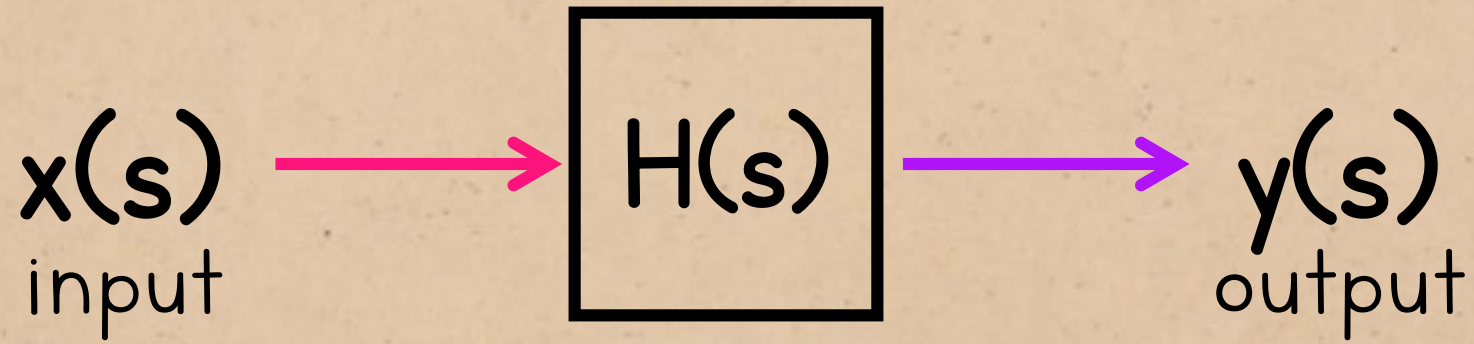
SYSTEMS

The basic unit of the block diagram is the system block.



SYSTEMS

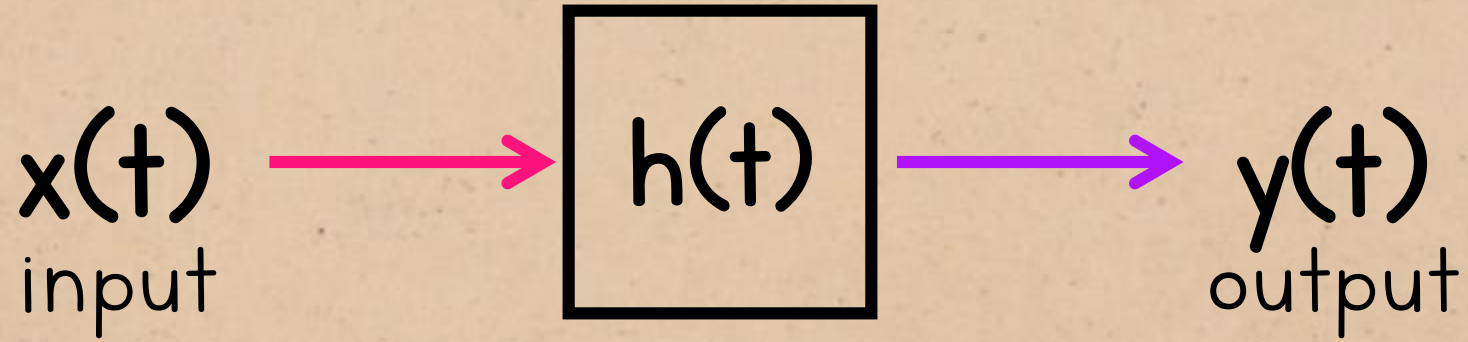
The output = the input multiplied
by the transfer function.



$$y(s) = H(s) x(s)$$

SYSTEMS

...or convolved, if you're working in
the time domain.

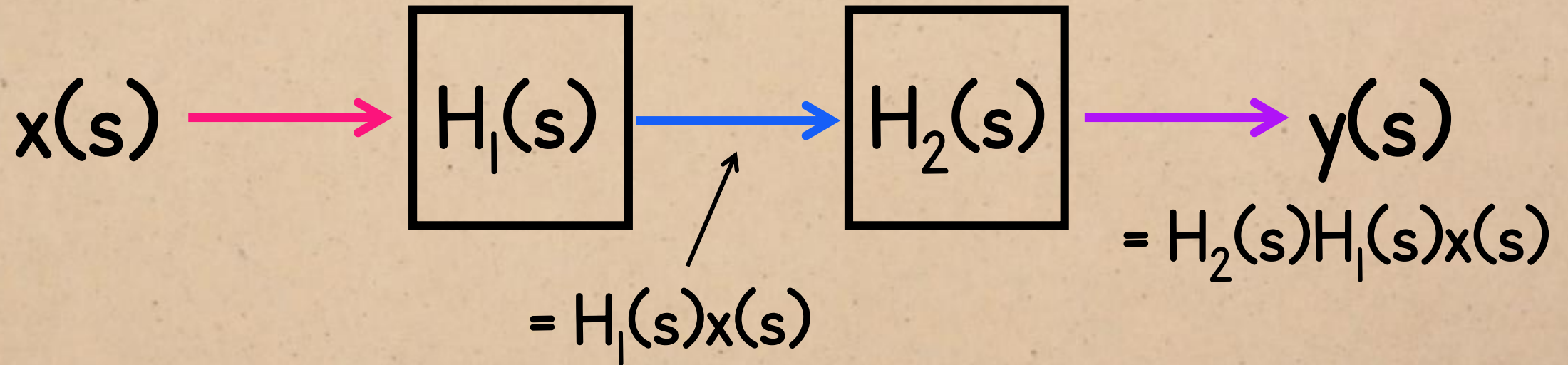


$$y(t) = h(t) * x(t)$$

(which we probably won't do very
often.)

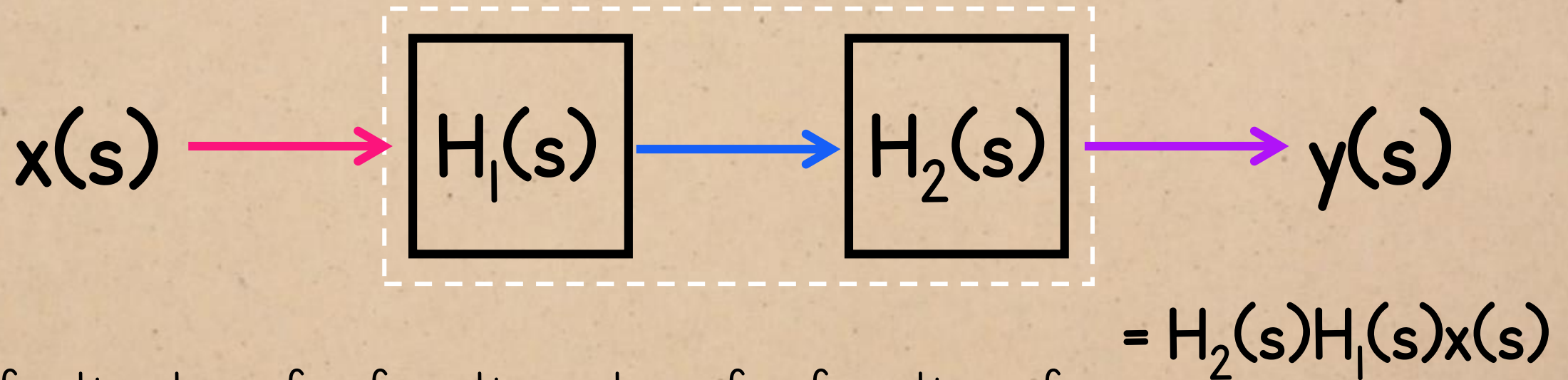
CASCADE CONNECTION

= one system connected after another, so the output of the first becomes the input to the next.



CASCADE CONNECTION

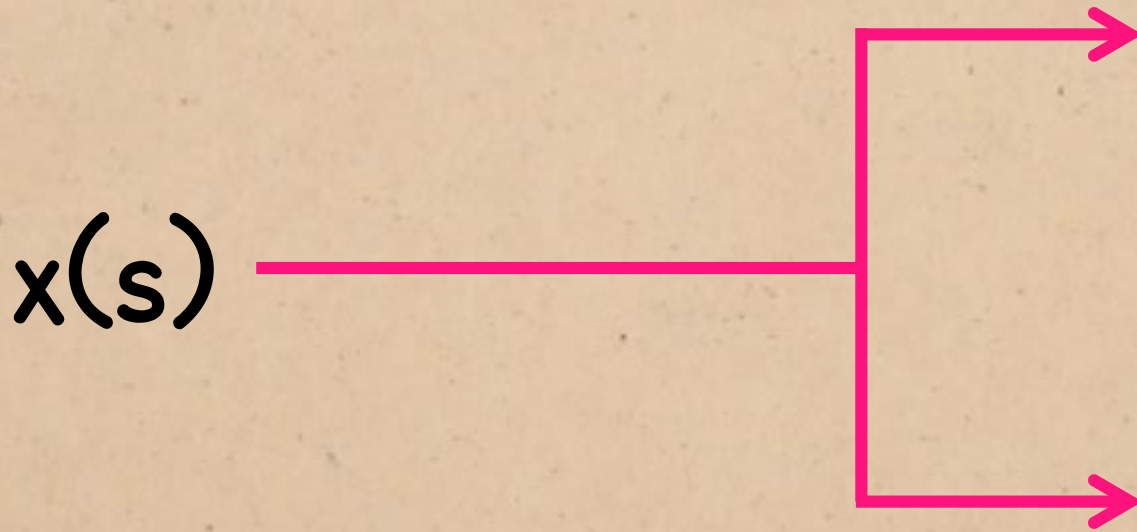
effective transfer function: $H(s) = H_2(s)H_1(s)$



Effective transfer function = transfer function of a single system that has the **same effect** as the combination of systems.

TAKE-OFF POINT

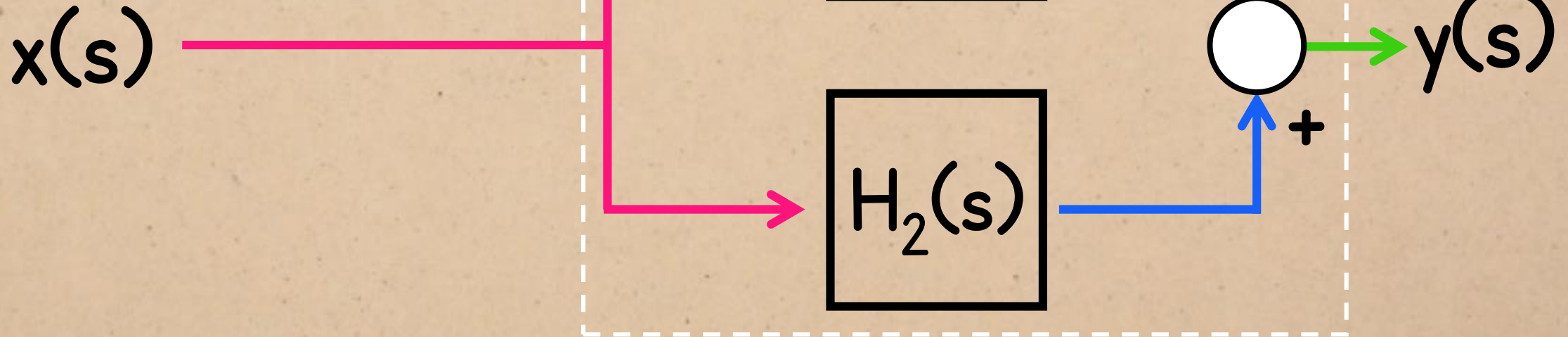
= point where a signal splits into different paths.



The signal will be the same anywhere along either of these arrows as it is at the start.

PARALLEL CONNECTION

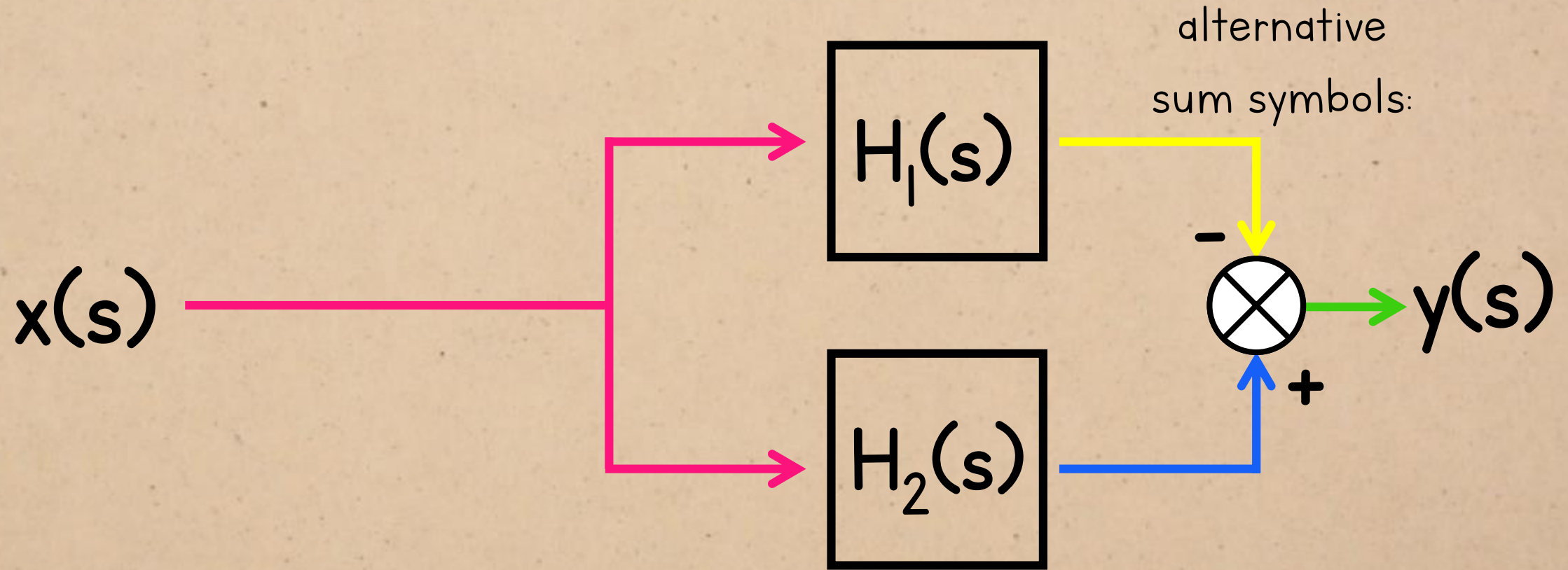
= two systems along separate paths that join at a summing junction.



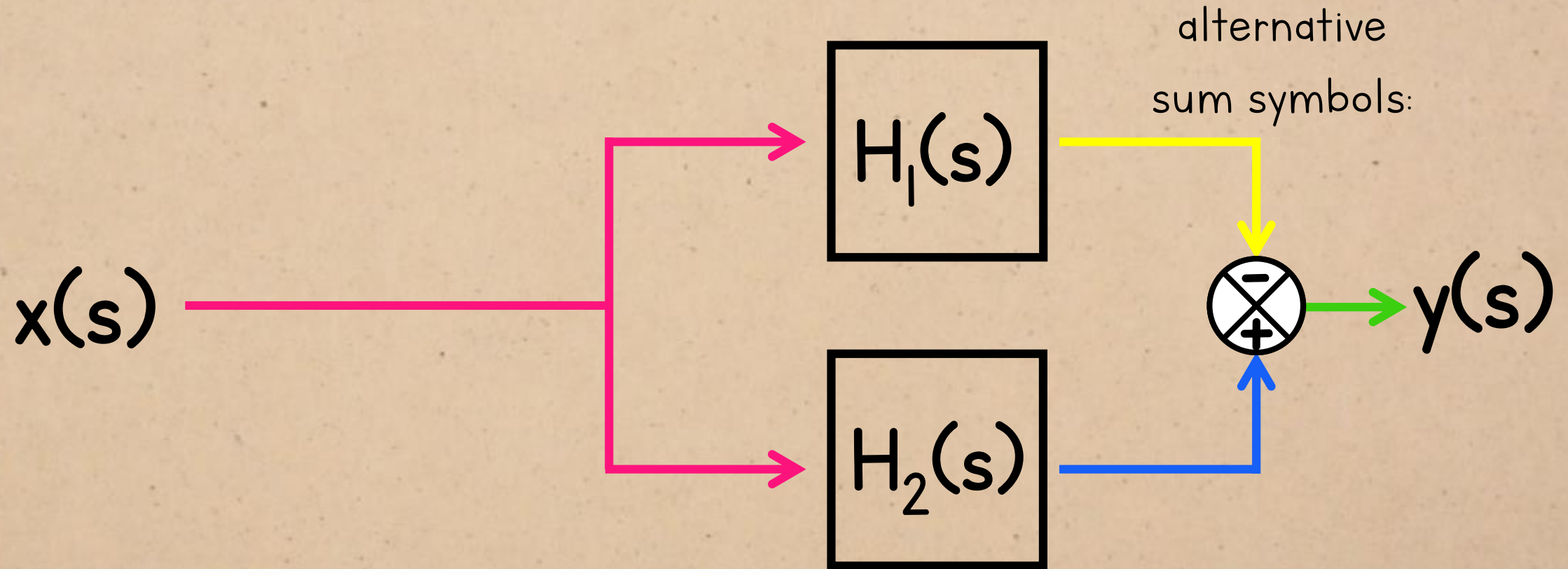
effective transfer function:

$$H(s) = H_1(s) + H_2(s)$$

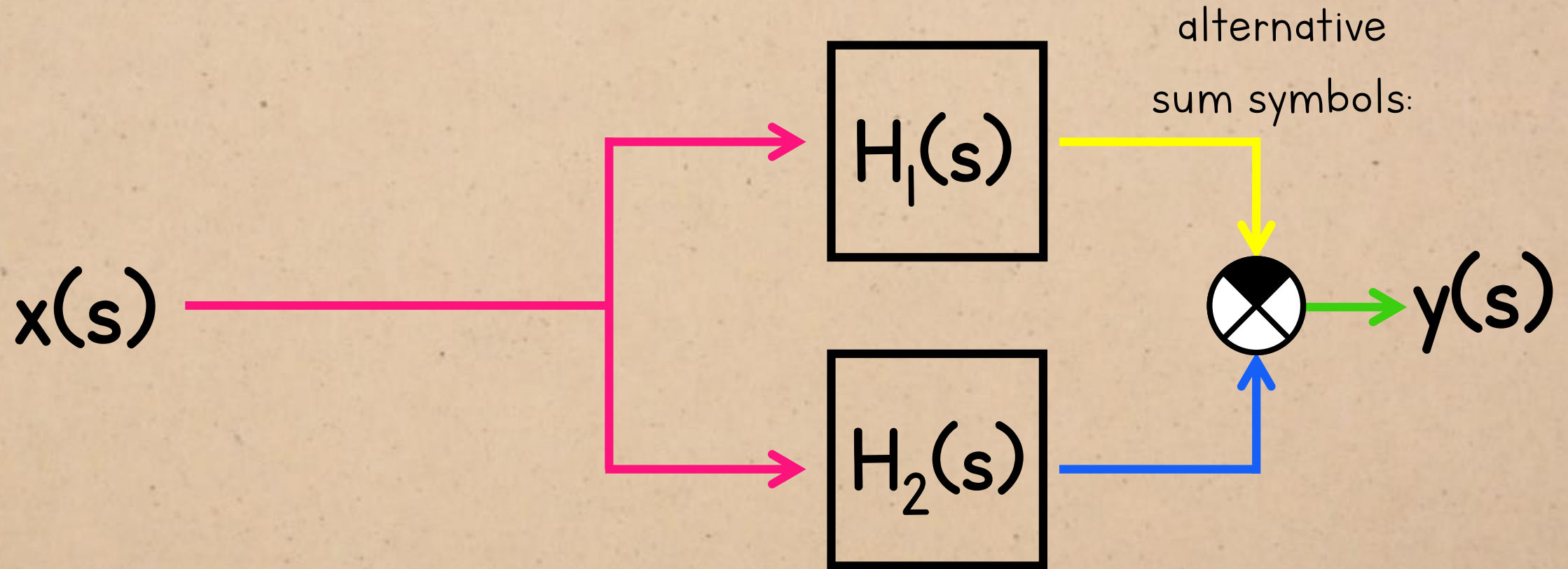
PARALLEL CONNECTION



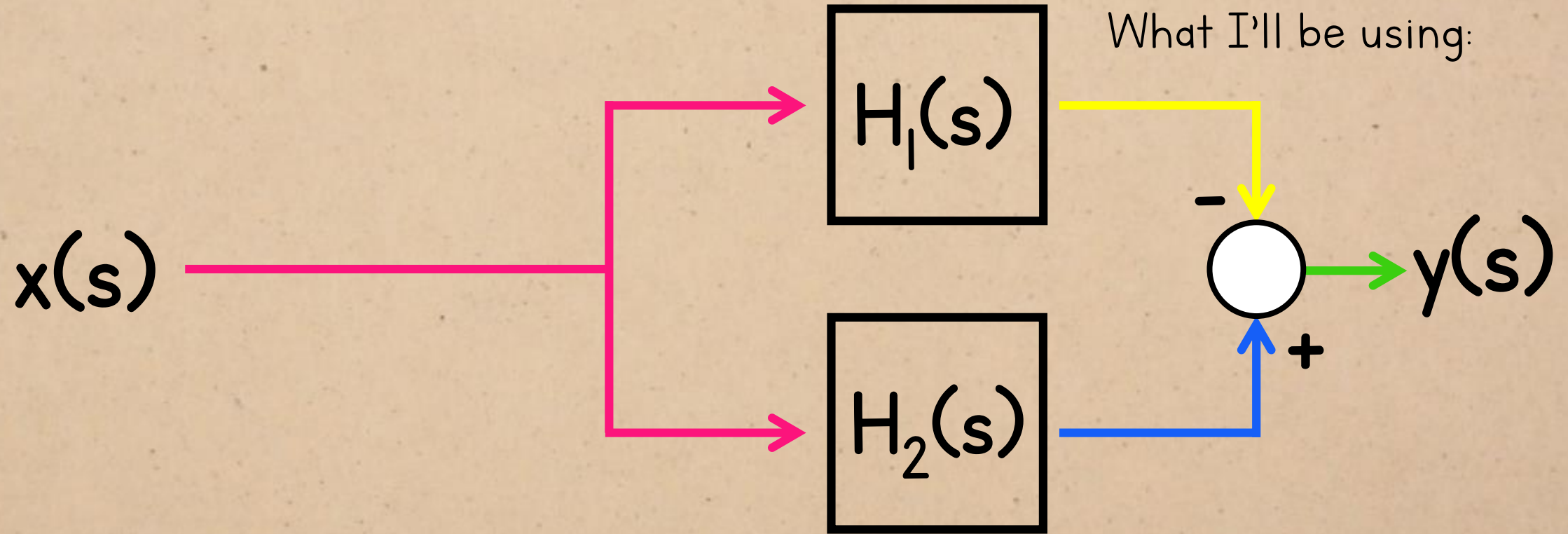
PARALLEL CONNECTION



PARALLEL CONNECTION

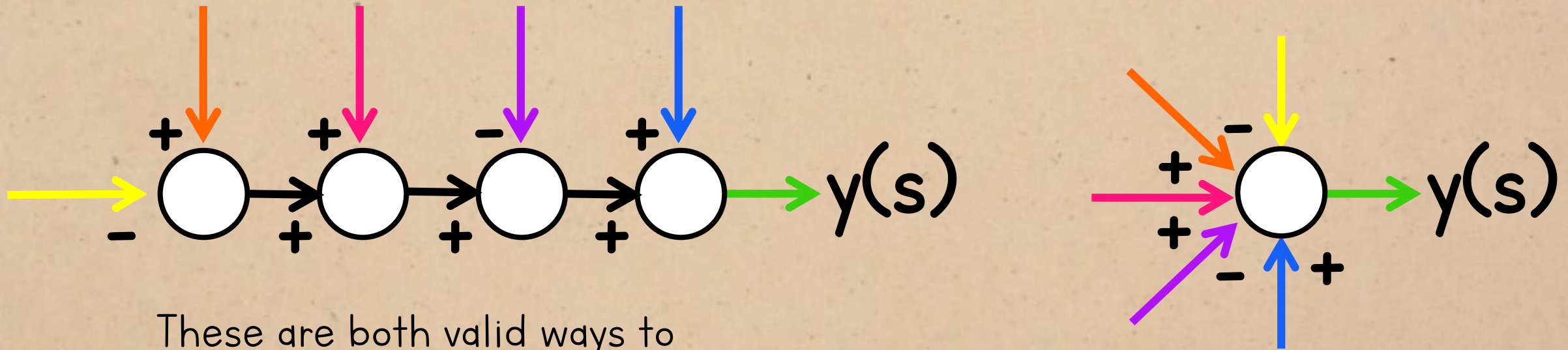


PARALLEL CONNECTION



PARALLEL CONNECTION

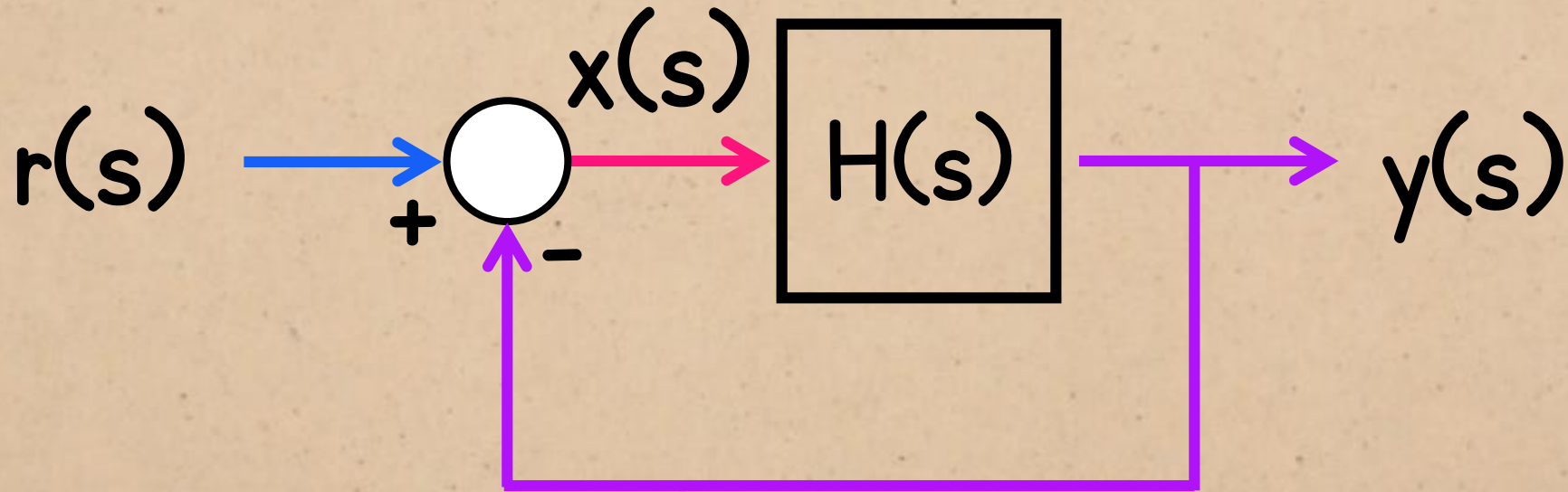
representing sums with >2 signals:



These are both valid ways to represent sums, but I find this one easier to read.

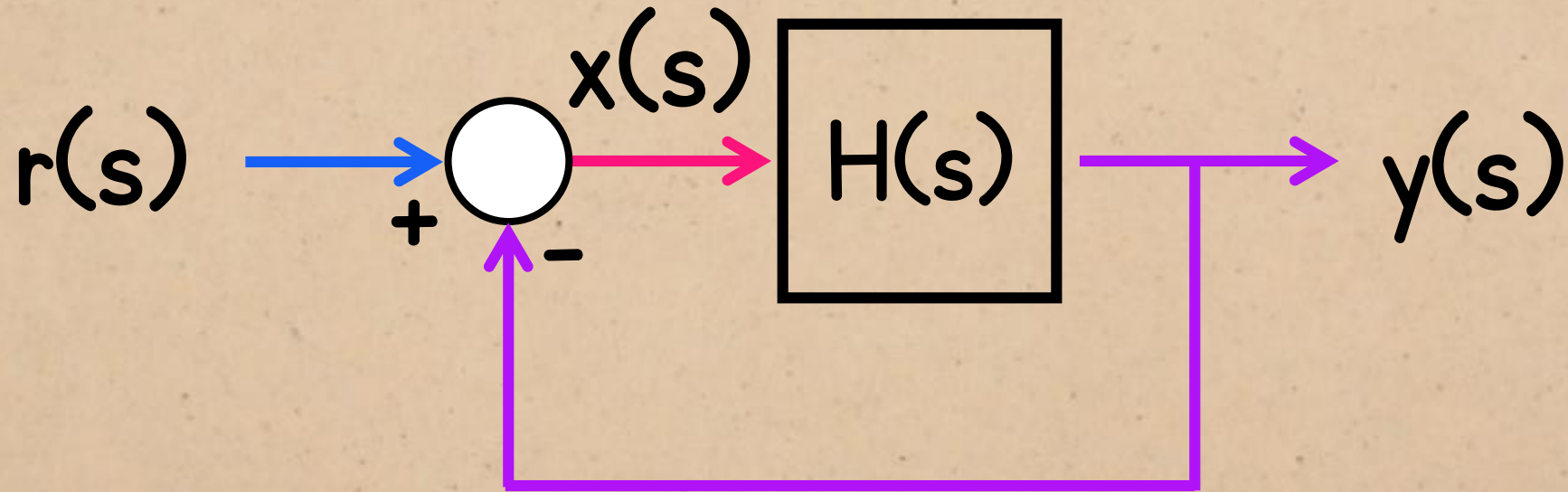
LOOP

= connecting the output of a system to its input.



FEEDBACK

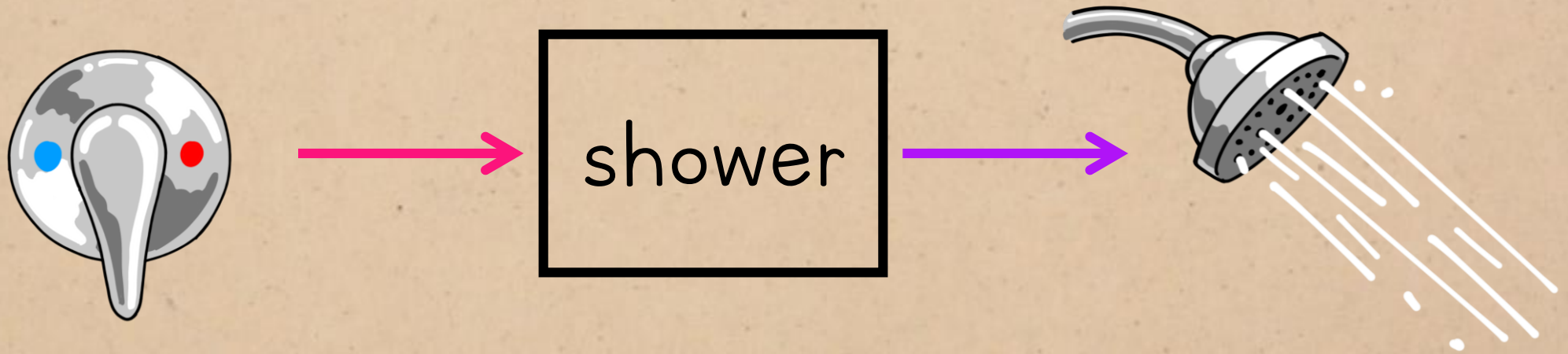
Feedback = the ability of a system to monitor its own output.



Putting a system in feedback gives it the ability to regulate itself by responding to its present output.

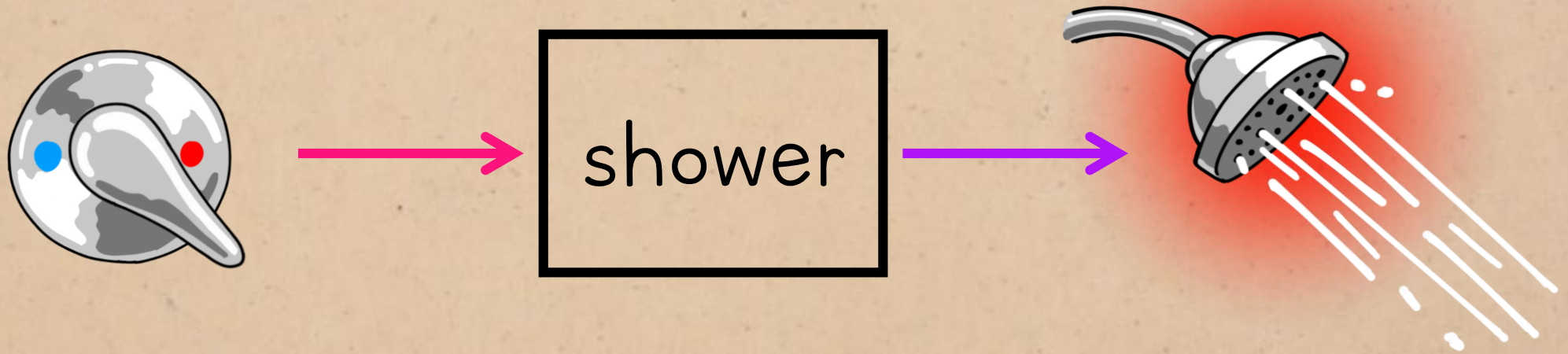
FEEDBACK

The shower is a simple example of where we use feedback in daily life:



FEEDBACK

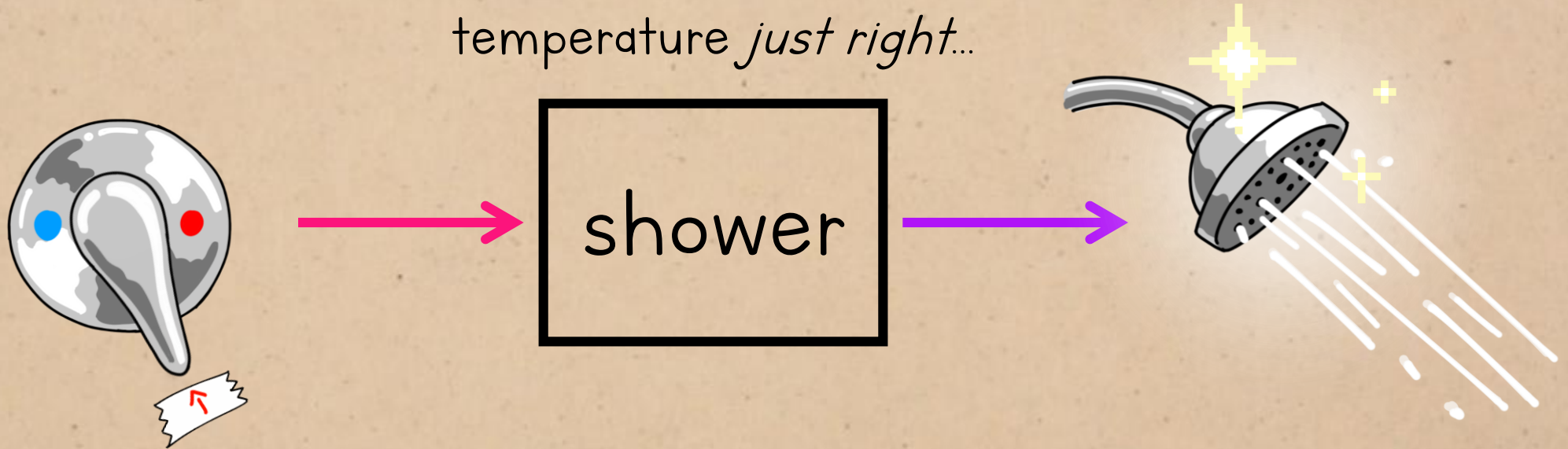
If you just turn the tap to a position, you have no way of knowing whether the water is at the right temperature before you get in.



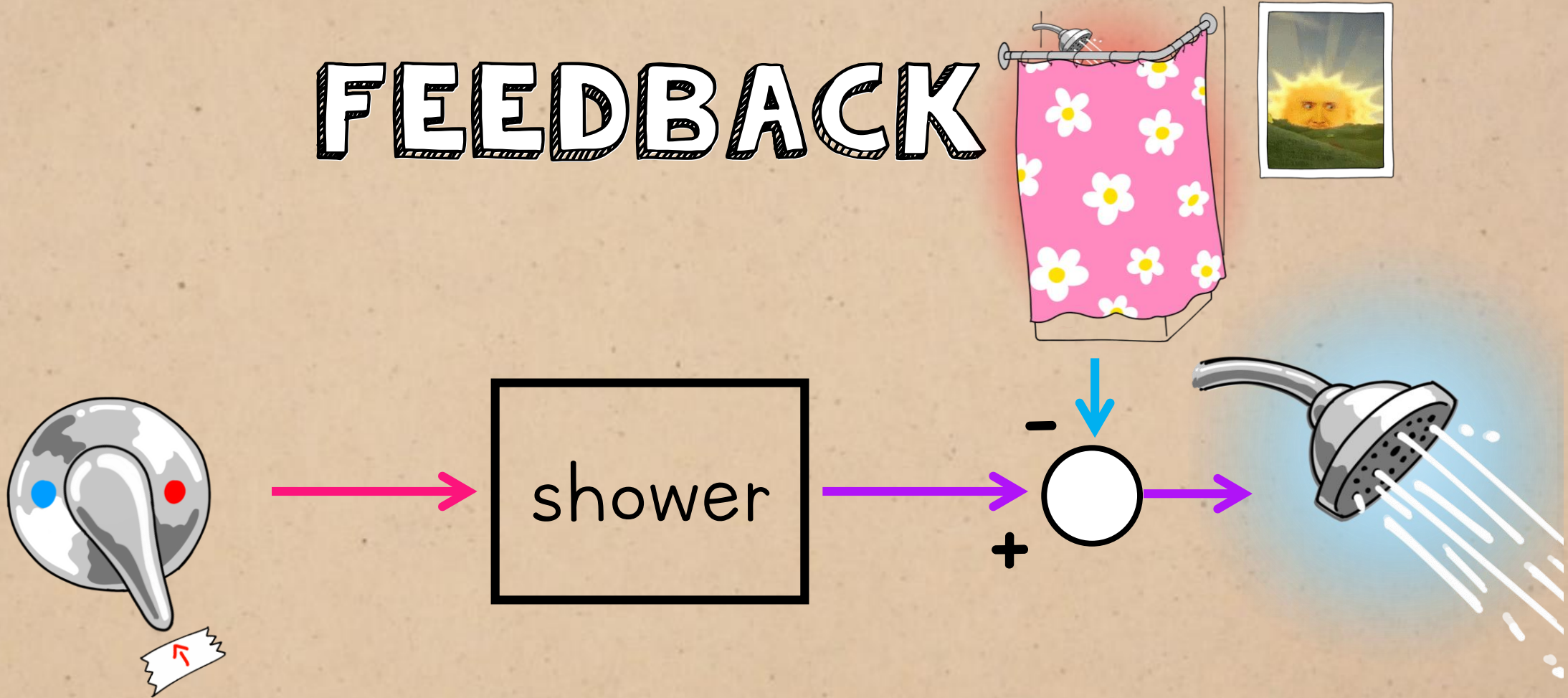
This is the equivalent of using an **open-loop** system.

FEEDBACK

Even if you know the exact position that gets the water temperature *just right*...



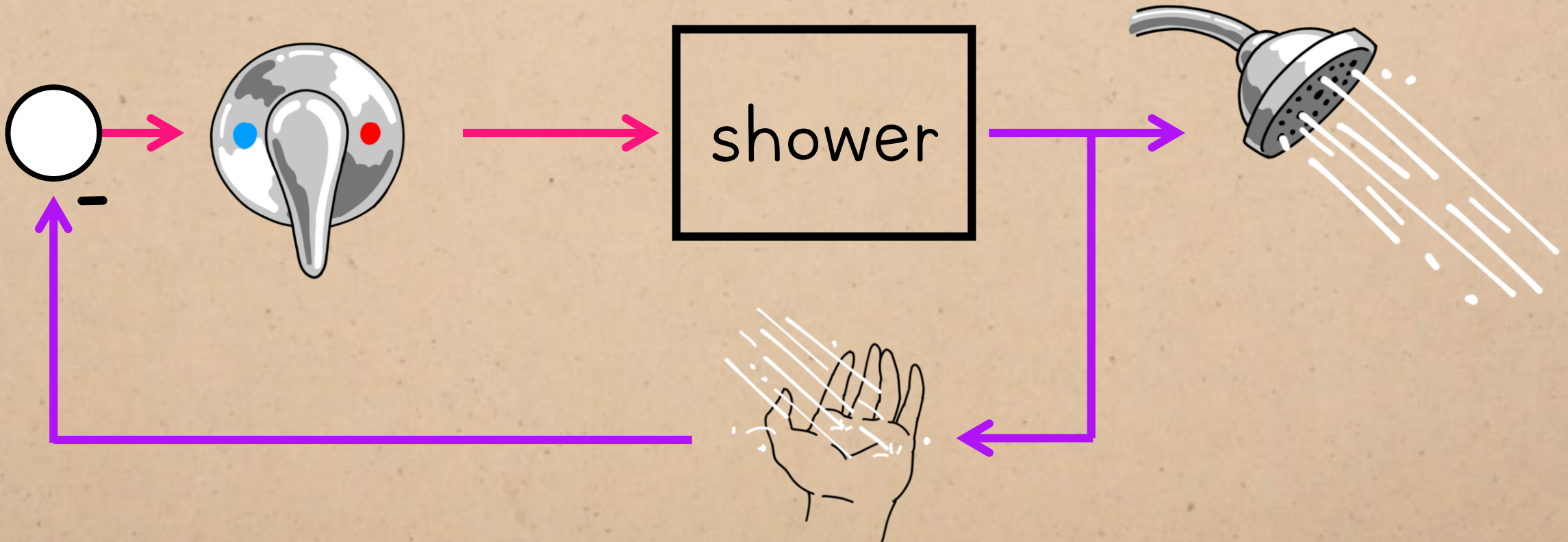
FEEDBACK



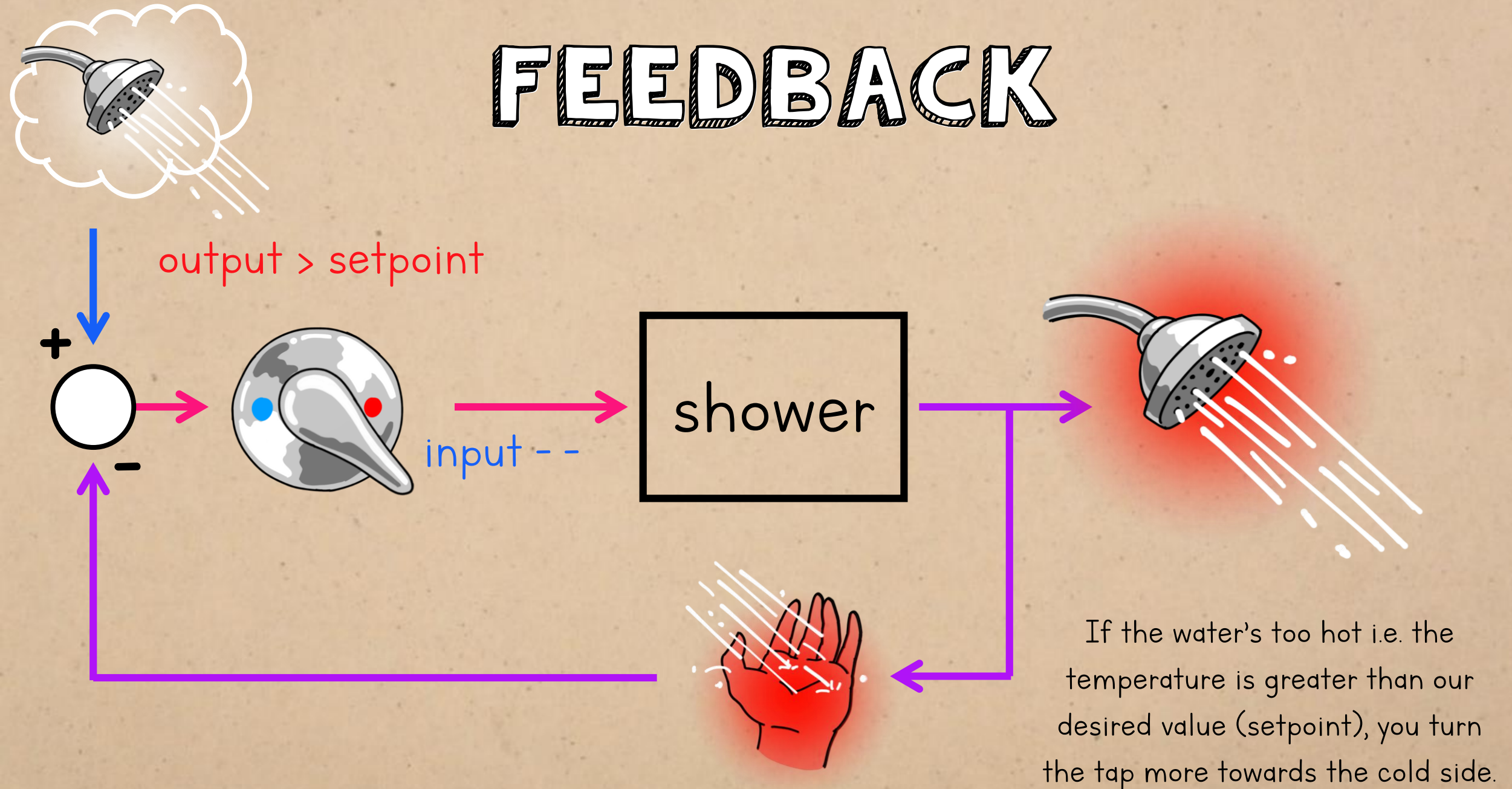
A disturbance, such as your roommate waking up earlier and using most of the hot water, could cause the temperature to come out different from what you expect.

FEEDBACK

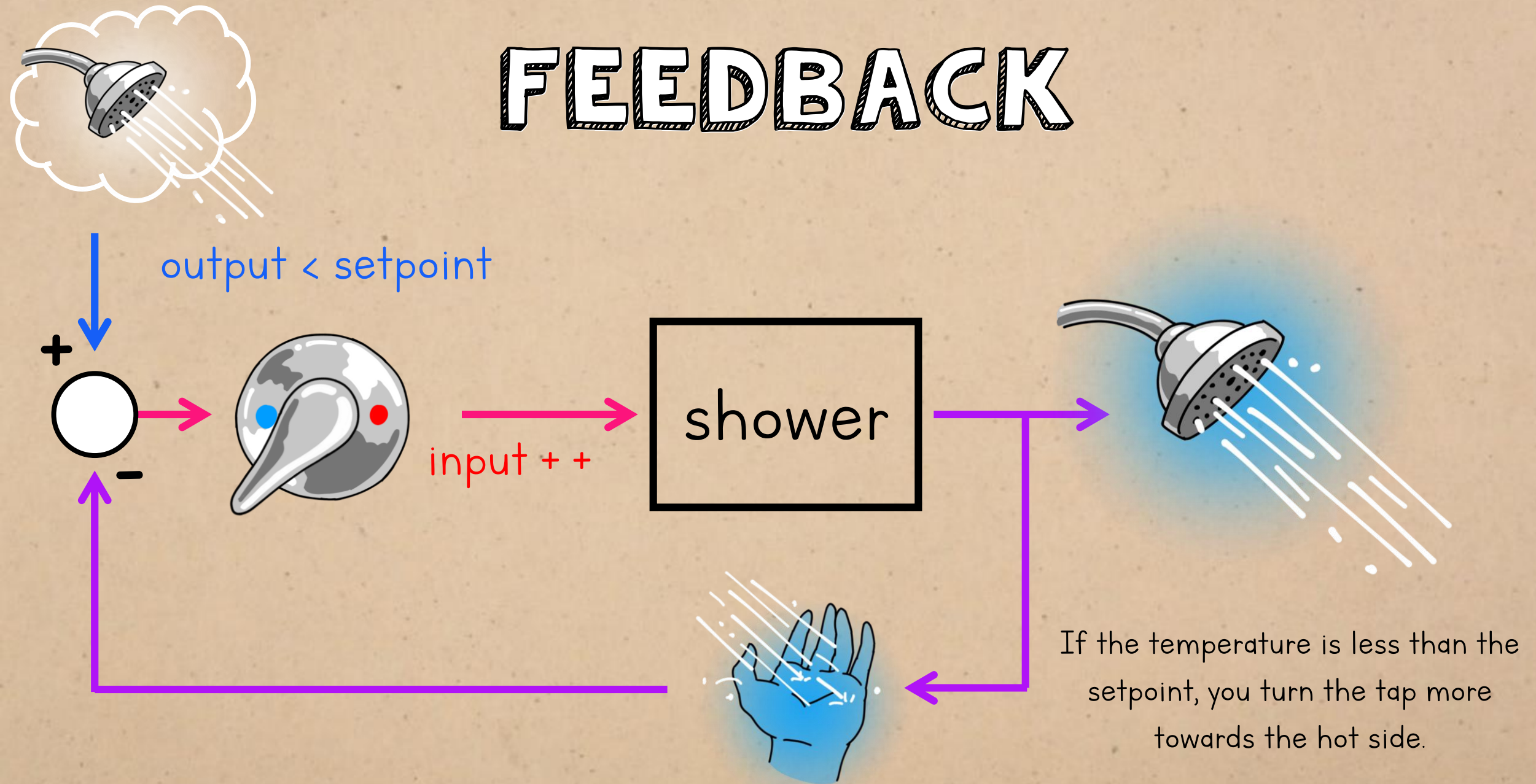
...so, we close the loop by adding a sensor to detect the output i.e. feeling the water



FEEDBACK

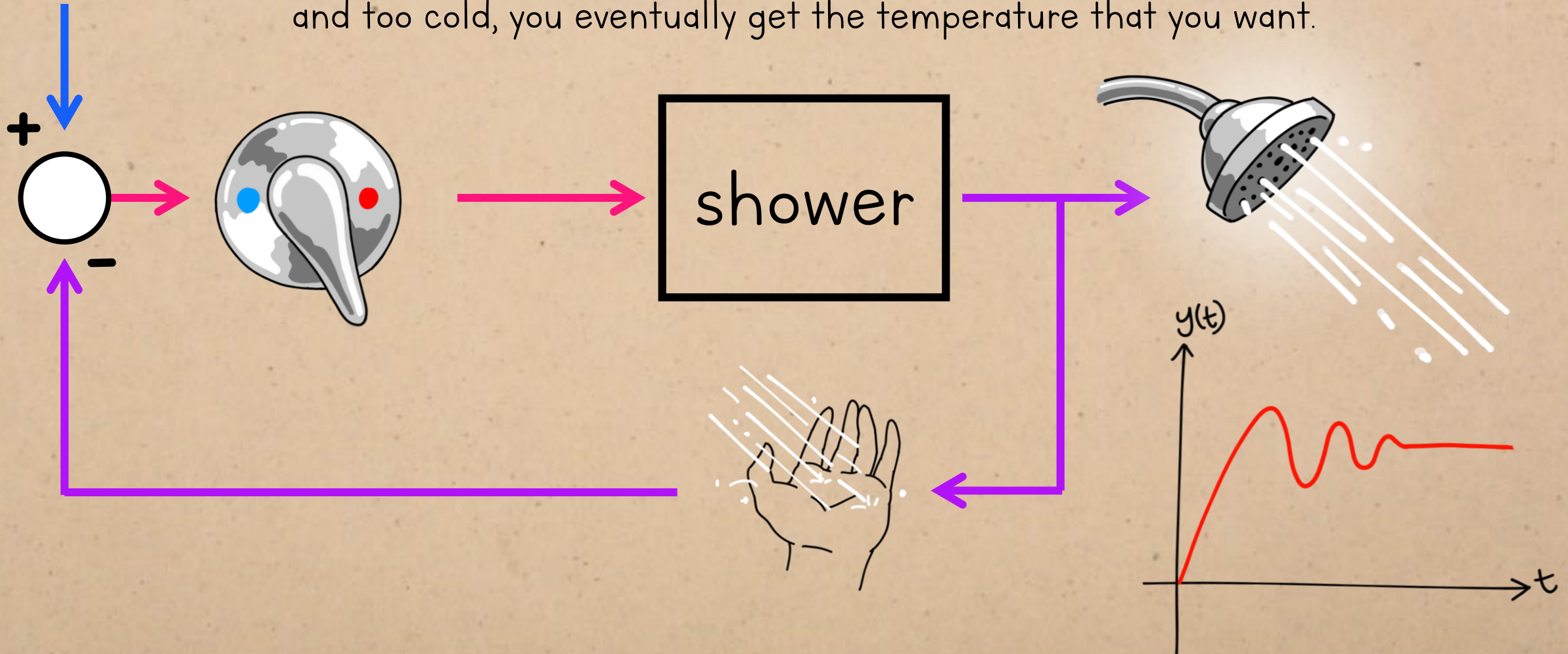


FEEDBACK



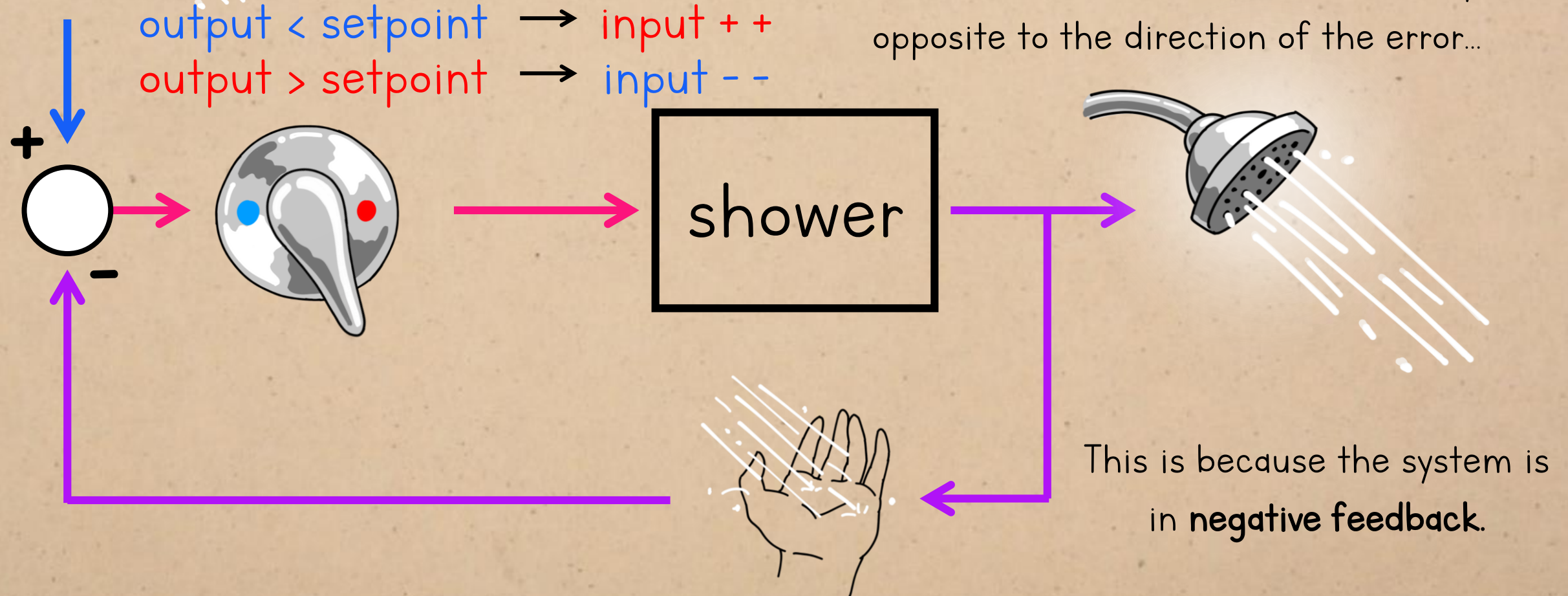
FEEDBACK

Even though it might oscillate a few times between states of too hot and too cold, you eventually get the temperature that you want.



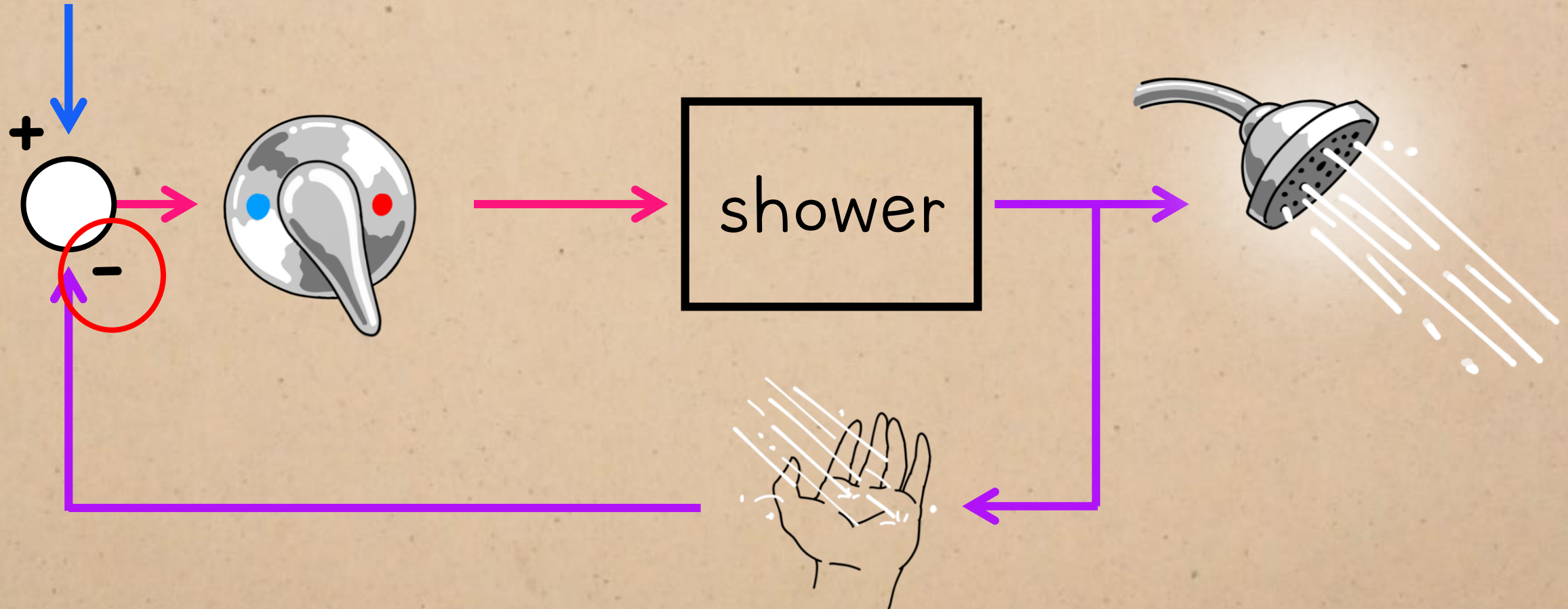
FEEDBACK

Notice that the corrections were always opposite to the direction of the error...



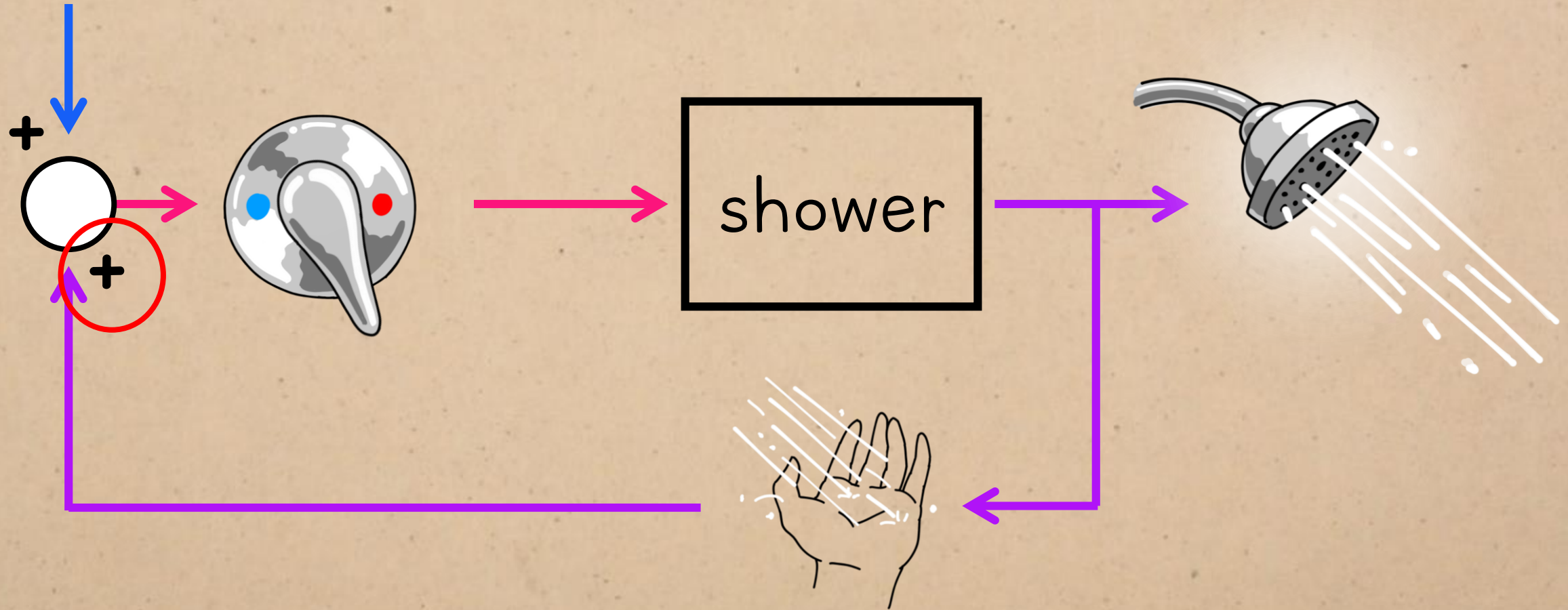
FEEDBACK

negative feedback: output is subtracted from setpoint



FEEDBACK

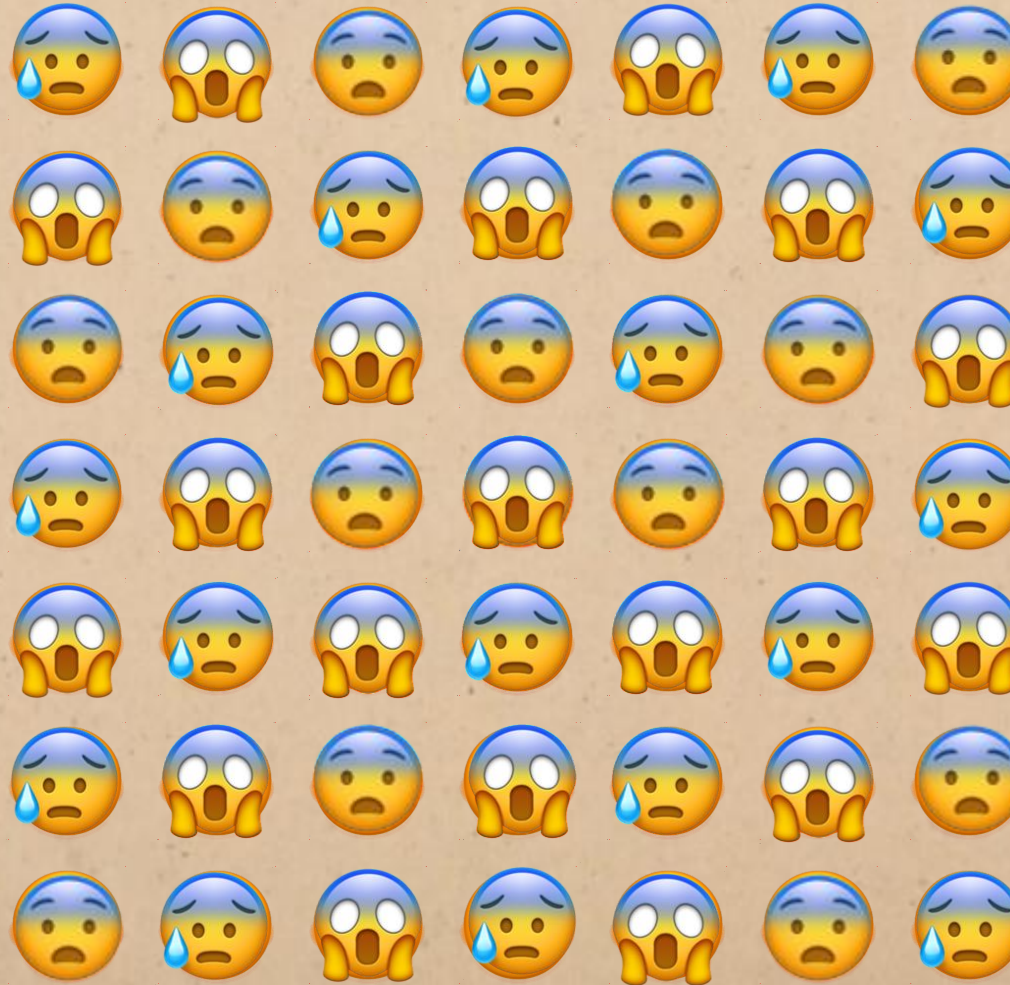
positive feedback: output is added to setpoint



FEEDBACK

positive feedback: output is subtracted from setpoint

Positive feedback
might be useful for
modelling an
outbreak of mass
hysteria...

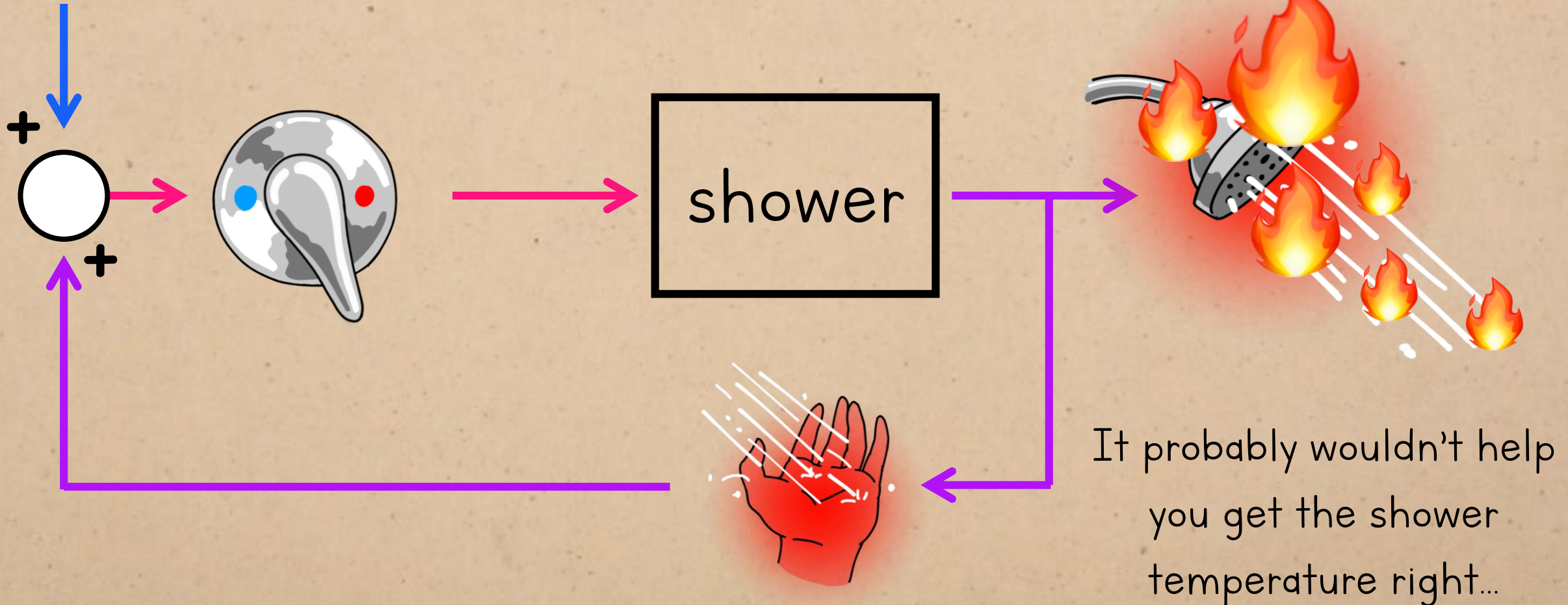


But it tends to
reinforce, rather than
regulate deviations
from the ideal, so we
won't see it often in
control.

FEEDBACK

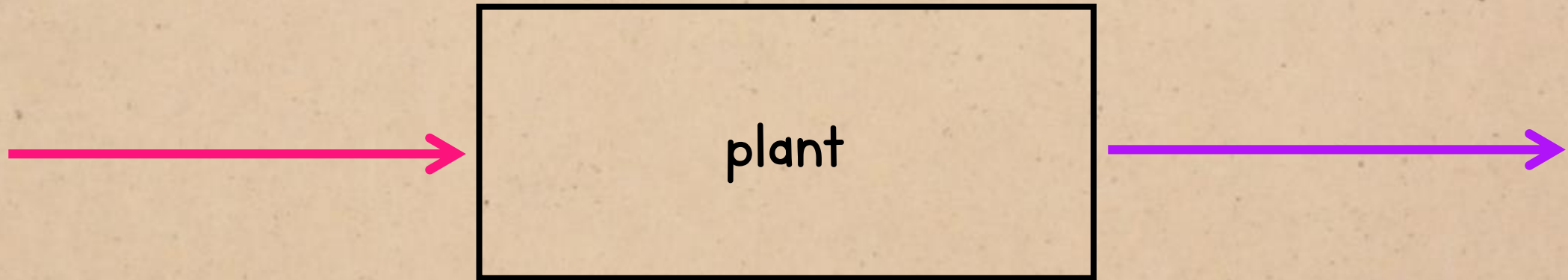


positive feedback: output is subtracted from setpoint



CONTROL SYSTEM

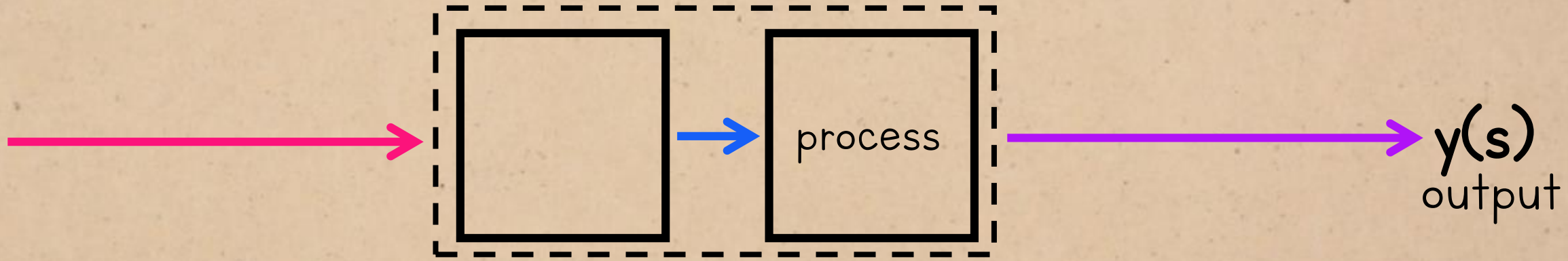
plant = thing we're trying to control



CONTROL SYSTEM

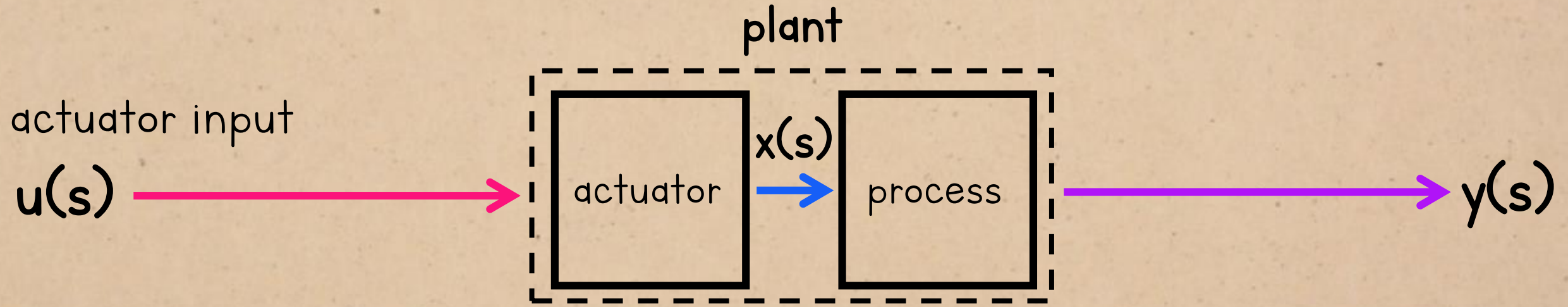
process = part of the plant that produces the output

plant



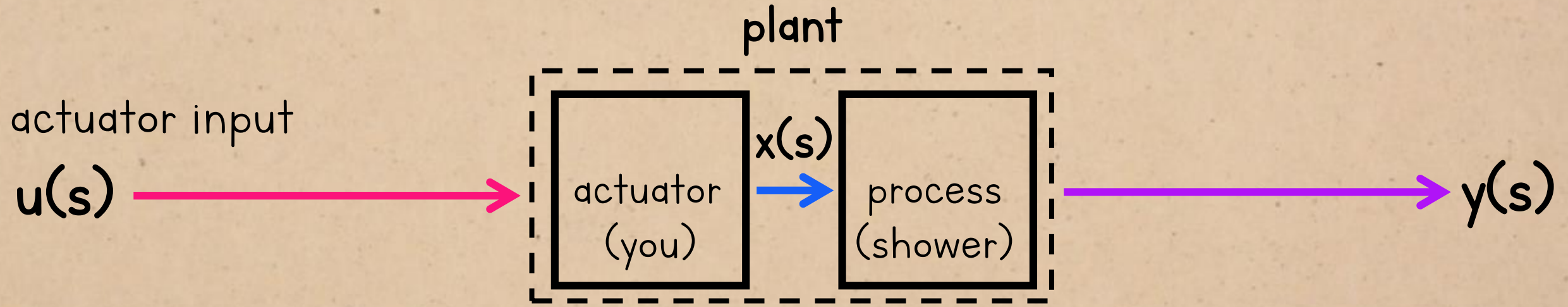
CONTROL SYSTEM

actuator = drives the process in response to the input



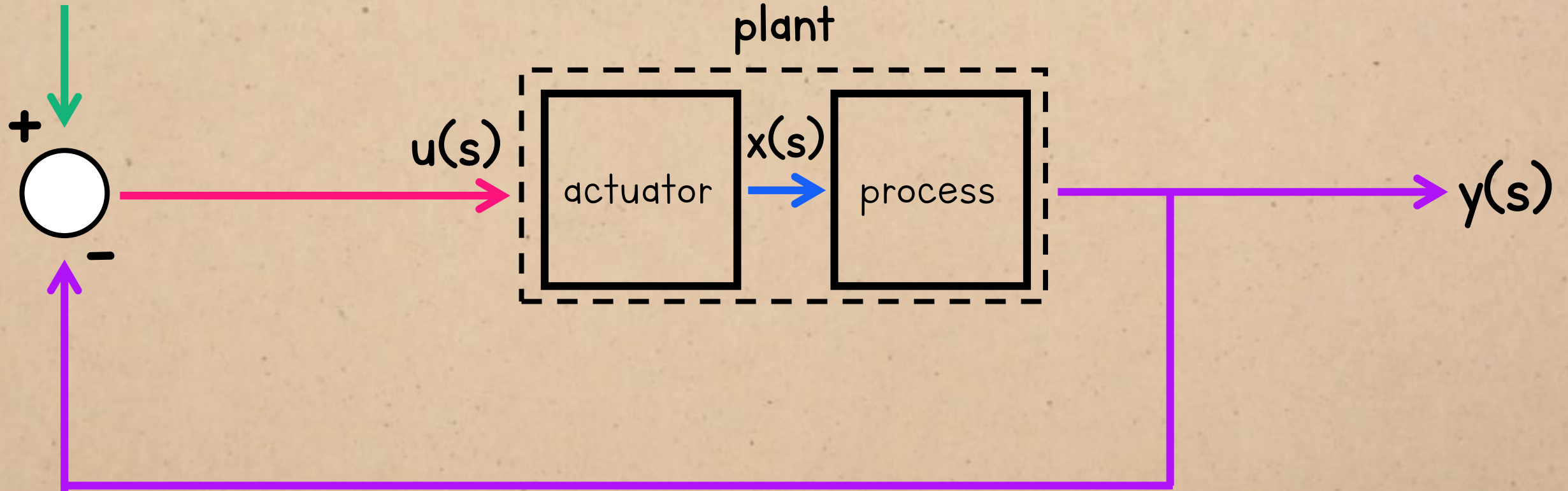
CONTROL SYSTEM

actuator = drives the process in response to the input



In the example, the shower is the process, which produces water temperature as an output, and you are the actuator, since you are the one controlling the input to the process (the tap position).

CONTROL SYSTEM

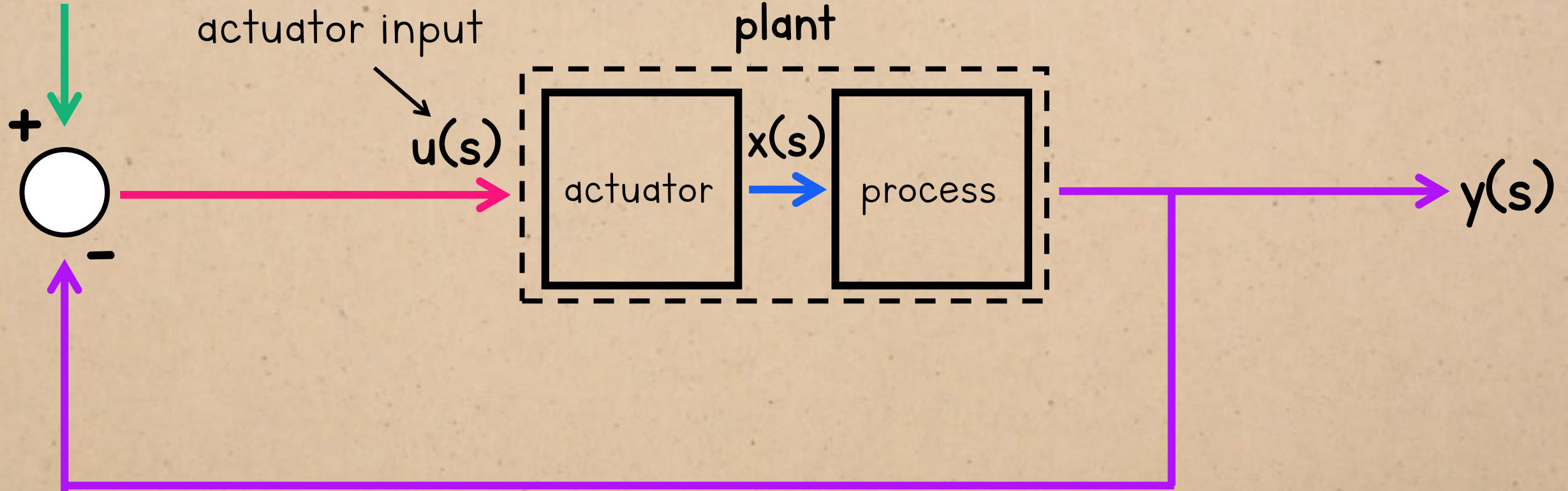


CONTROL SYSTEM

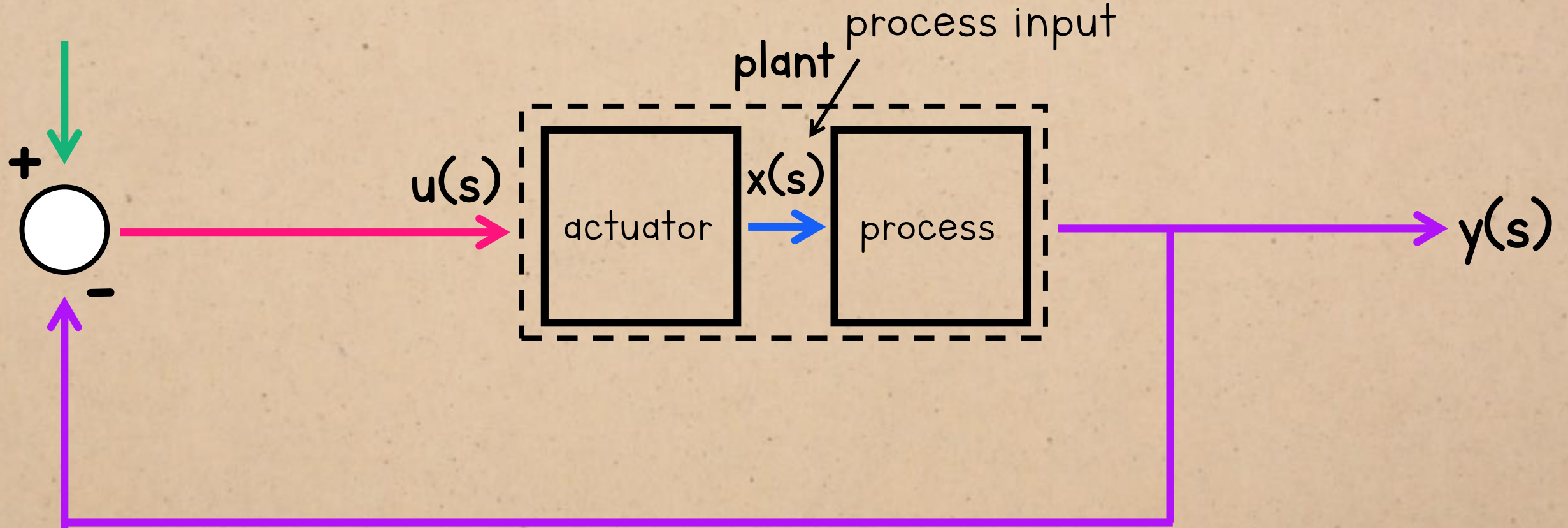
"input" =

actuator input

plant



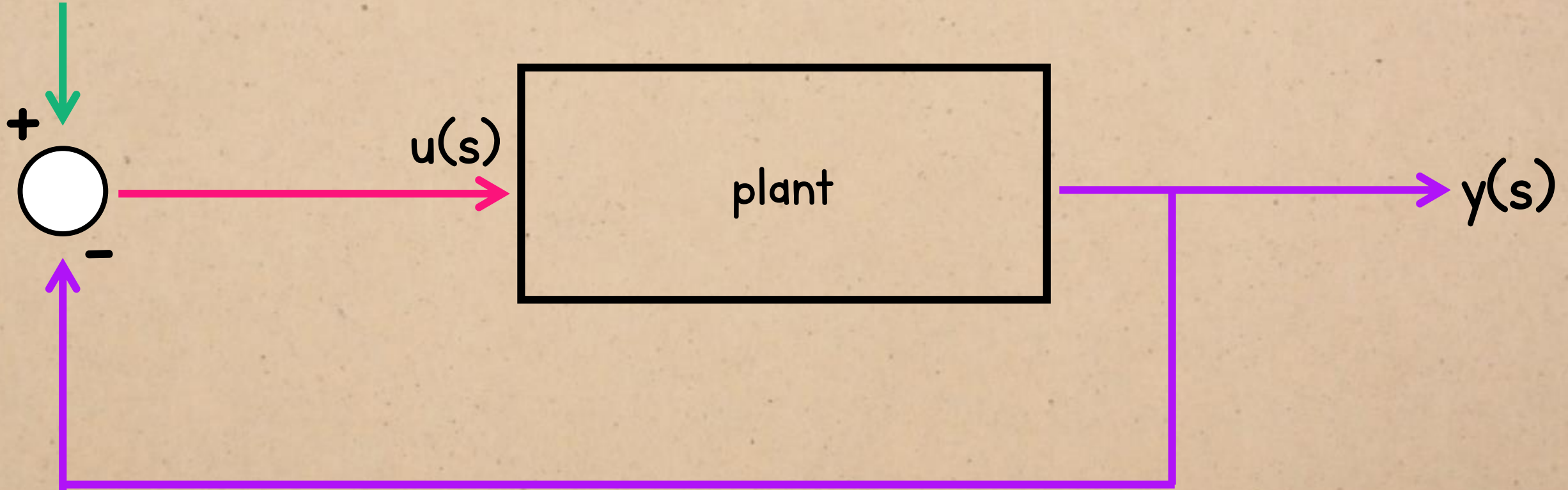
CONTROL SYSTEM



CONTROL SYSTEM

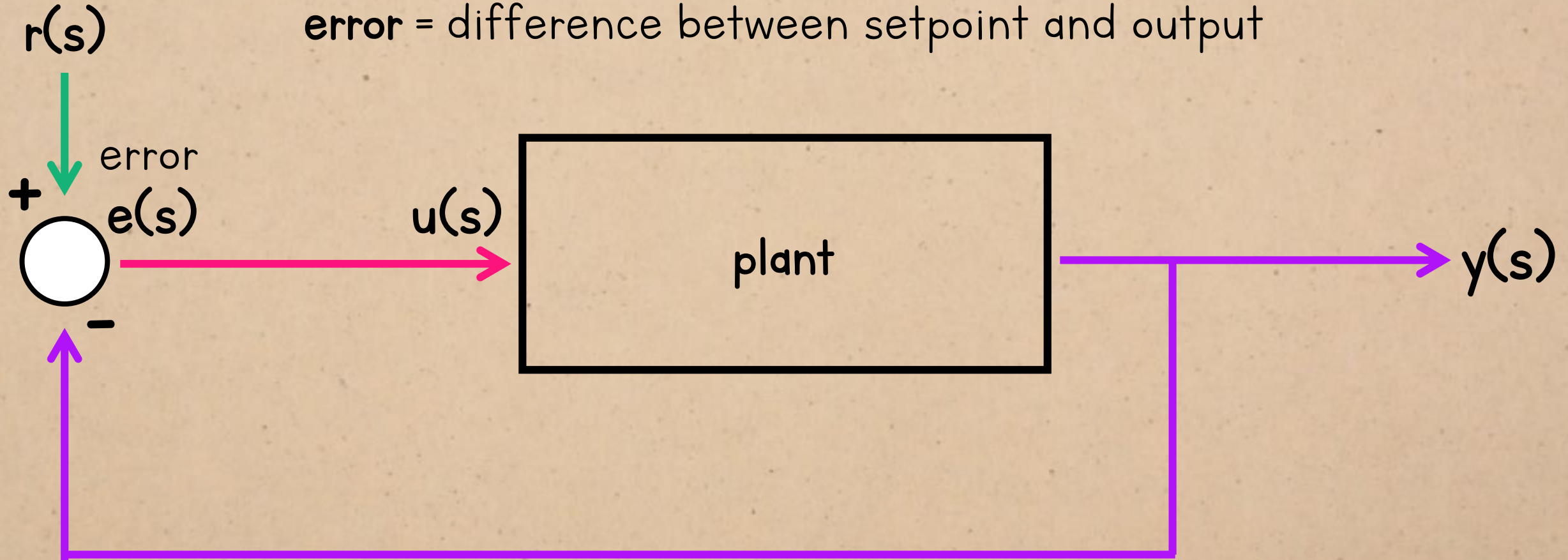
setpoint = desired value for the output

setpoint
 $r(s)$



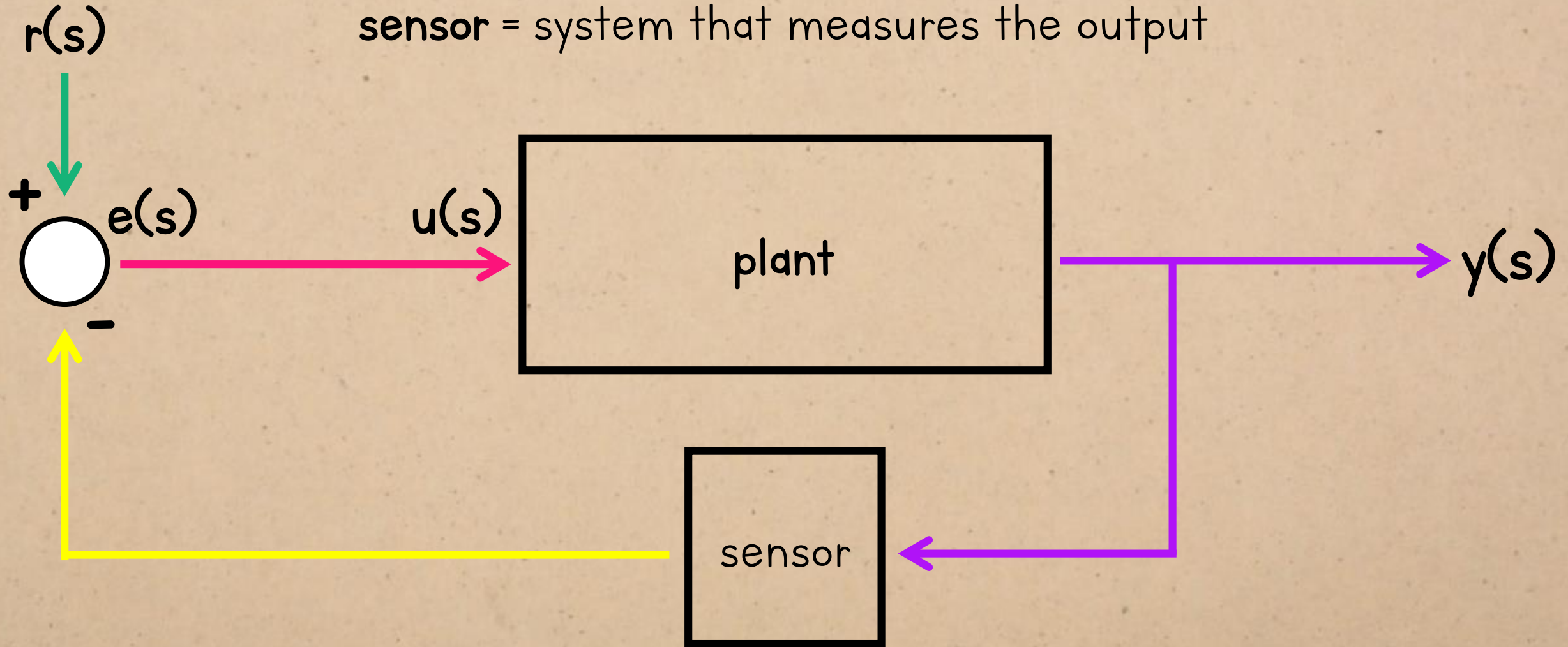
CONTROL SYSTEM

error = difference between setpoint and output



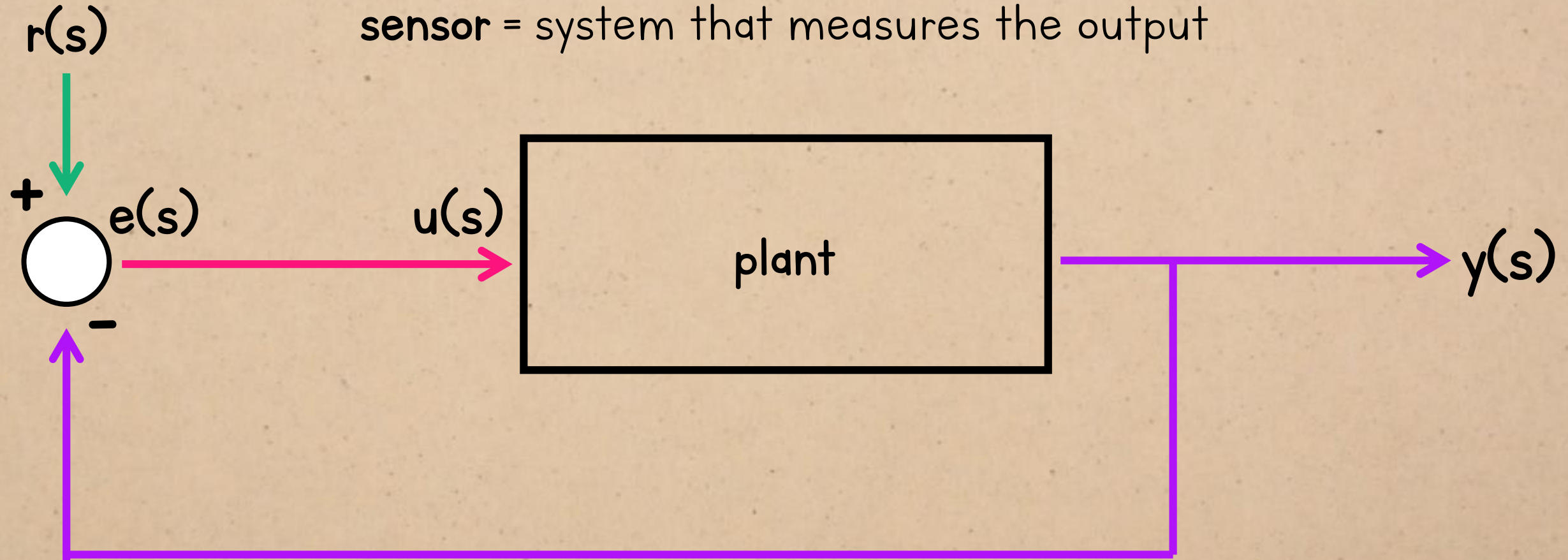
CONTROL SYSTEM

sensor = system that measures the output



CONTROL SYSTEM

sensor = system that measures the output

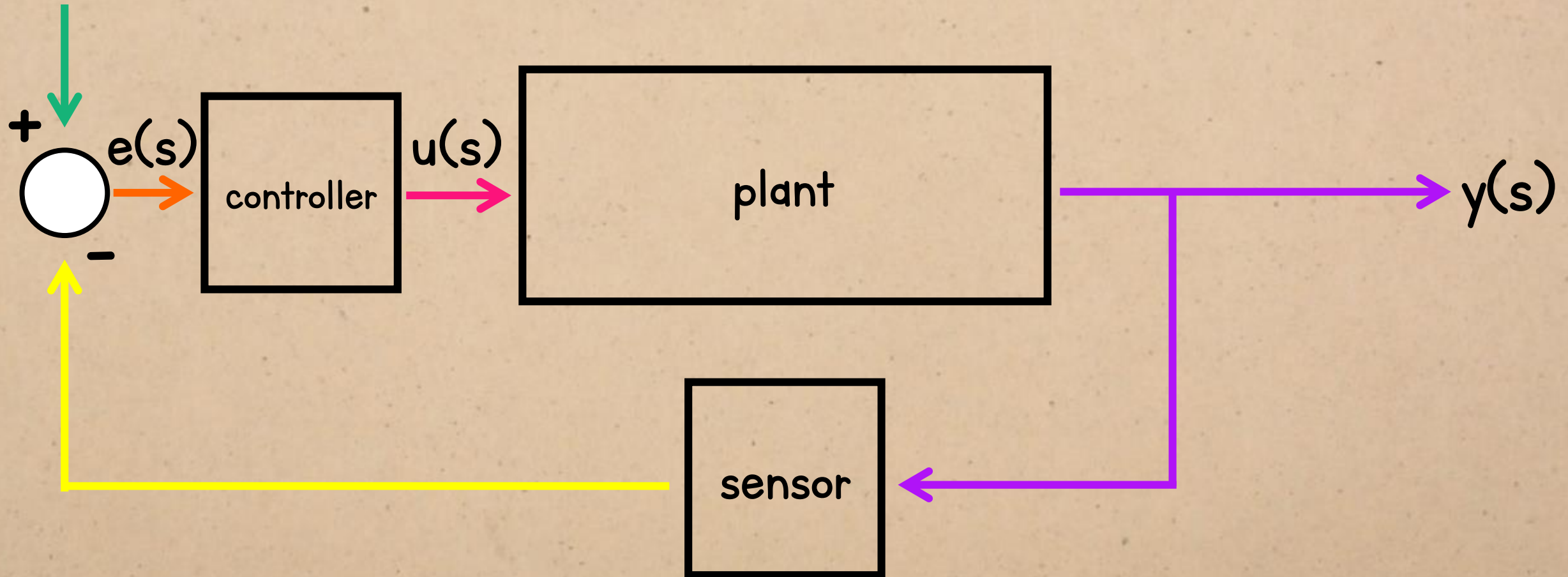


no sensor block = *perfect* sensing

i.e. output of the sensor is exactly the true output

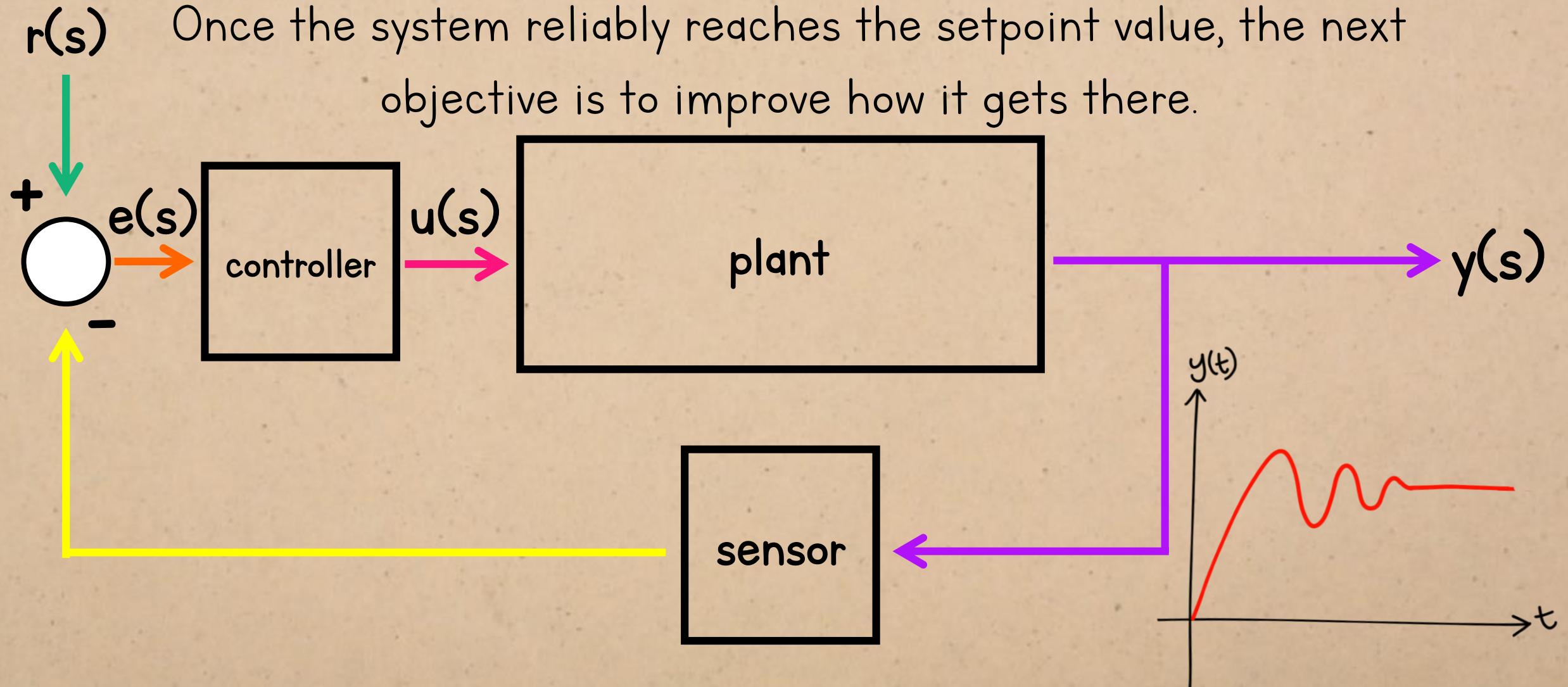
CONTROL SYSTEM

$r(s)$ controller = system that adjusts plant input in response to error

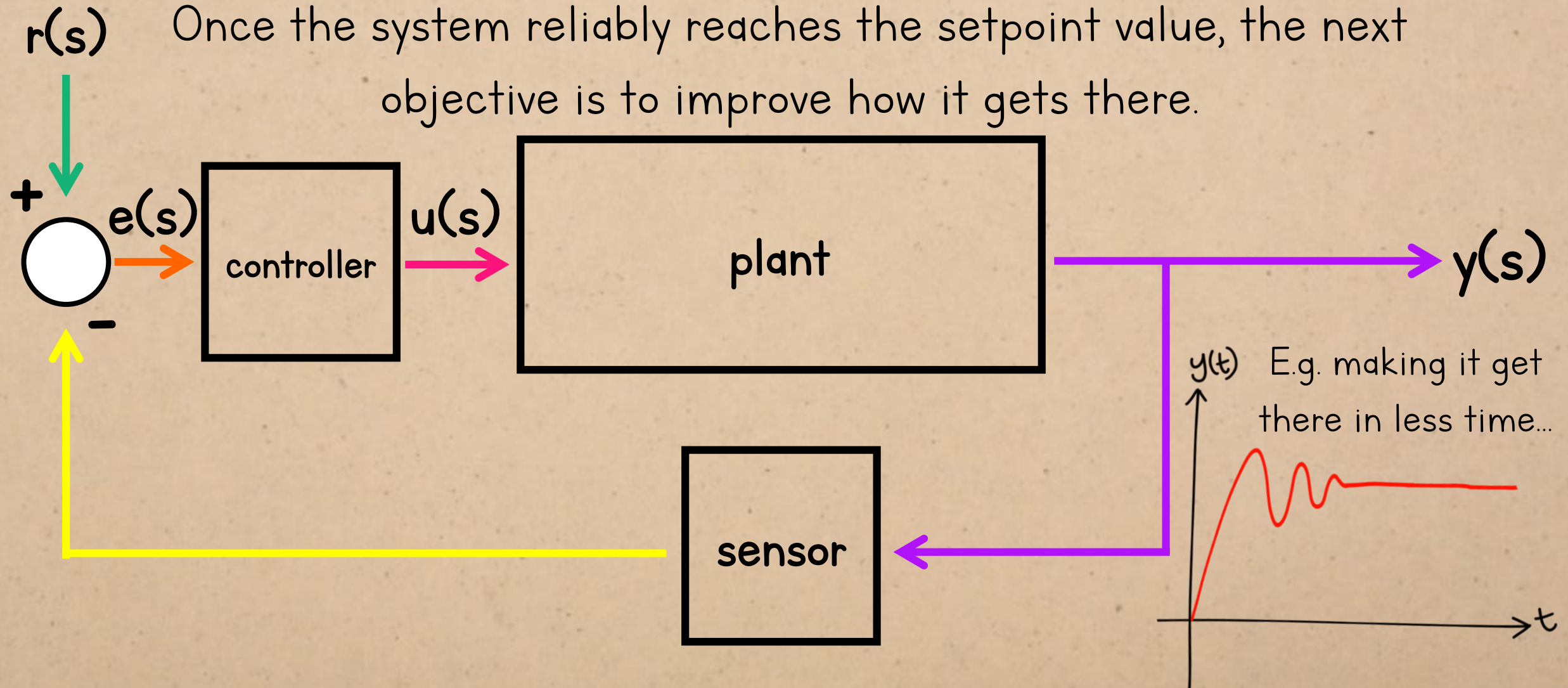


If you don't like the definition from the first lecture, you can think of control engineering as "the science and art of choosing inputs to get the outputs that we want."

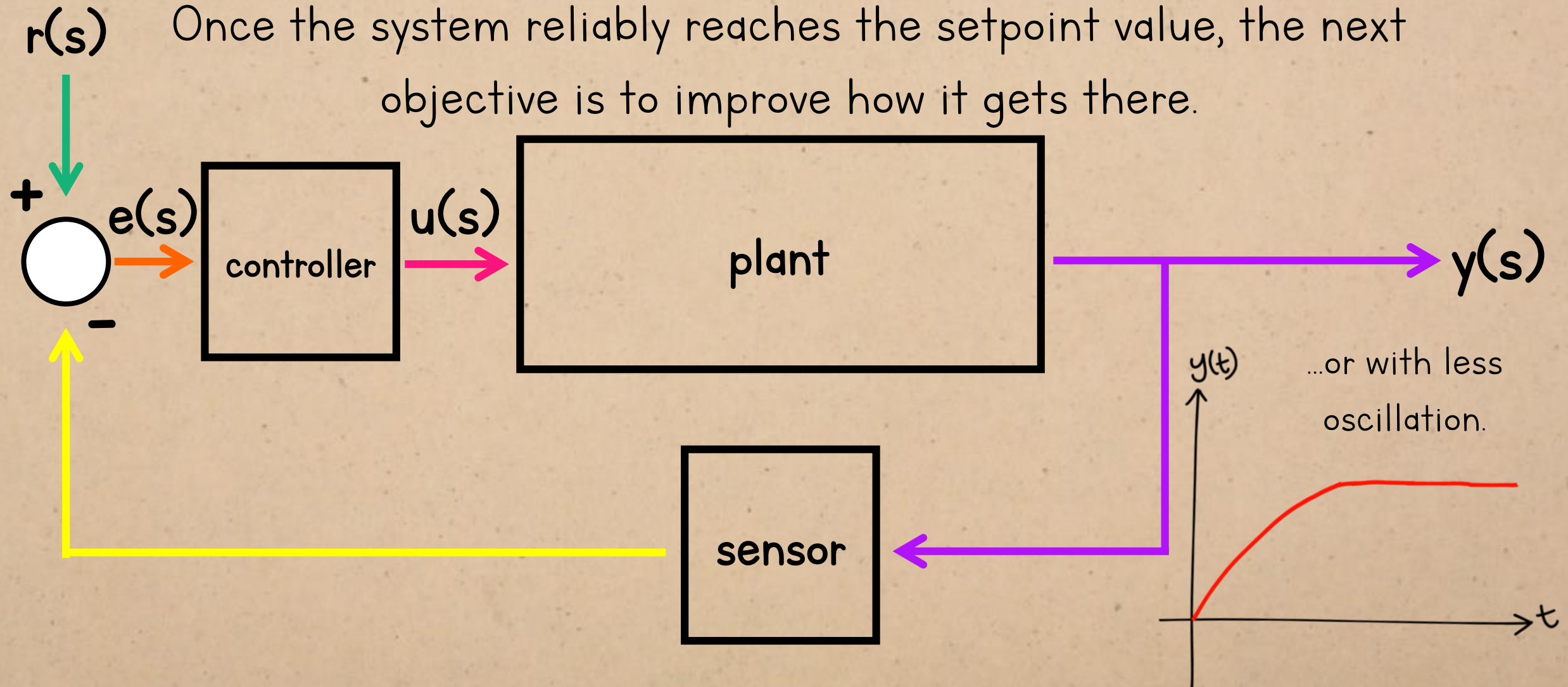
CONTROL SYSTEM



CONTROL SYSTEM

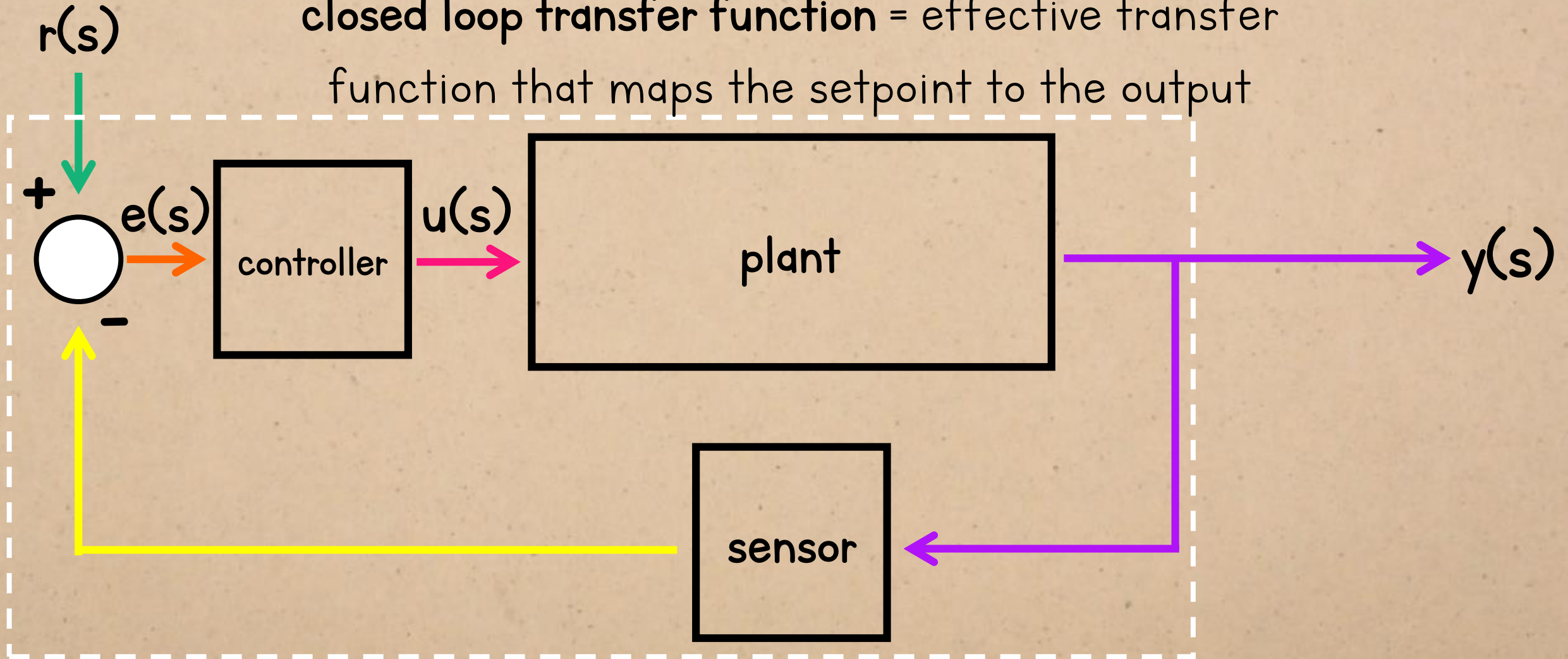


CONTROL SYSTEM



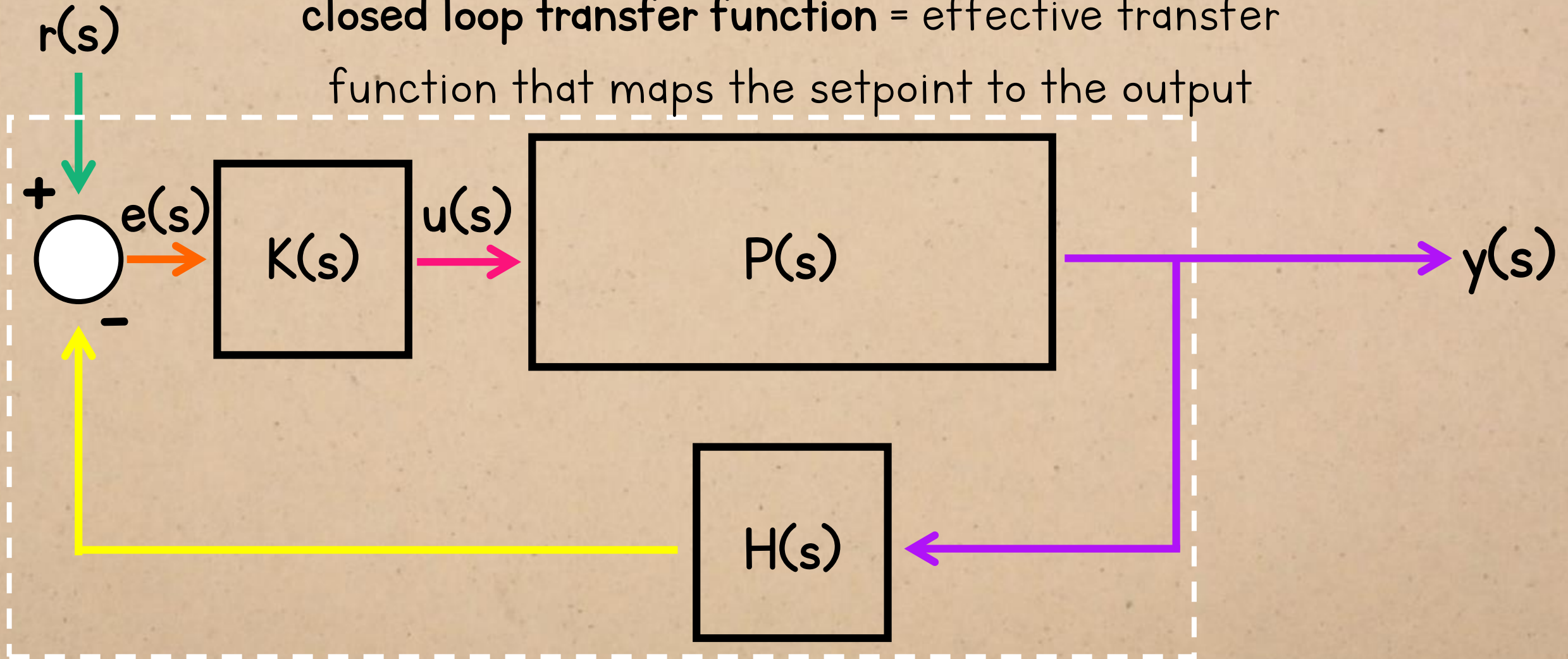
CONTROL SYSTEM

closed loop transfer function = effective transfer function that maps the setpoint to the output

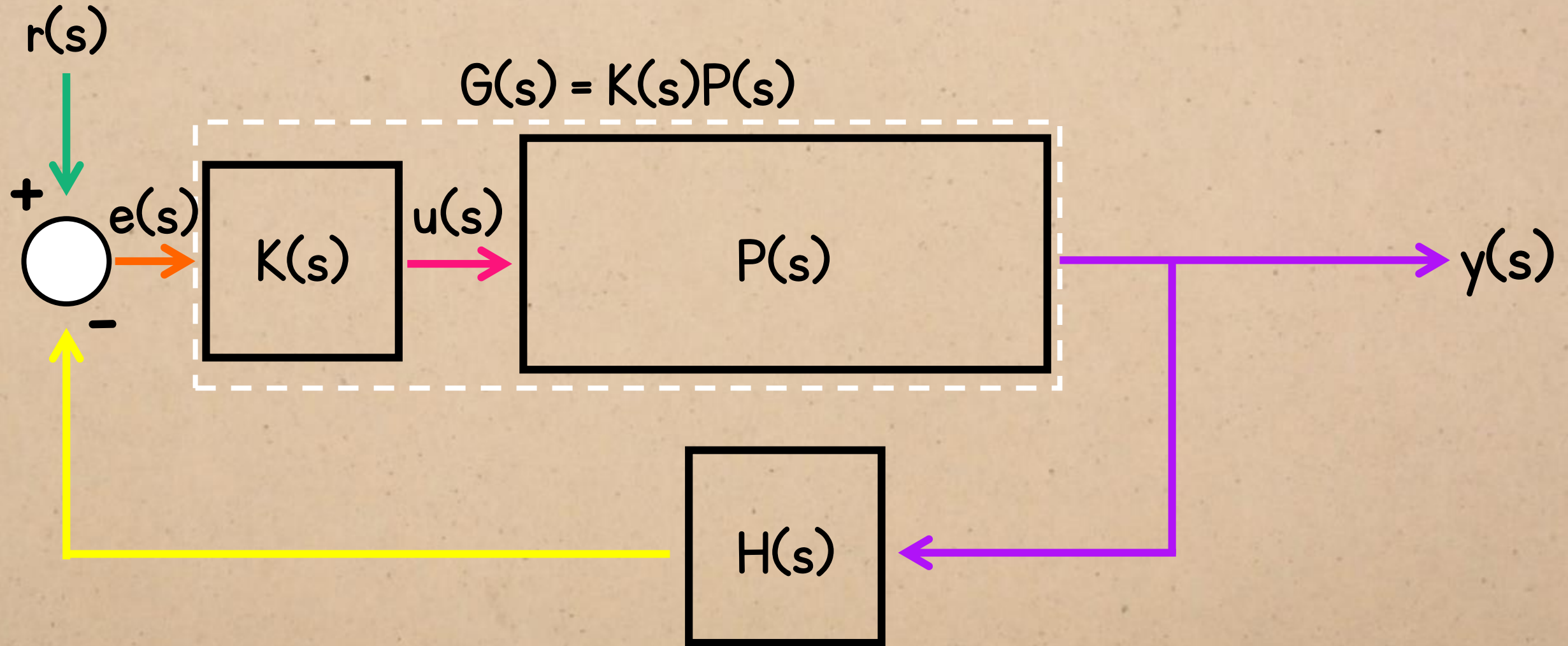


CONTROL SYSTEM

closed loop transfer function = effective transfer function that maps the setpoint to the output



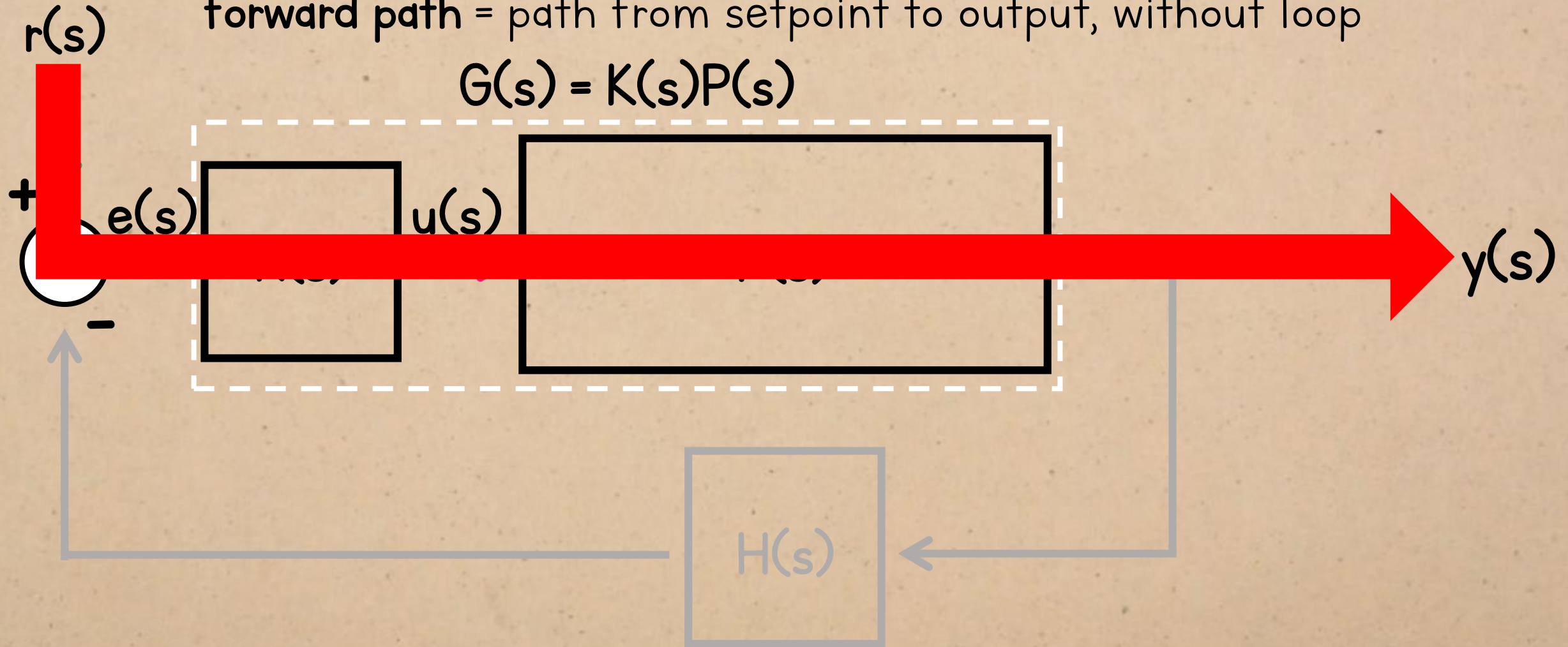
CONTROL SYSTEM



CONTROL SYSTEM

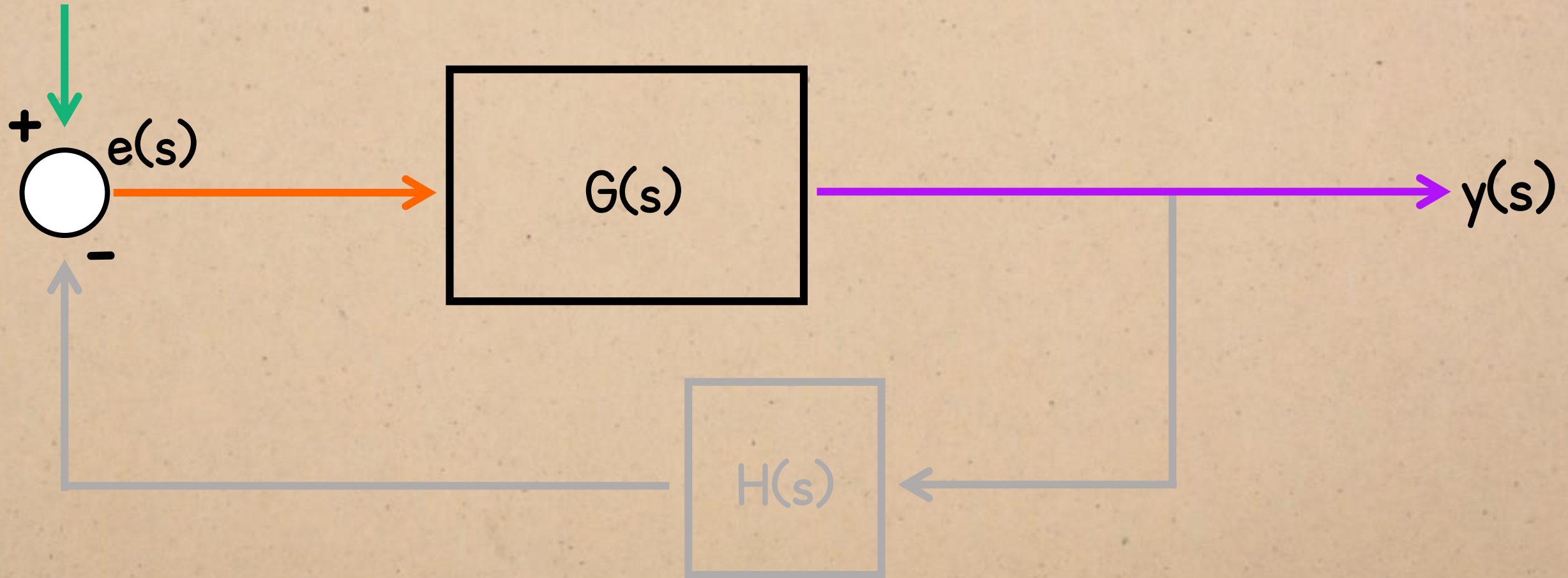
forward path = path from setpoint to output, without loop

$$G(s) = K(s)P(s)$$

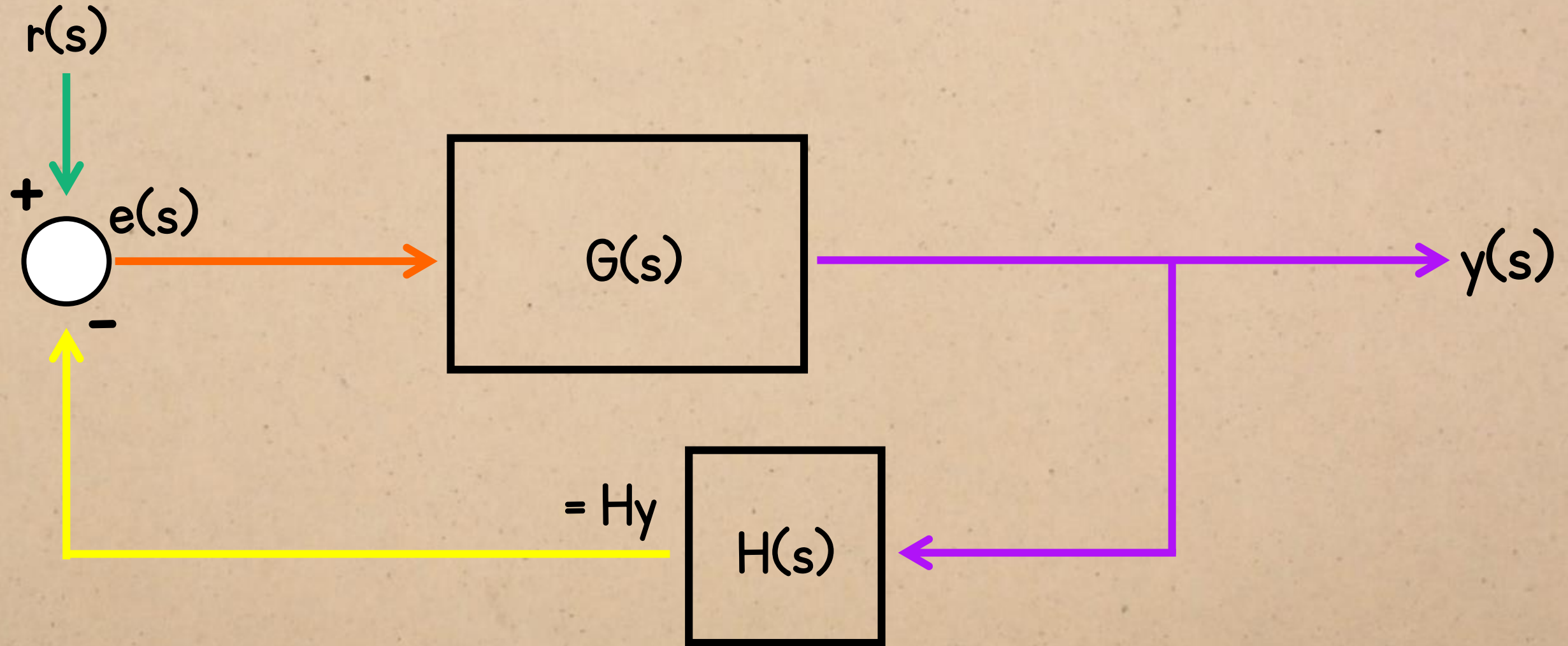


CONTROL SYSTEM

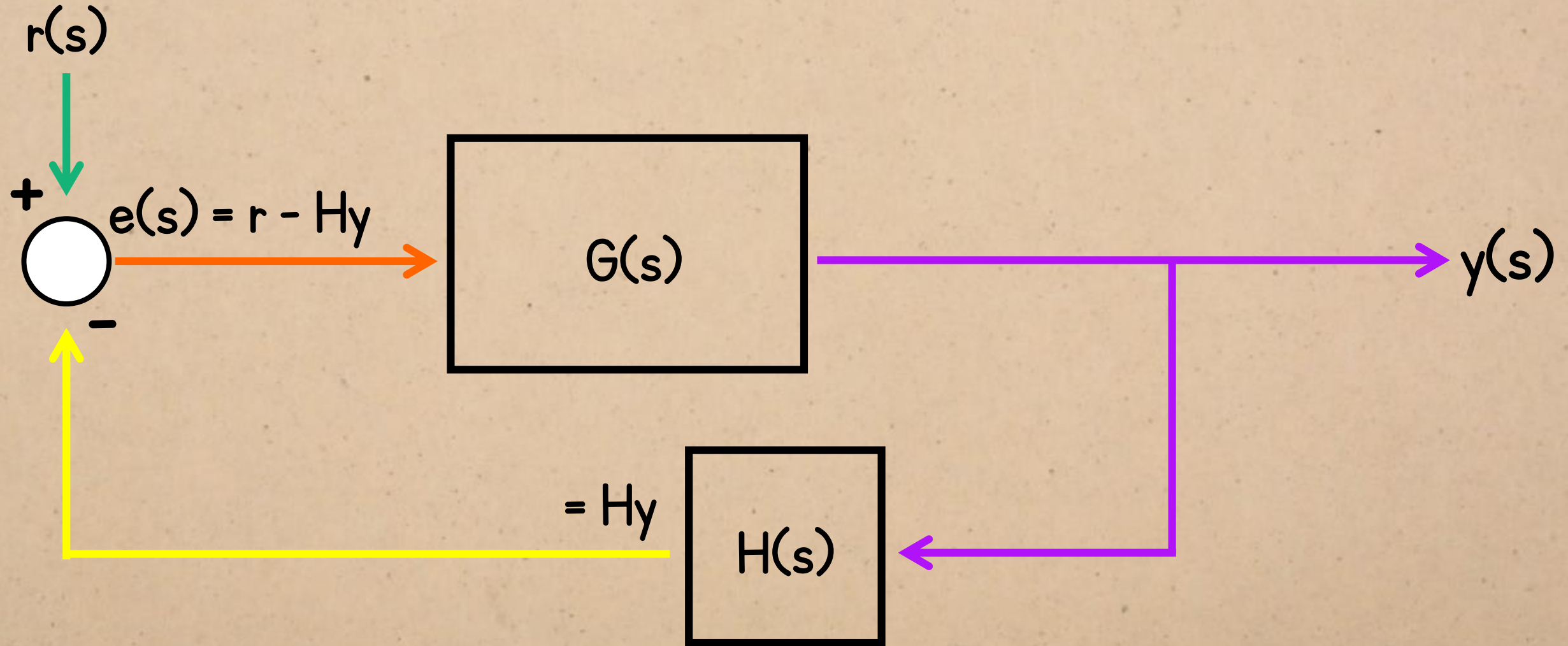
$r(s)$ open loop transfer function = transfer function of forward path



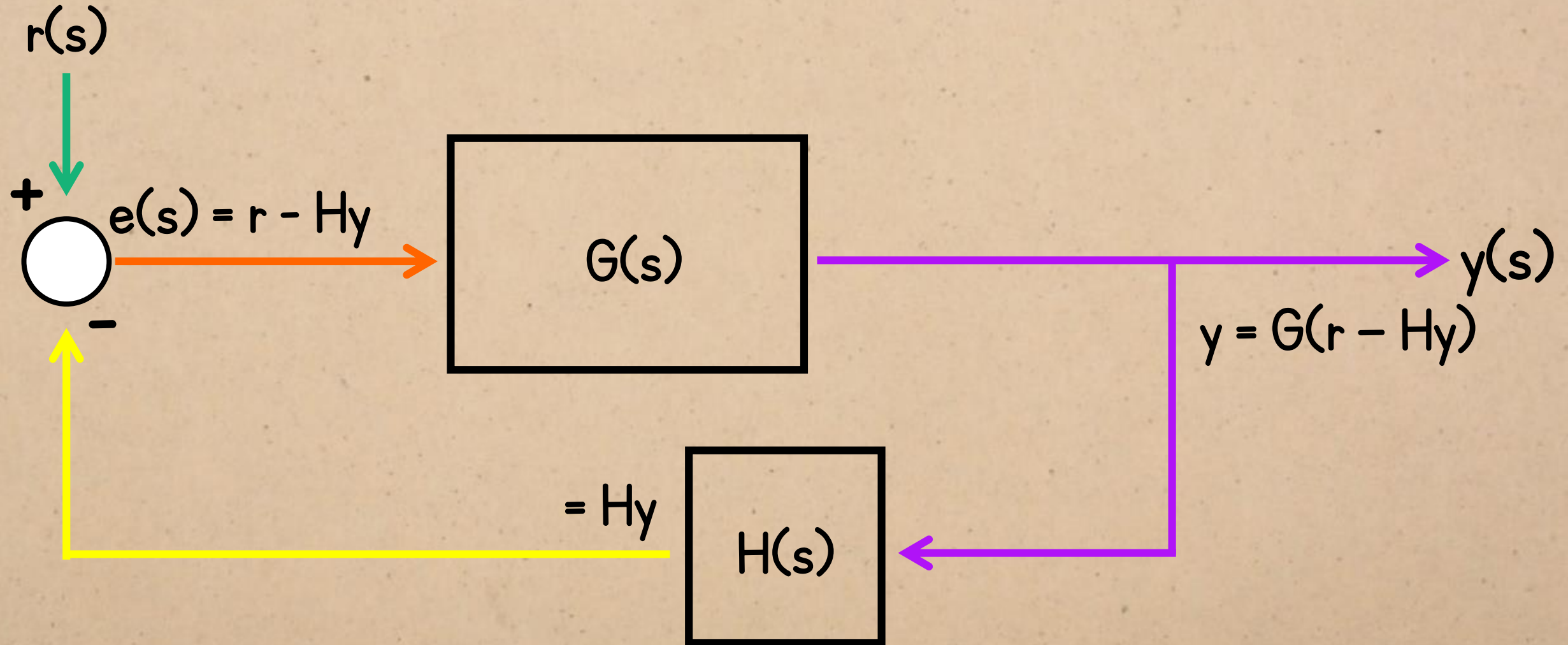
CONTROL SYSTEM



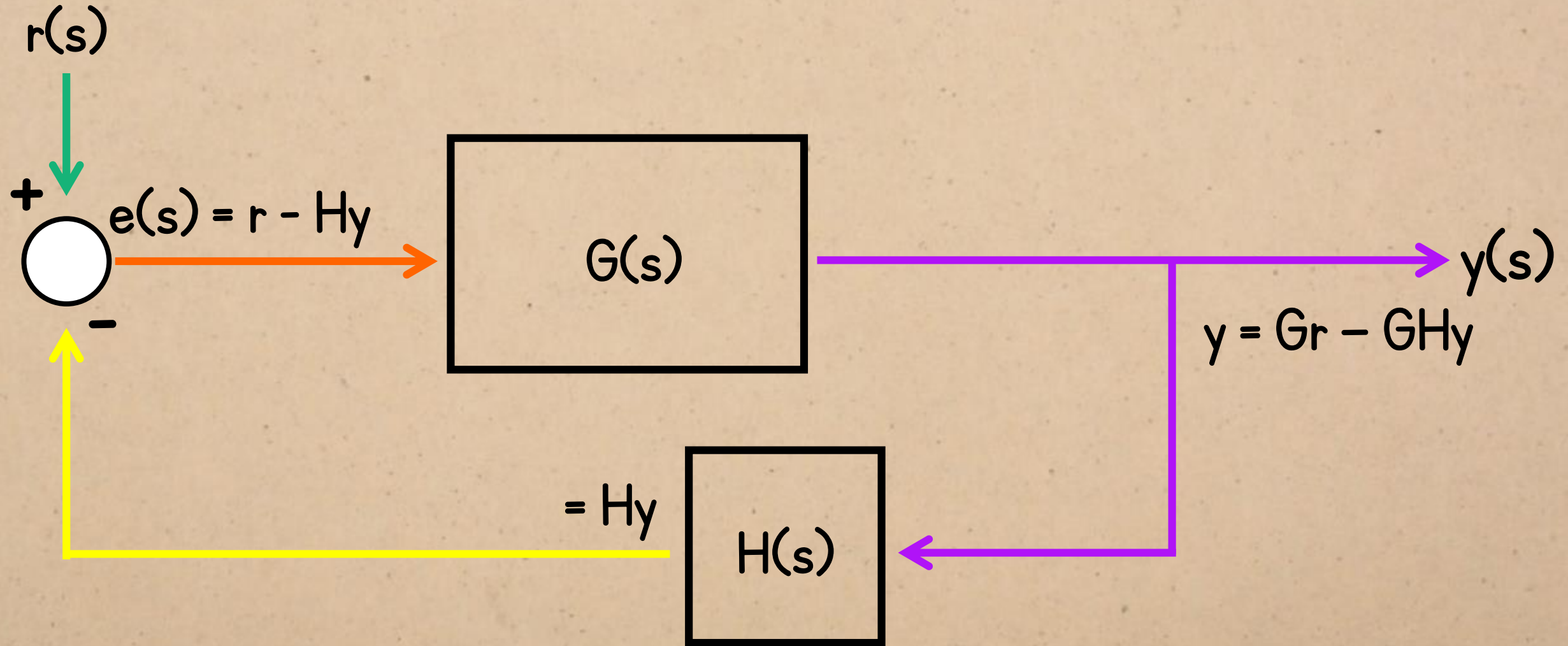
CONTROL SYSTEM



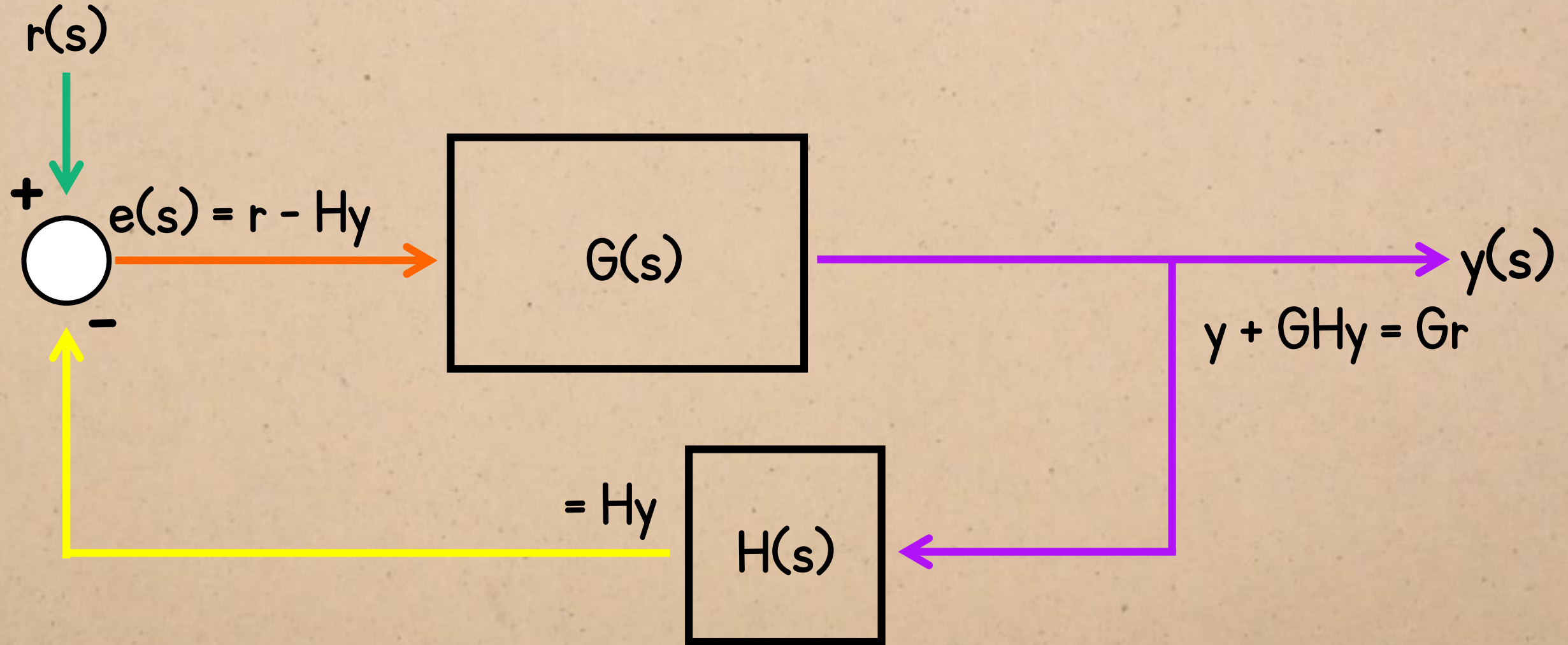
CONTROL SYSTEM



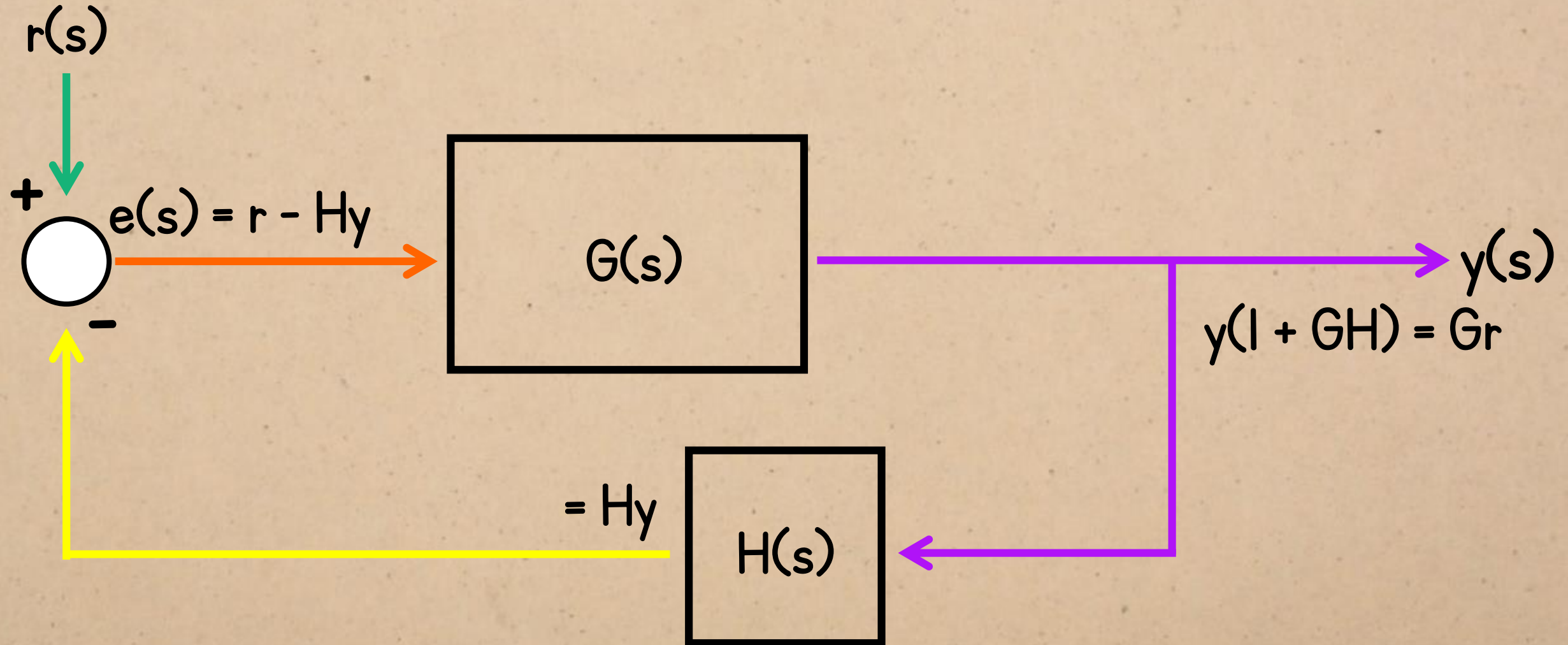
CONTROL SYSTEM



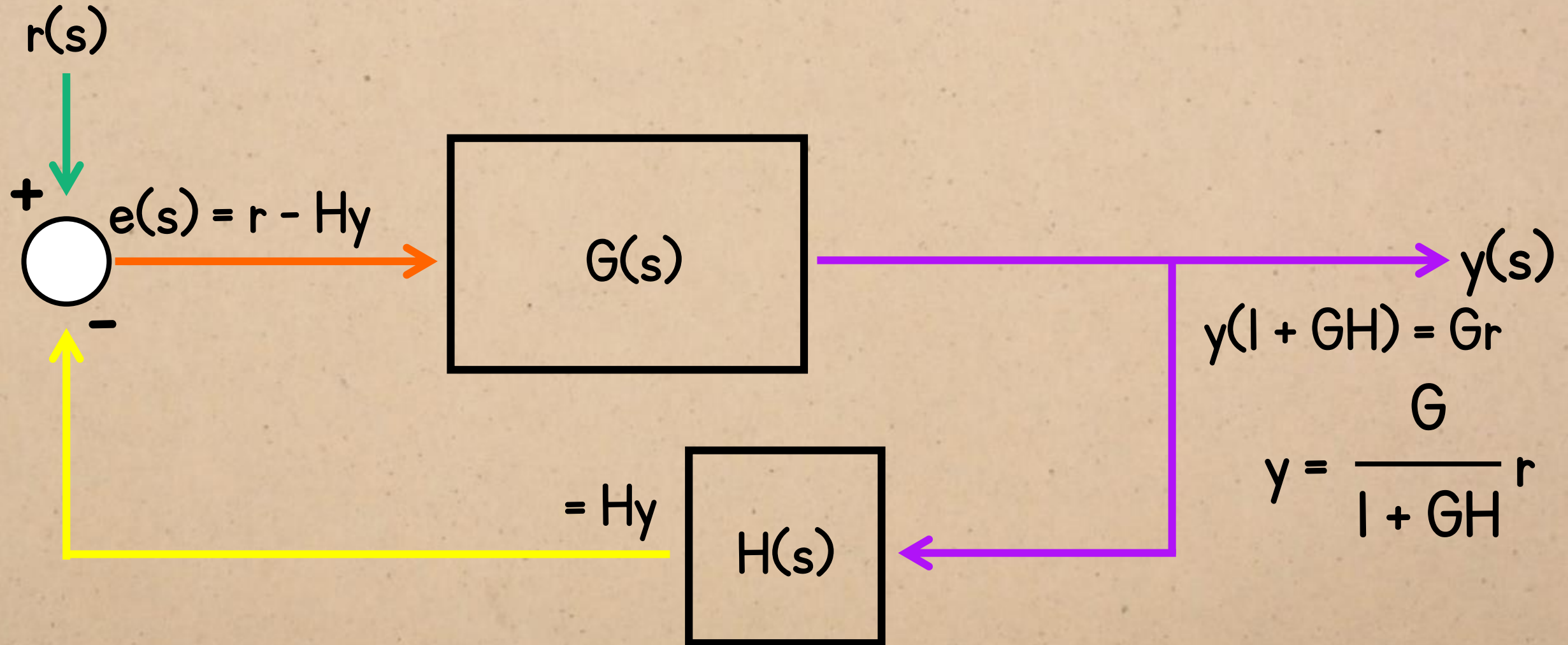
CONTROL SYSTEM



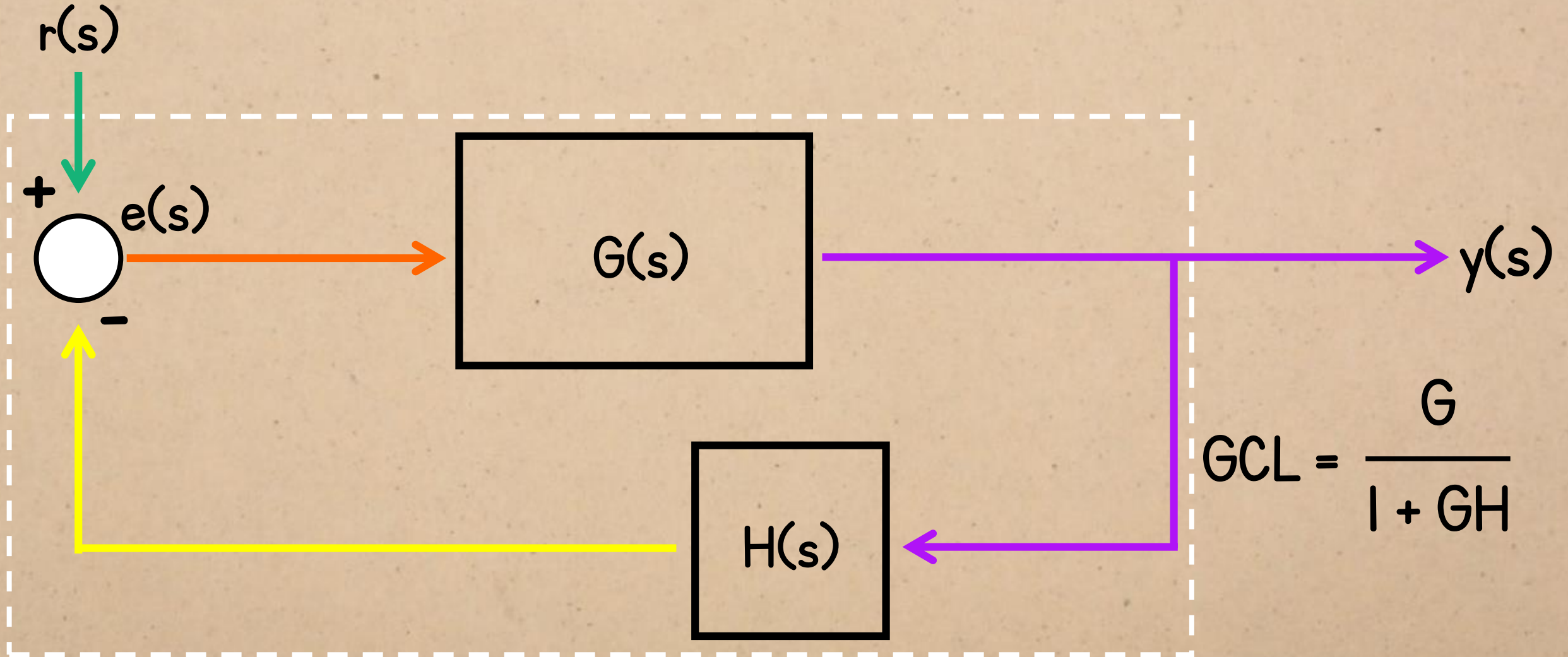
CONTROL SYSTEM



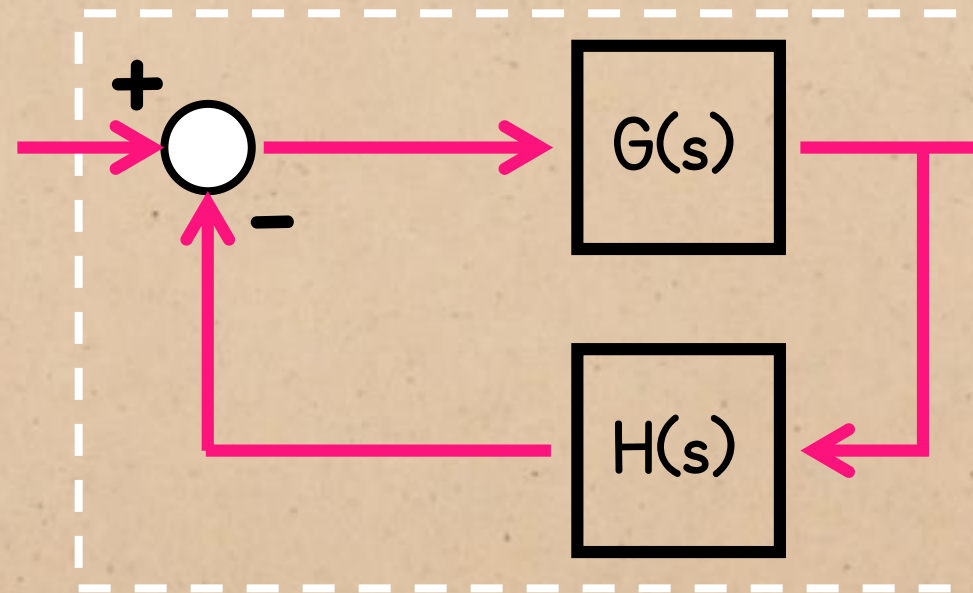
CONTROL SYSTEM



CONTROL SYSTEM



CONTROL SYSTEM

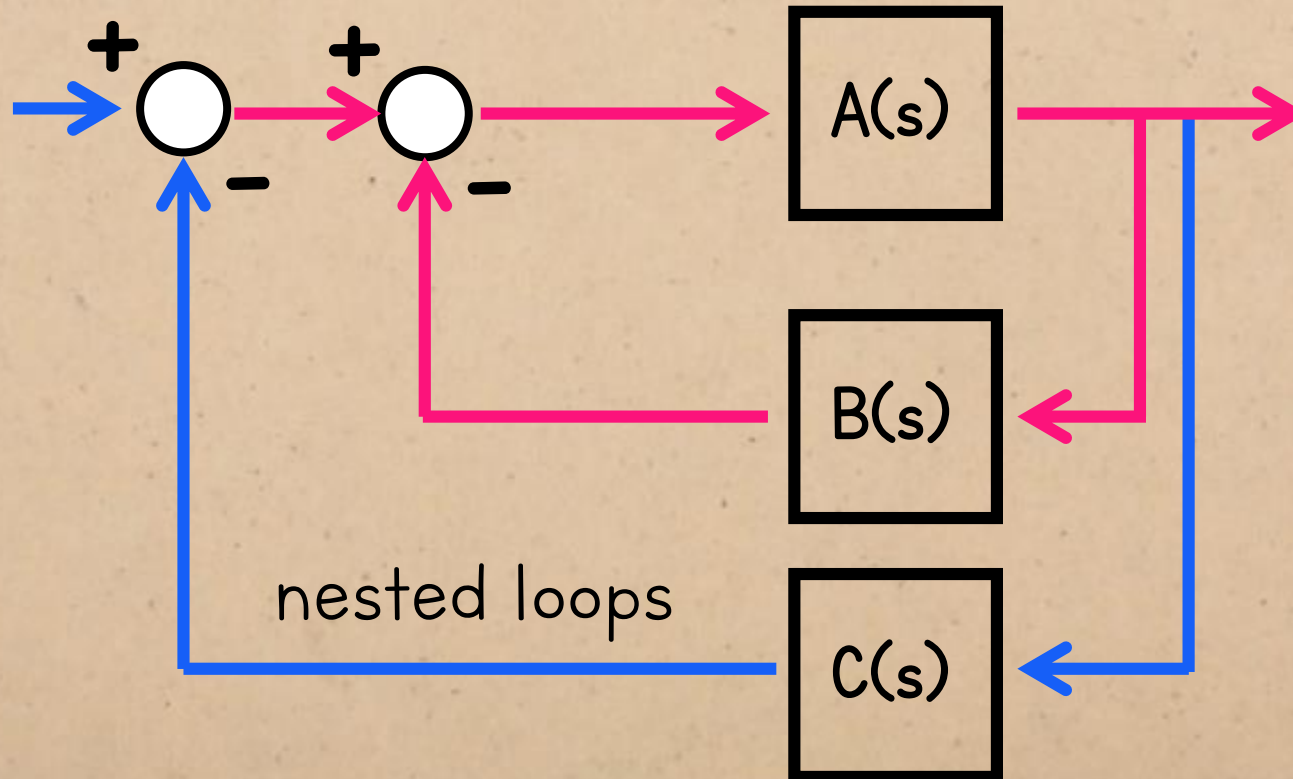


effective transfer
function for basic loop

$$GCL = \frac{G}{1 + GH}$$

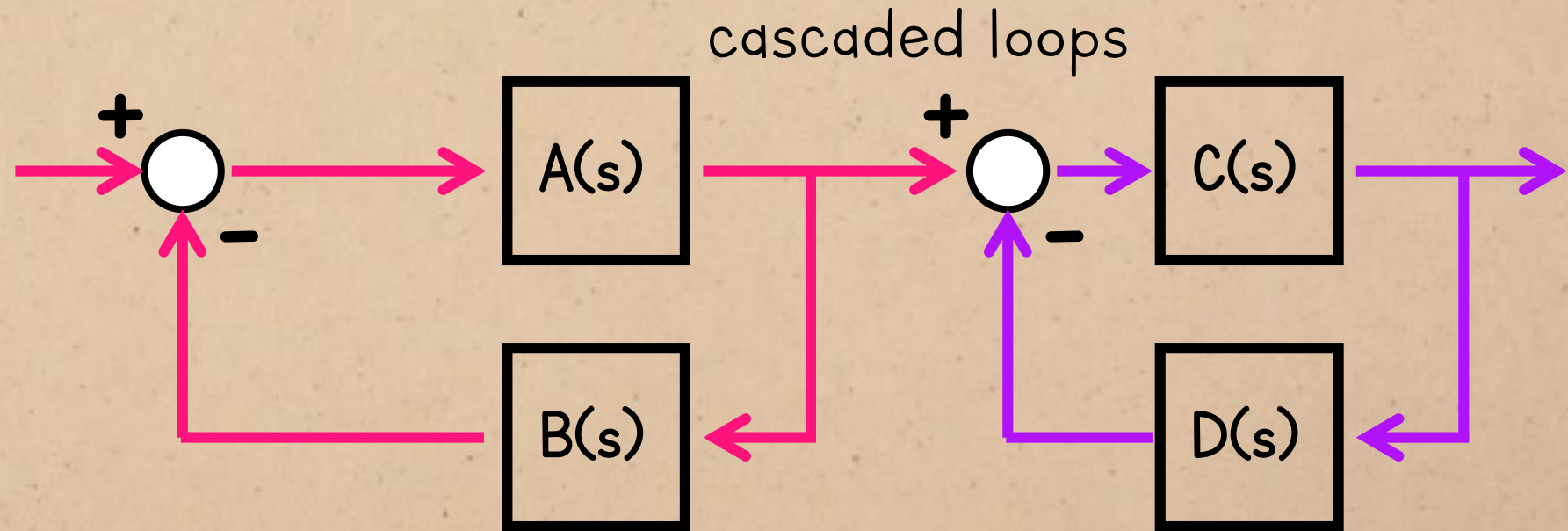
CONTROL SYSTEM

Some more complicated loop formations:

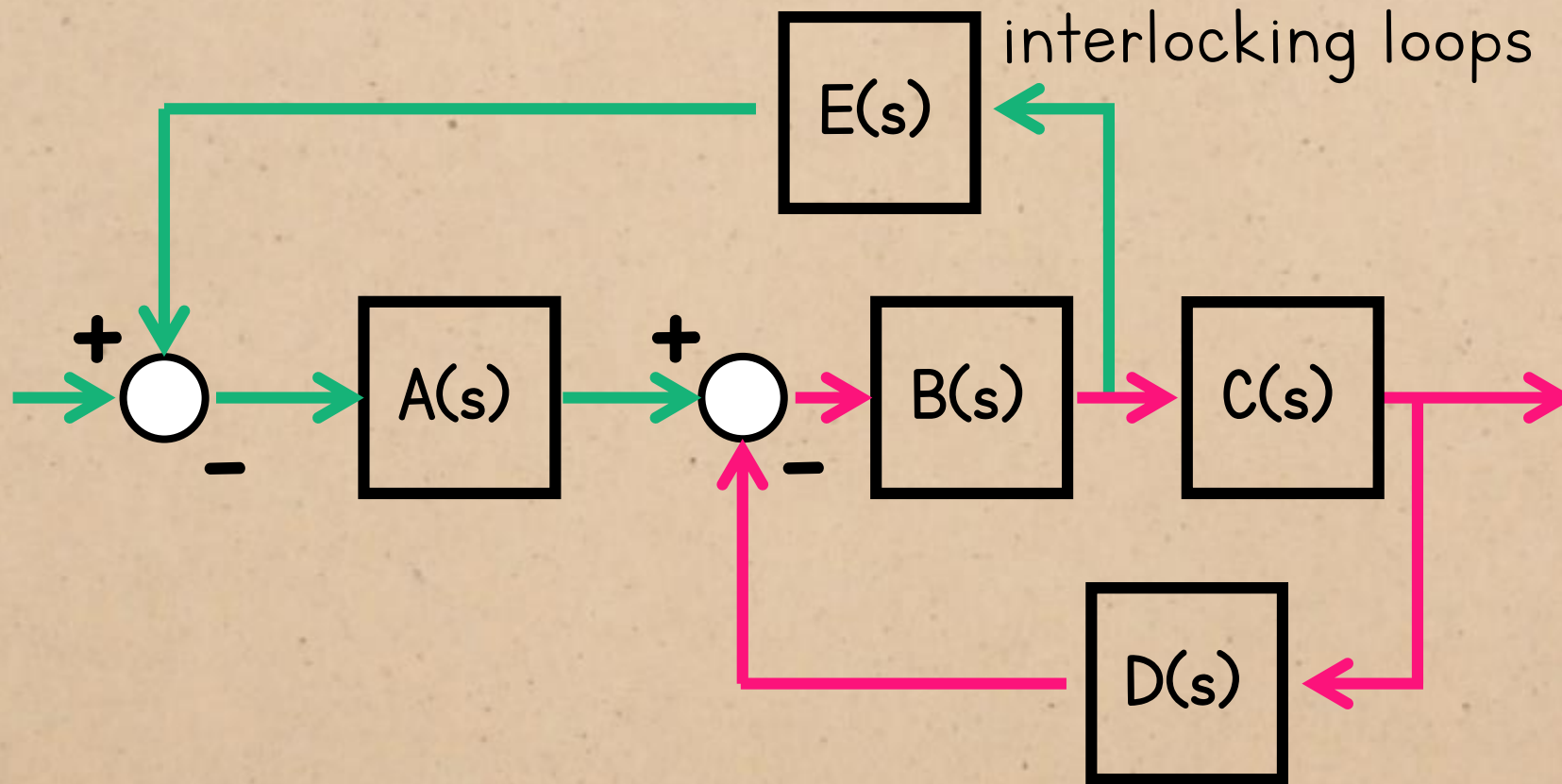


CONTROL SYSTEM

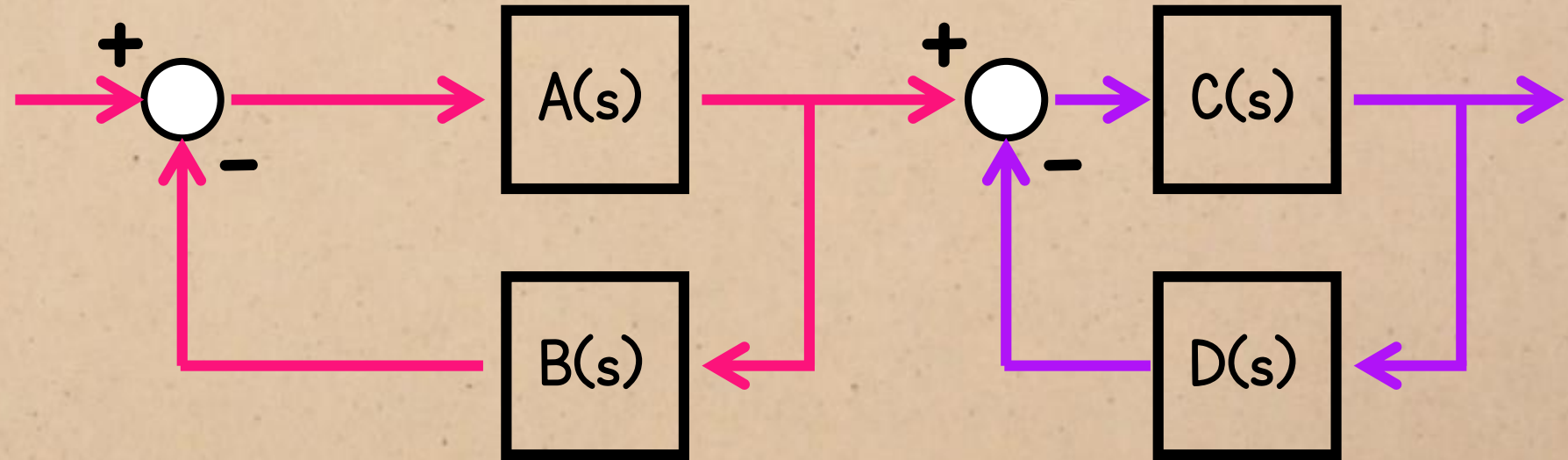
Some more complicated loop formations:



CONTROL SYSTEM



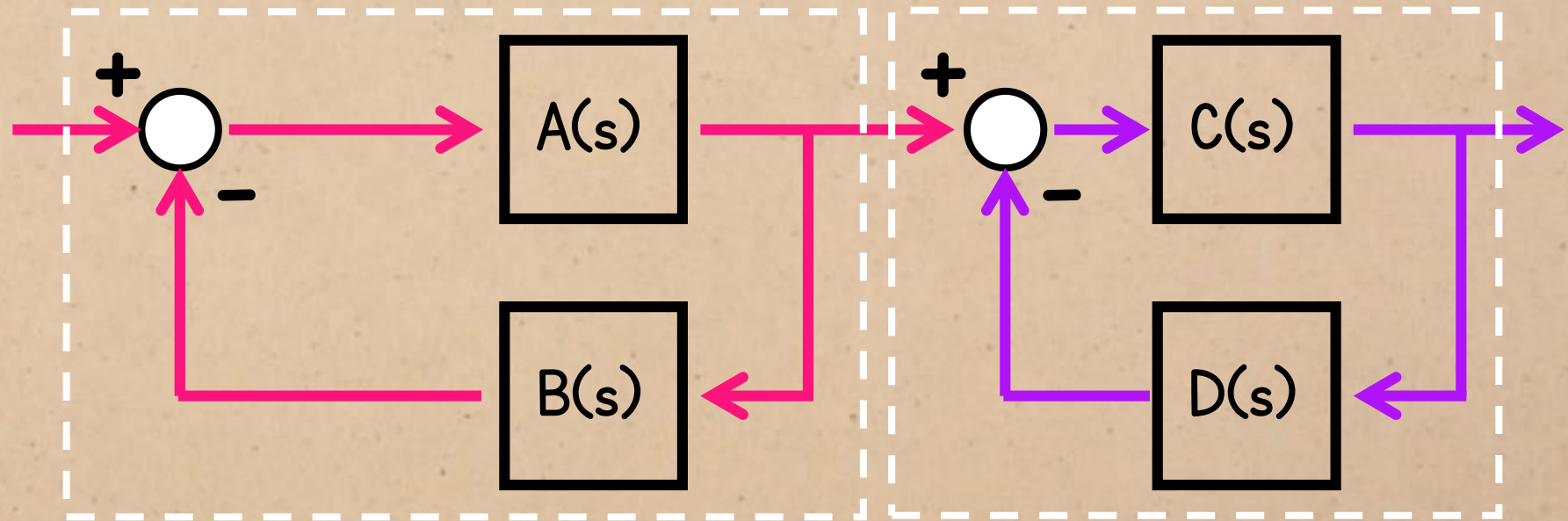
CASCADED LOOPS



CASCADED LOOPS

$$= \frac{A}{1 + AB}$$

$$= \frac{C}{1 + CD}$$

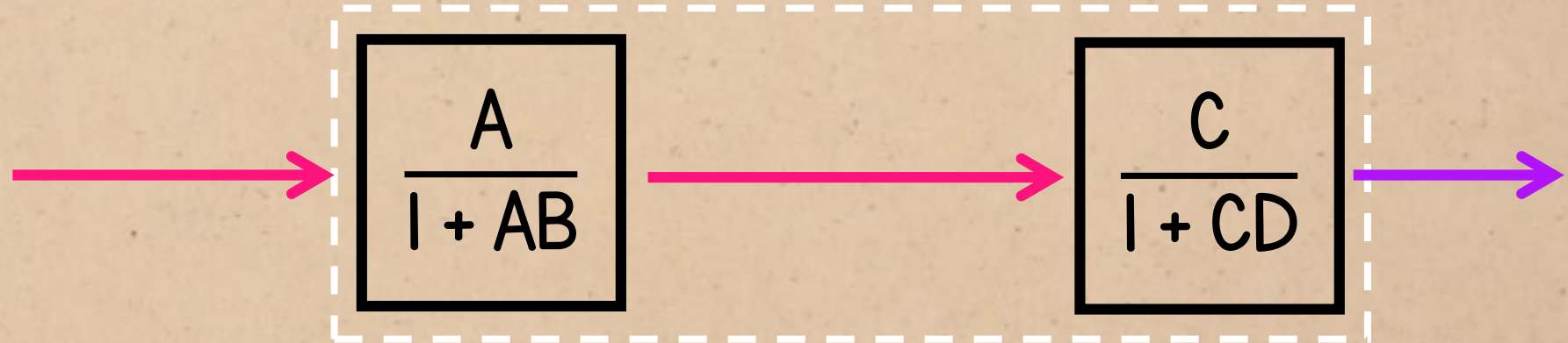


CASCADED LOOPS

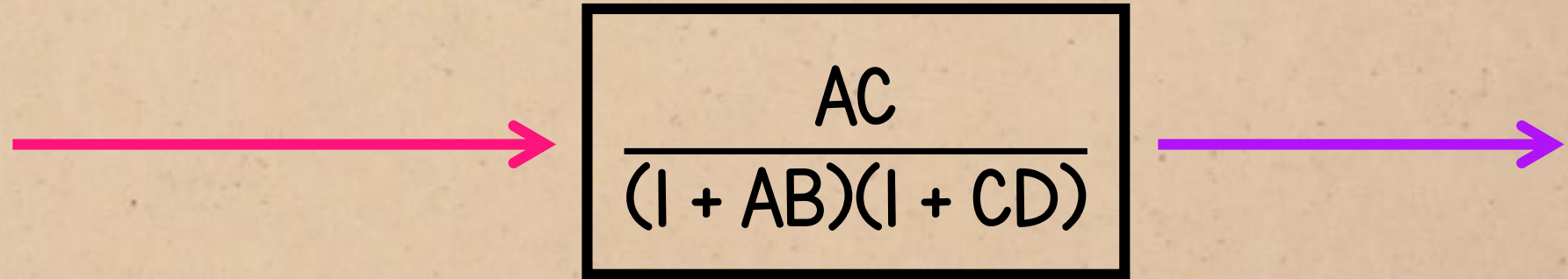


CASCADED LOOPS

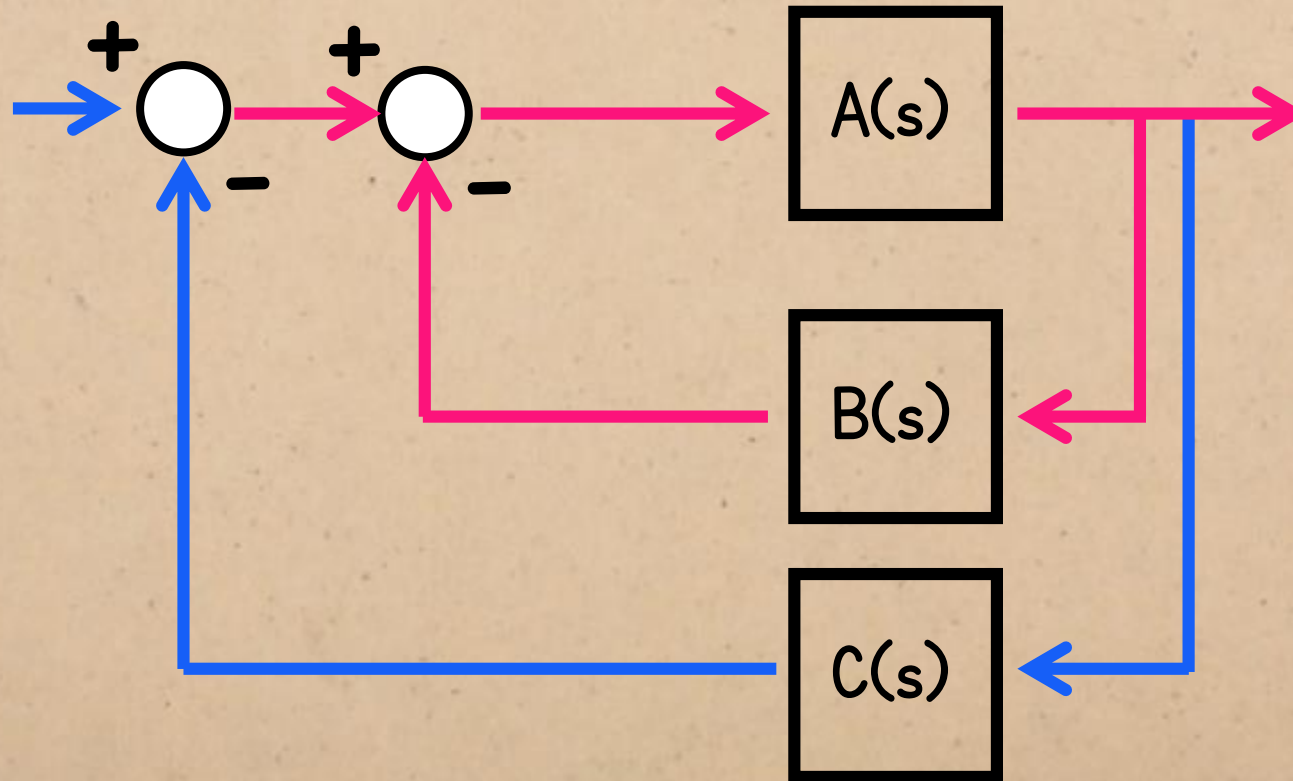
$$= \frac{A}{1 + AB} \times \frac{C}{1 + CD}$$



CASCADED LOOPS

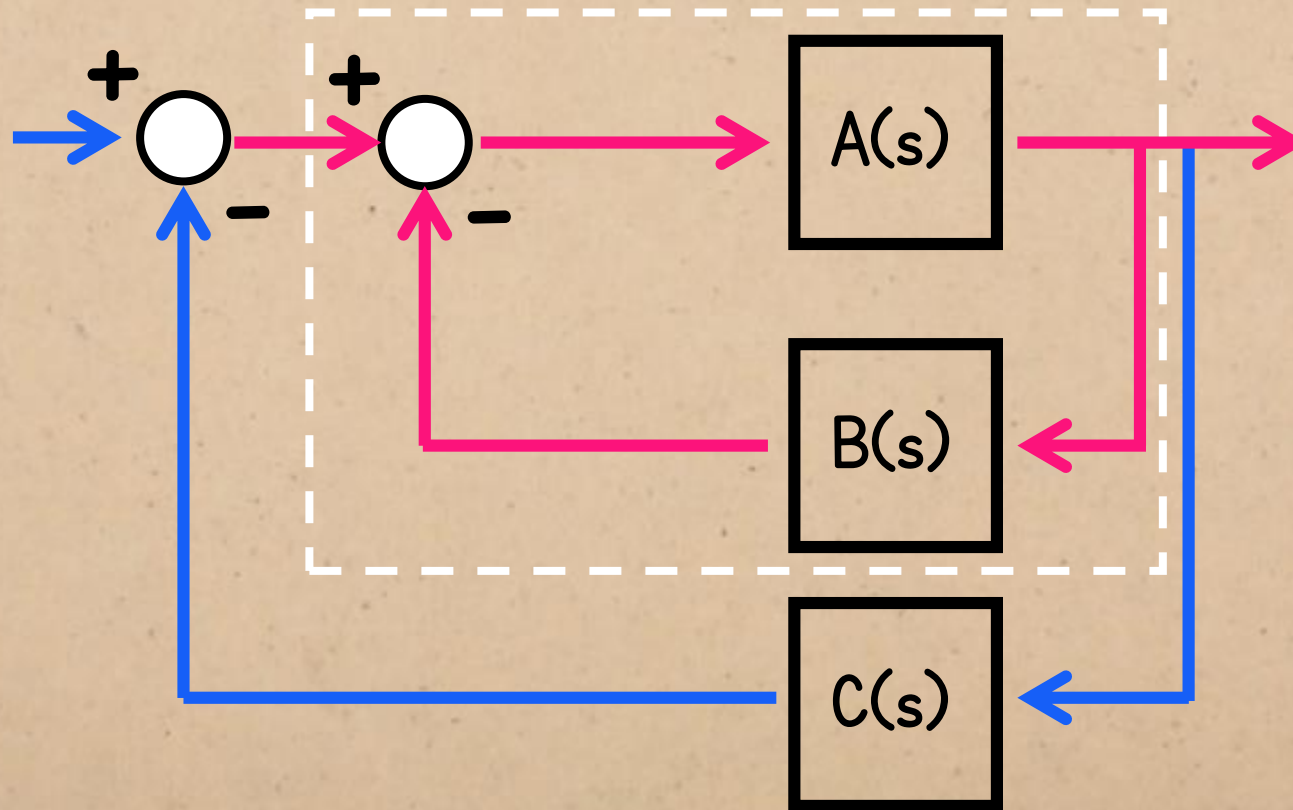


NESTED LOOPS

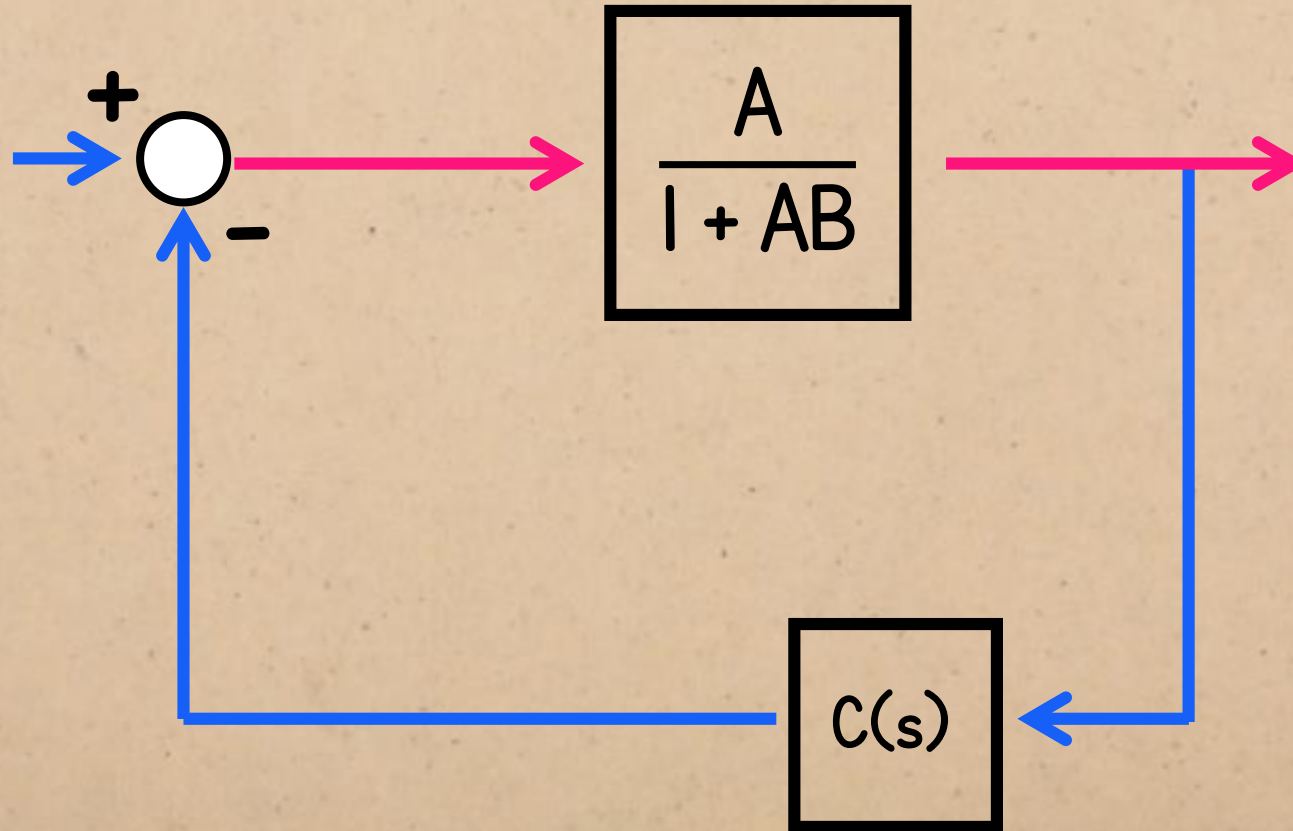


NESTED LOOPS

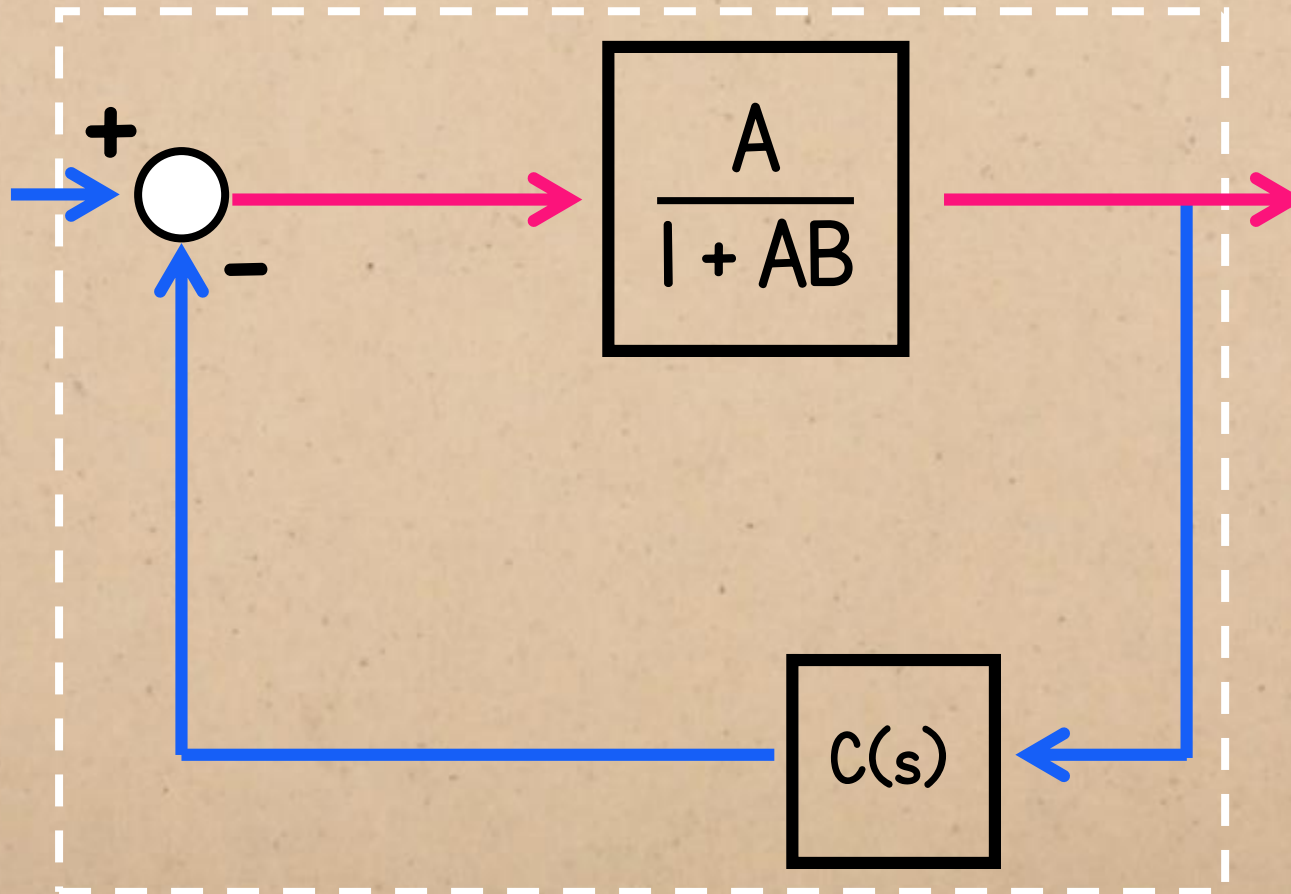
$$= \frac{A}{1 + AB}$$



NESTED LOOPS

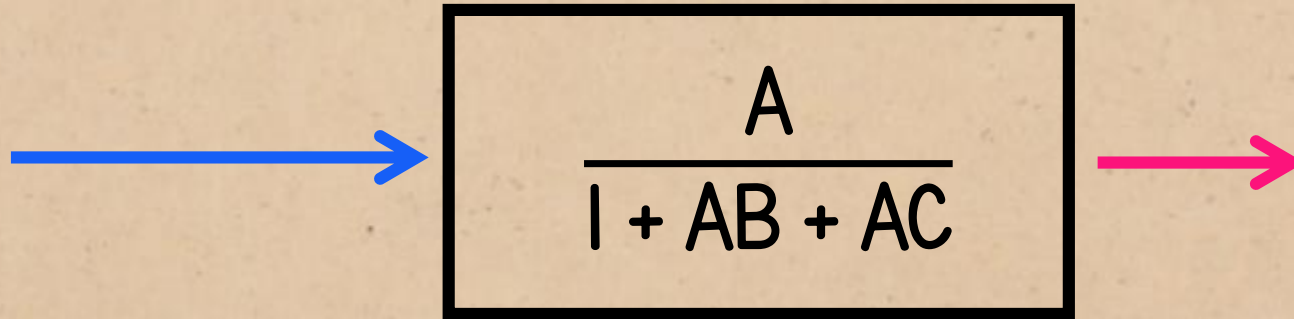


NESTED LOOPS

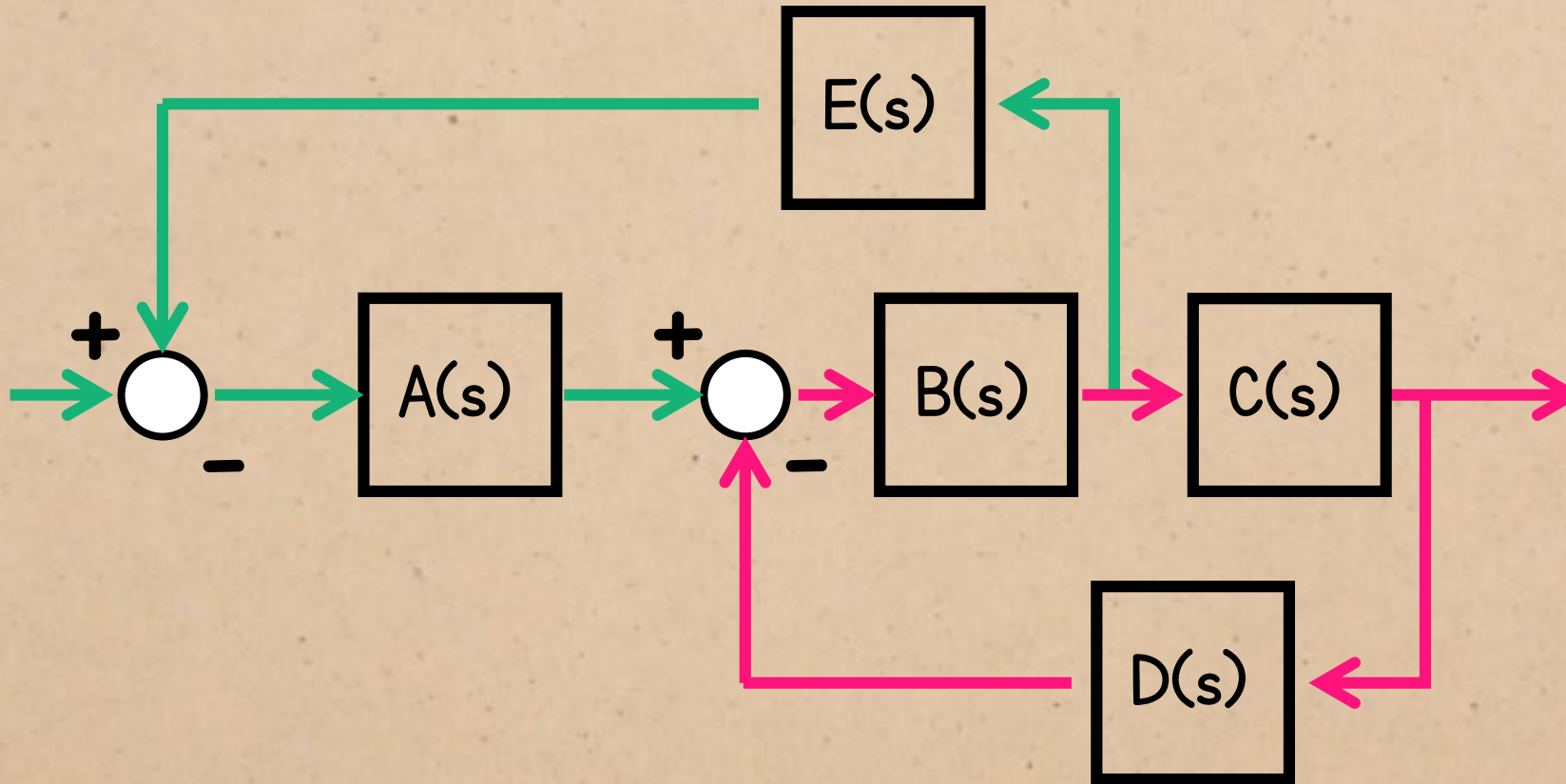


$$= \frac{\frac{A}{1 + AB}}{1 + \left(\frac{A}{1 + AB} \right) C}$$

NESTED LOOPS

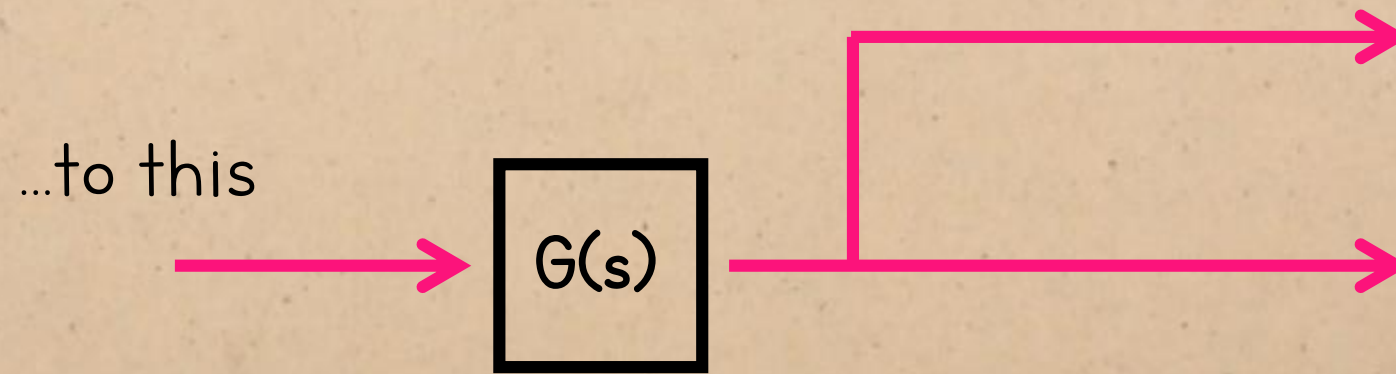
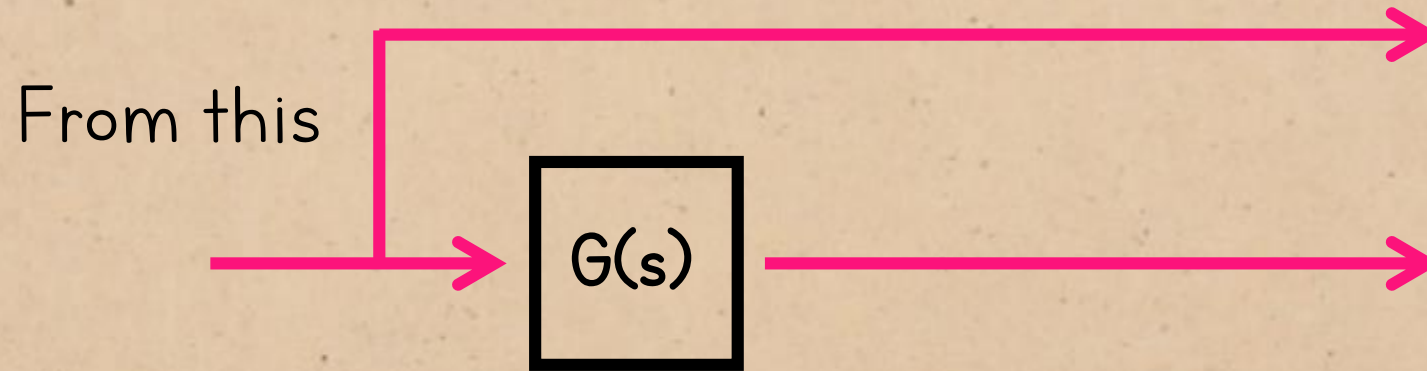


INTERLOCKING LOOPS



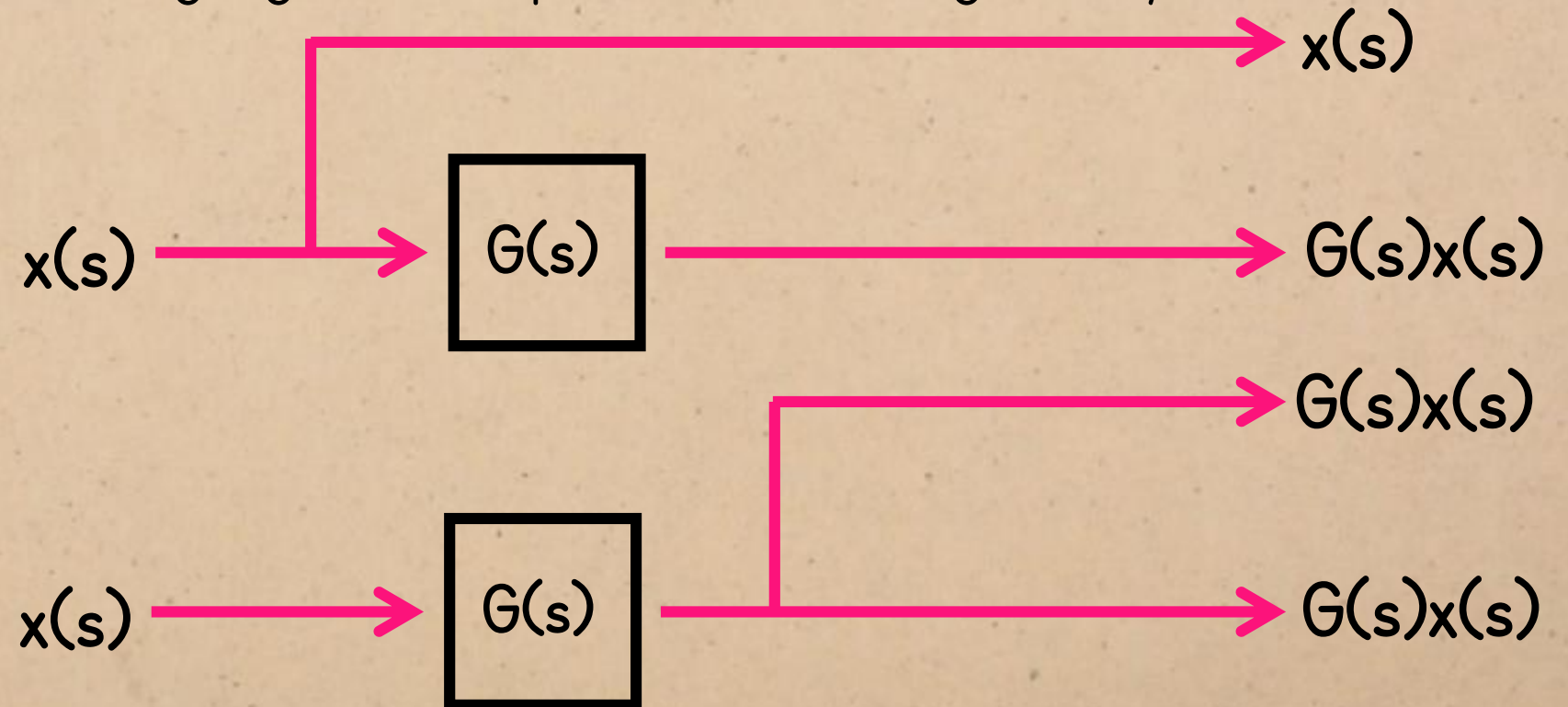
We need some new techniques to deal with interlocking loops...

First, **MOVING A
TAKE-OFF POINT**



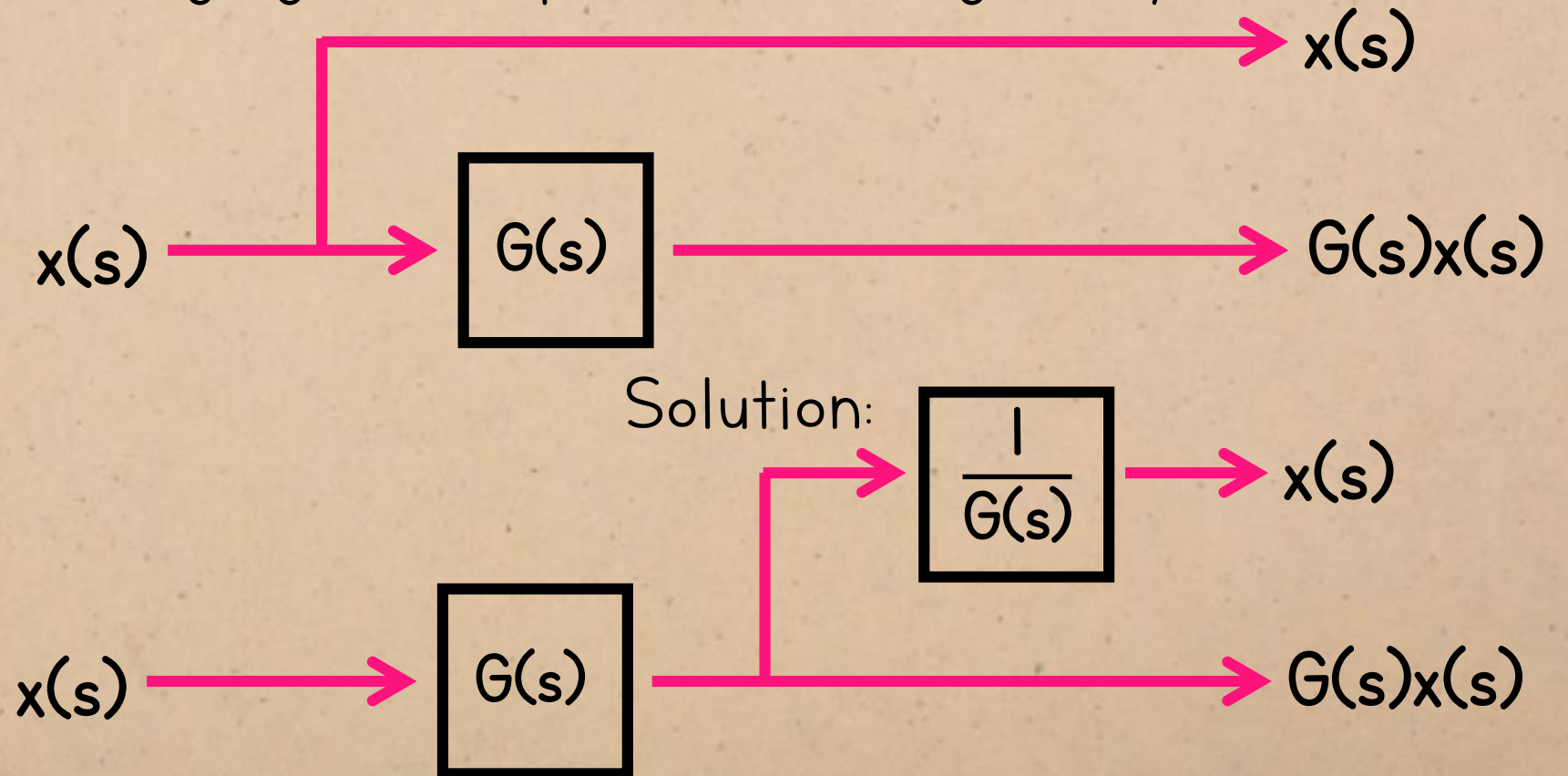
MOVING A TAKE-OFF POINT

...without changing the outputs of the original system.

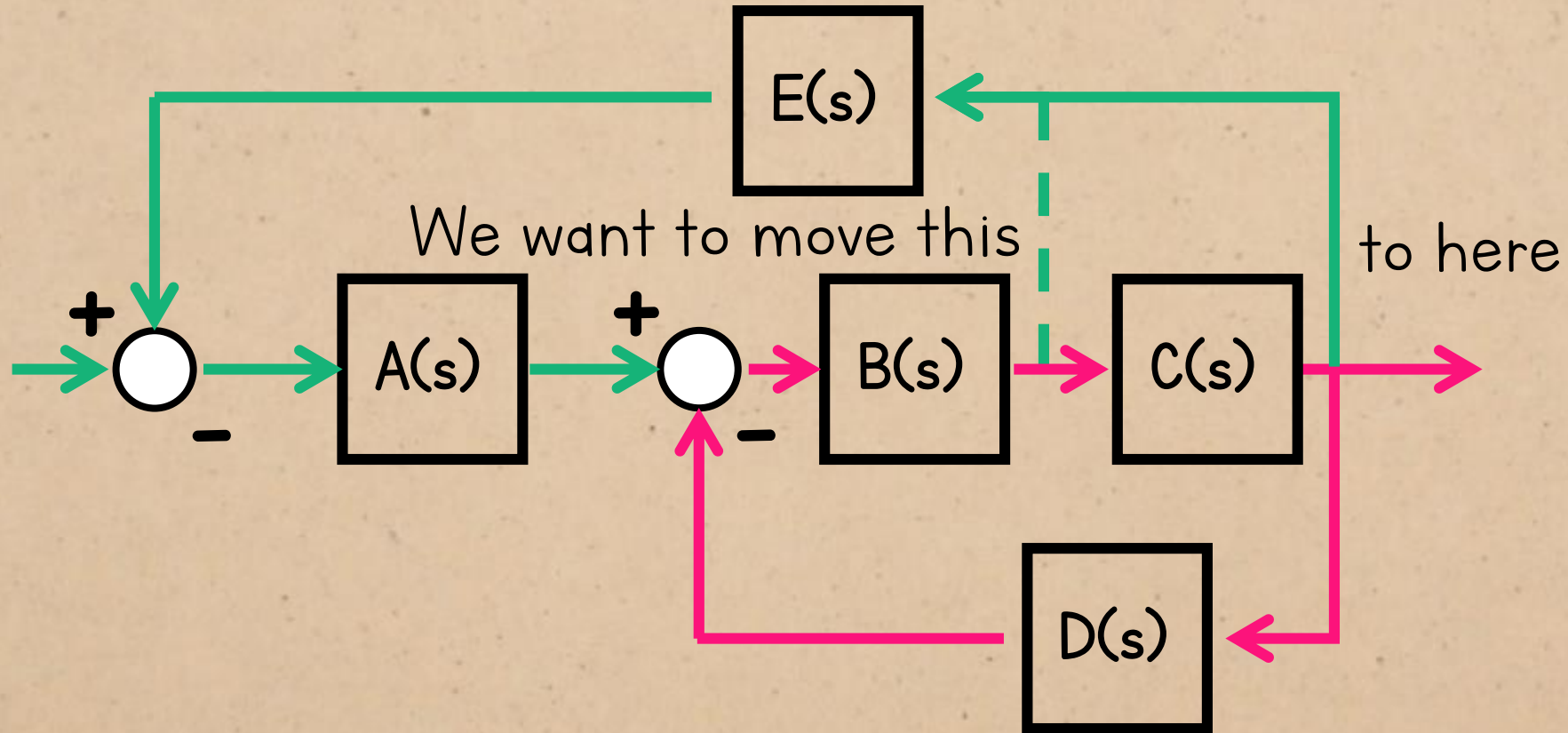


MOVING A TAKE-OFF POINT

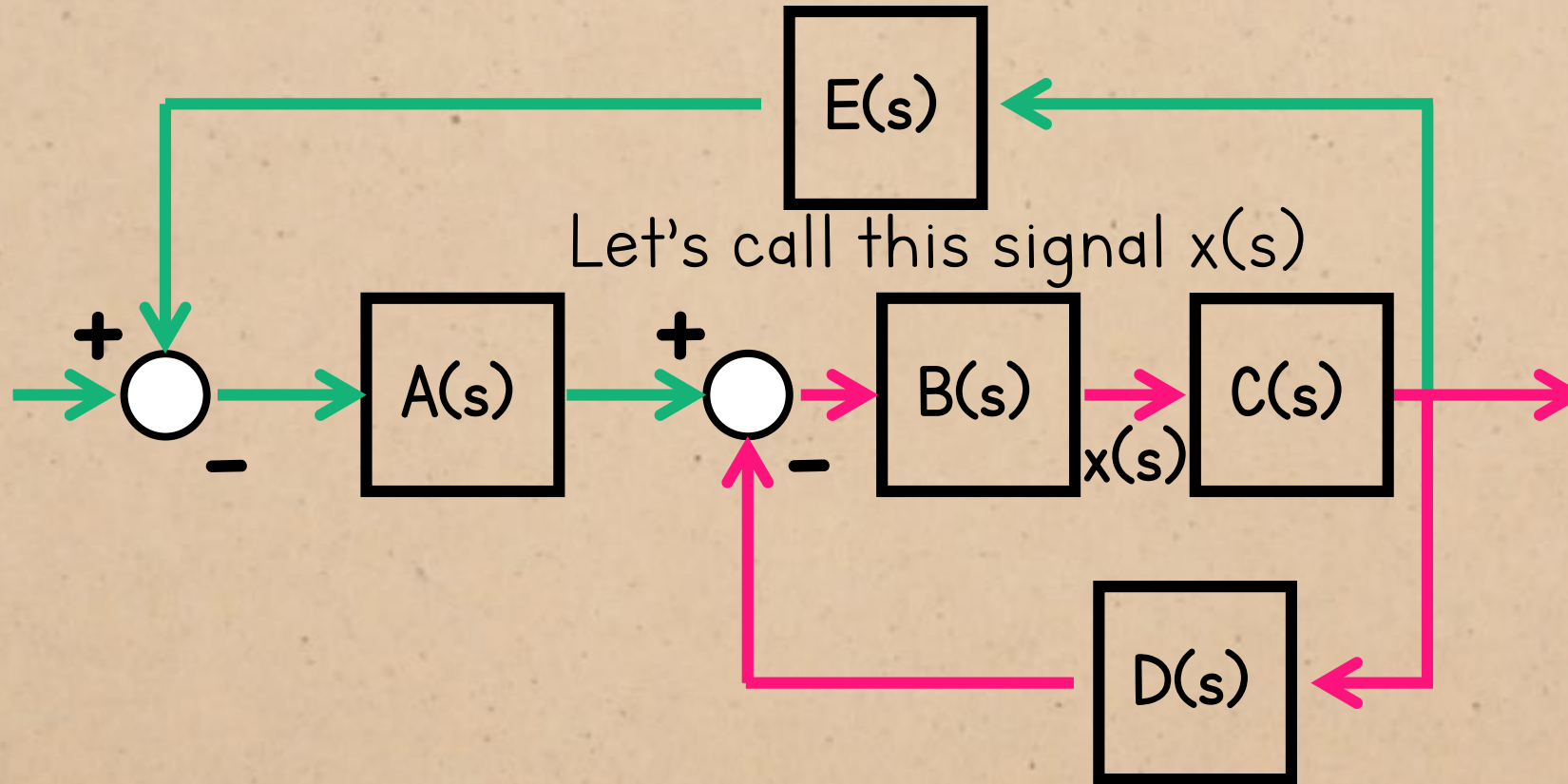
...without changing the outputs of the original system.



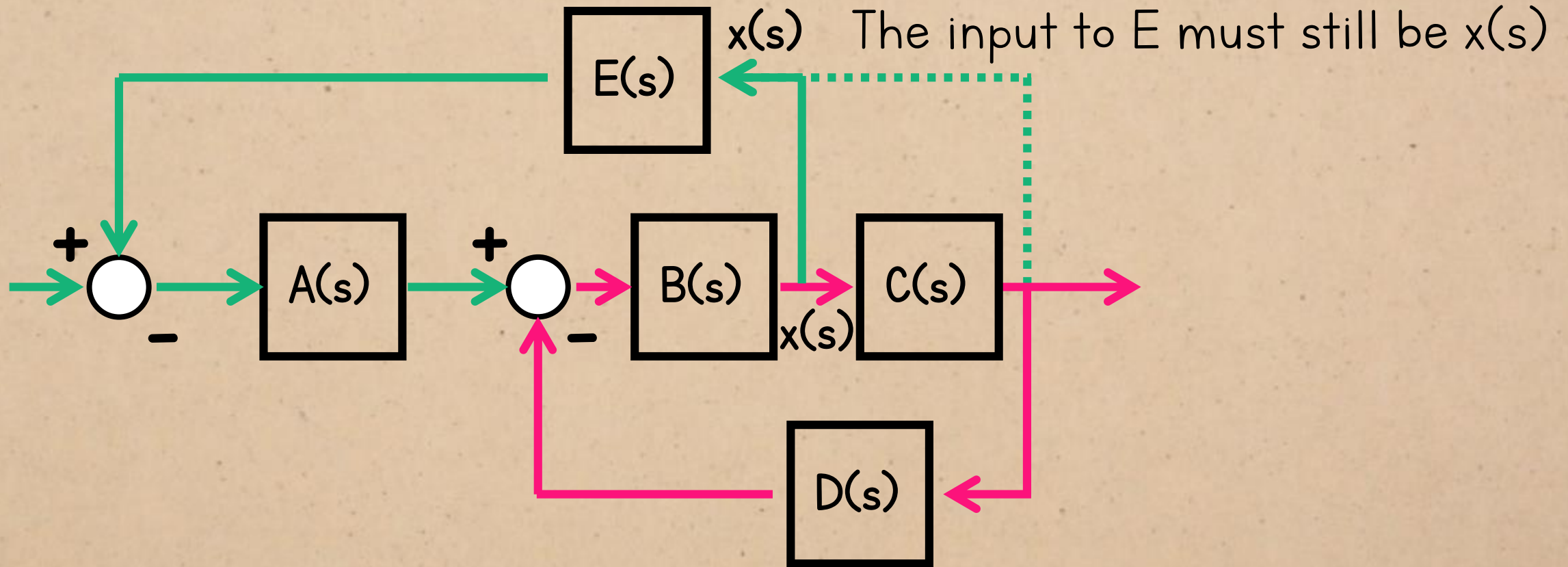
INTERLOCKING LOOPS



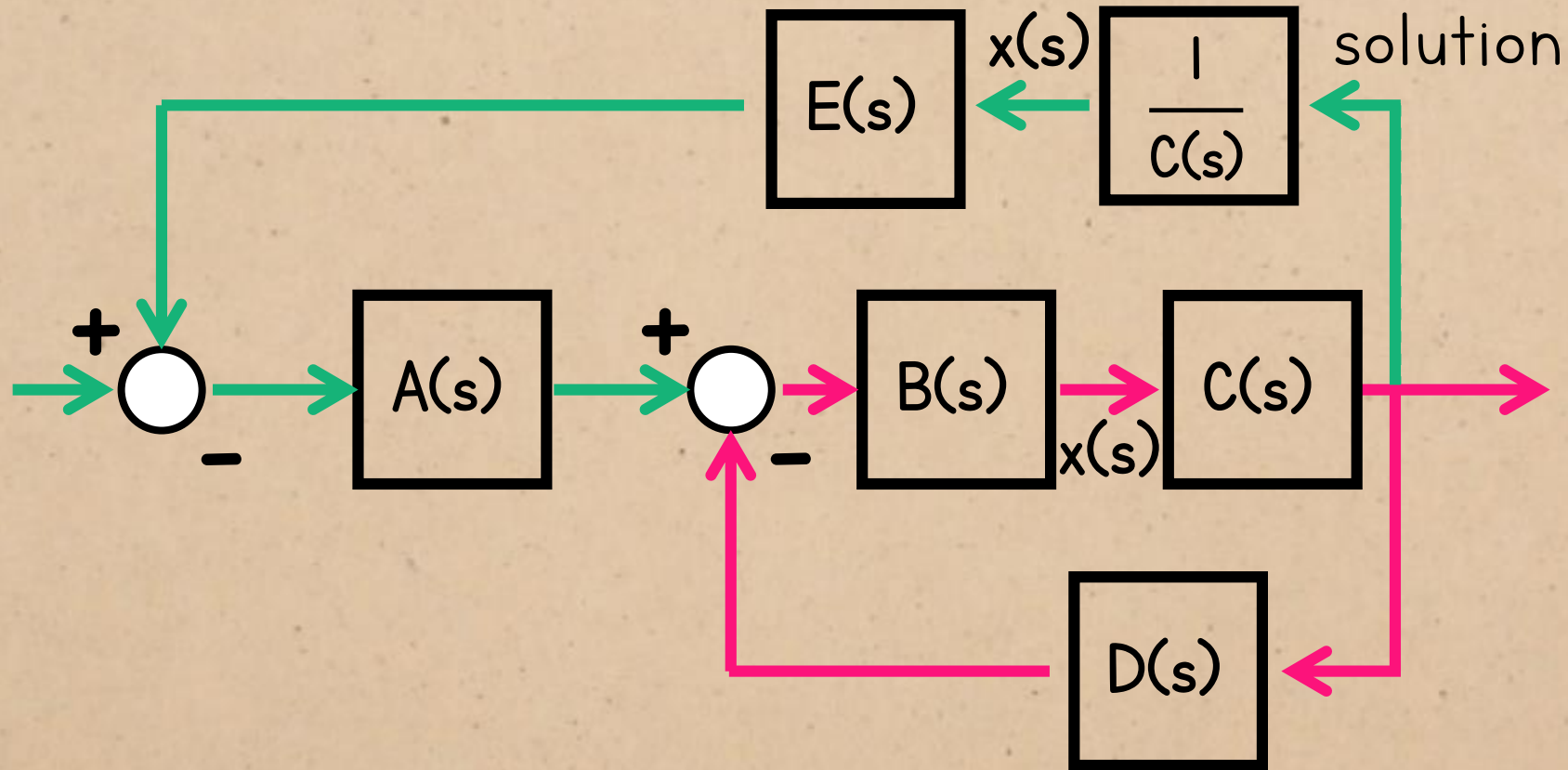
INTERLOCKING LOOPS



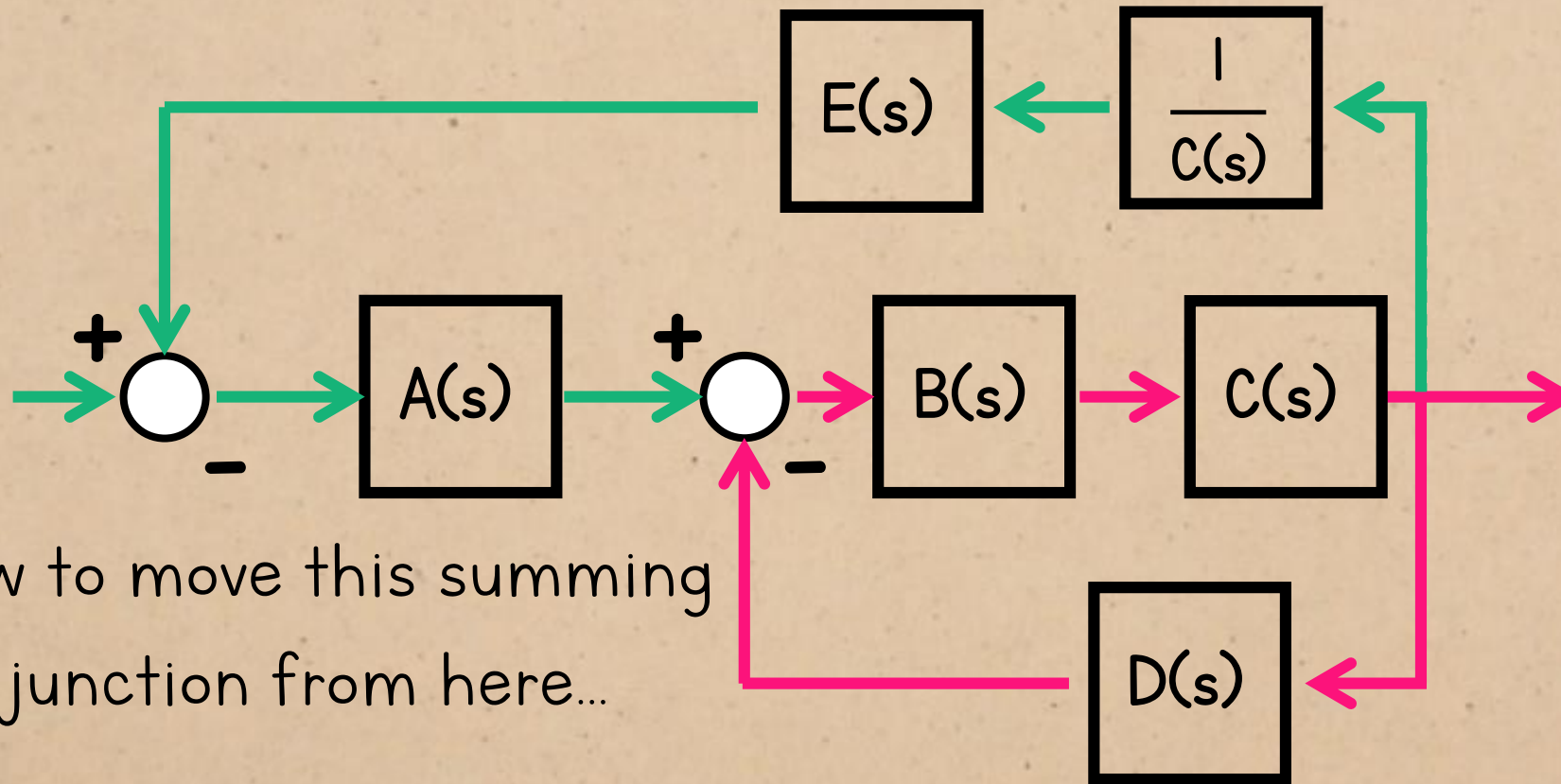
INTERLOCKING LOOPS



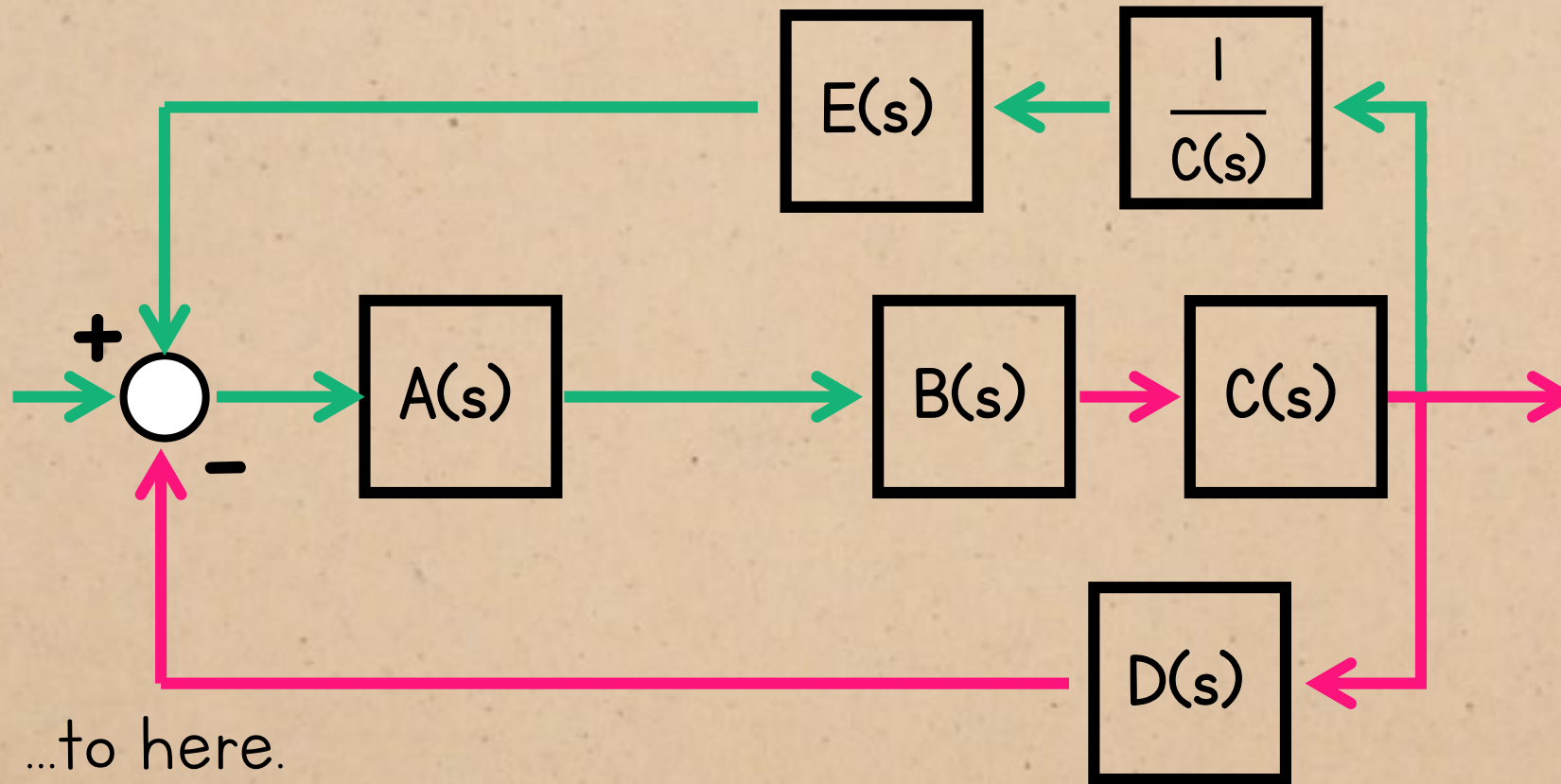
INTERLOCKING LOOPS



INTERLOCKING LOOPS

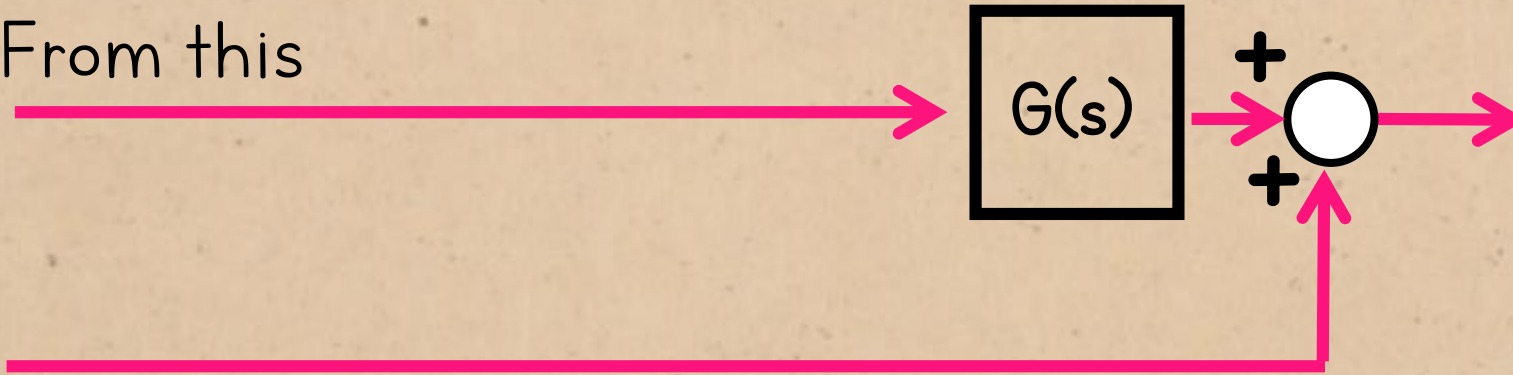


INTERLOCKING LOOPS

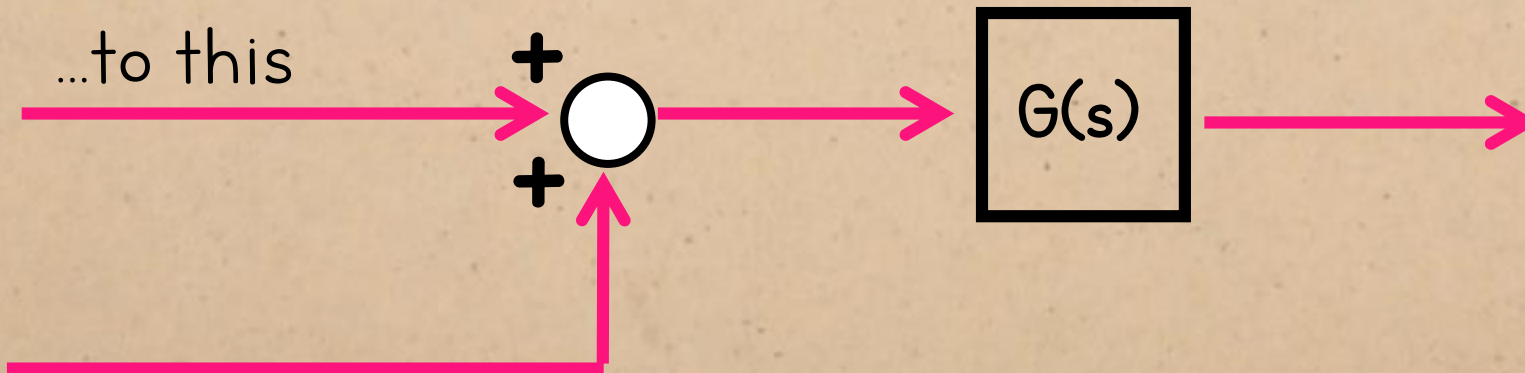


MOVING A SUMMING JUNCTION

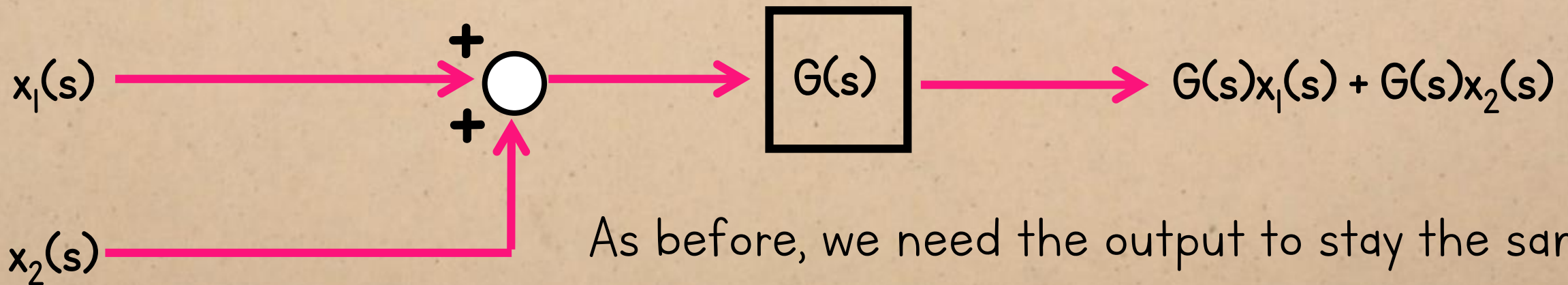
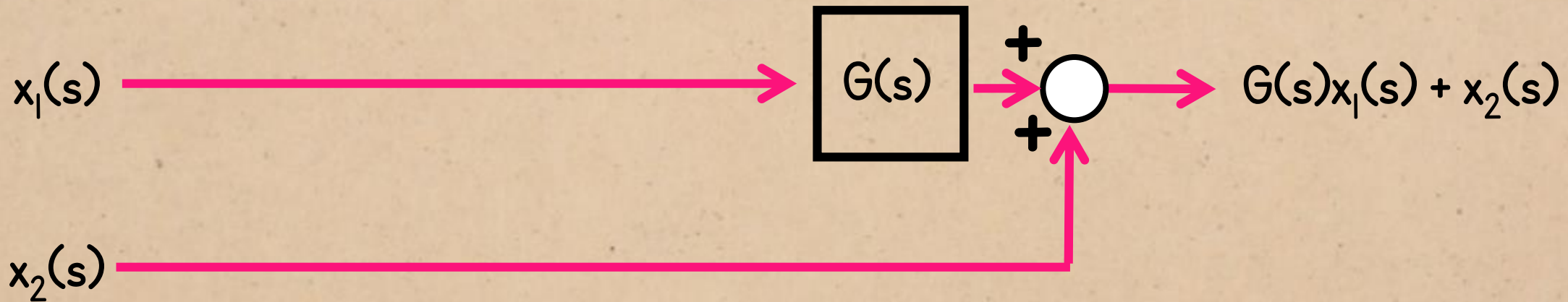
From this



...to this

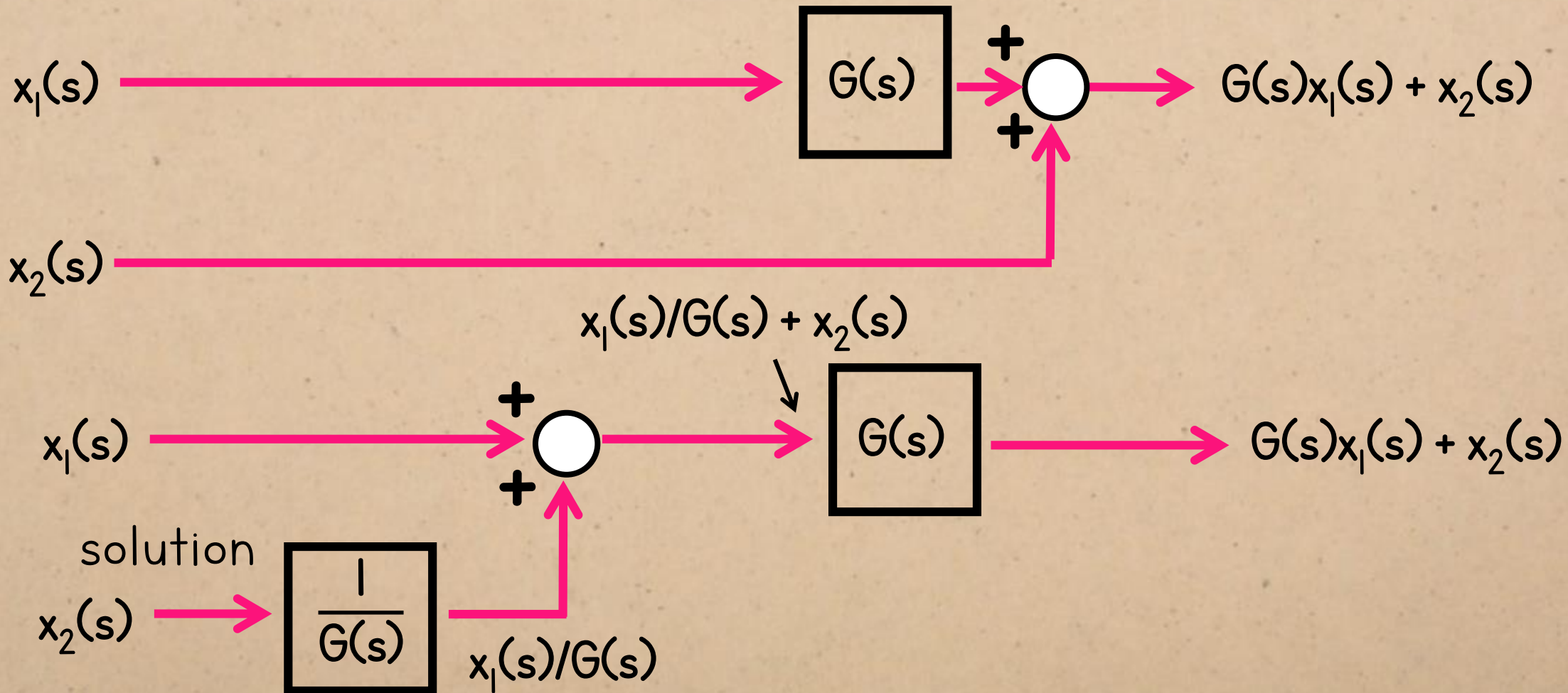


MOVING A SUMMING JUNCTION

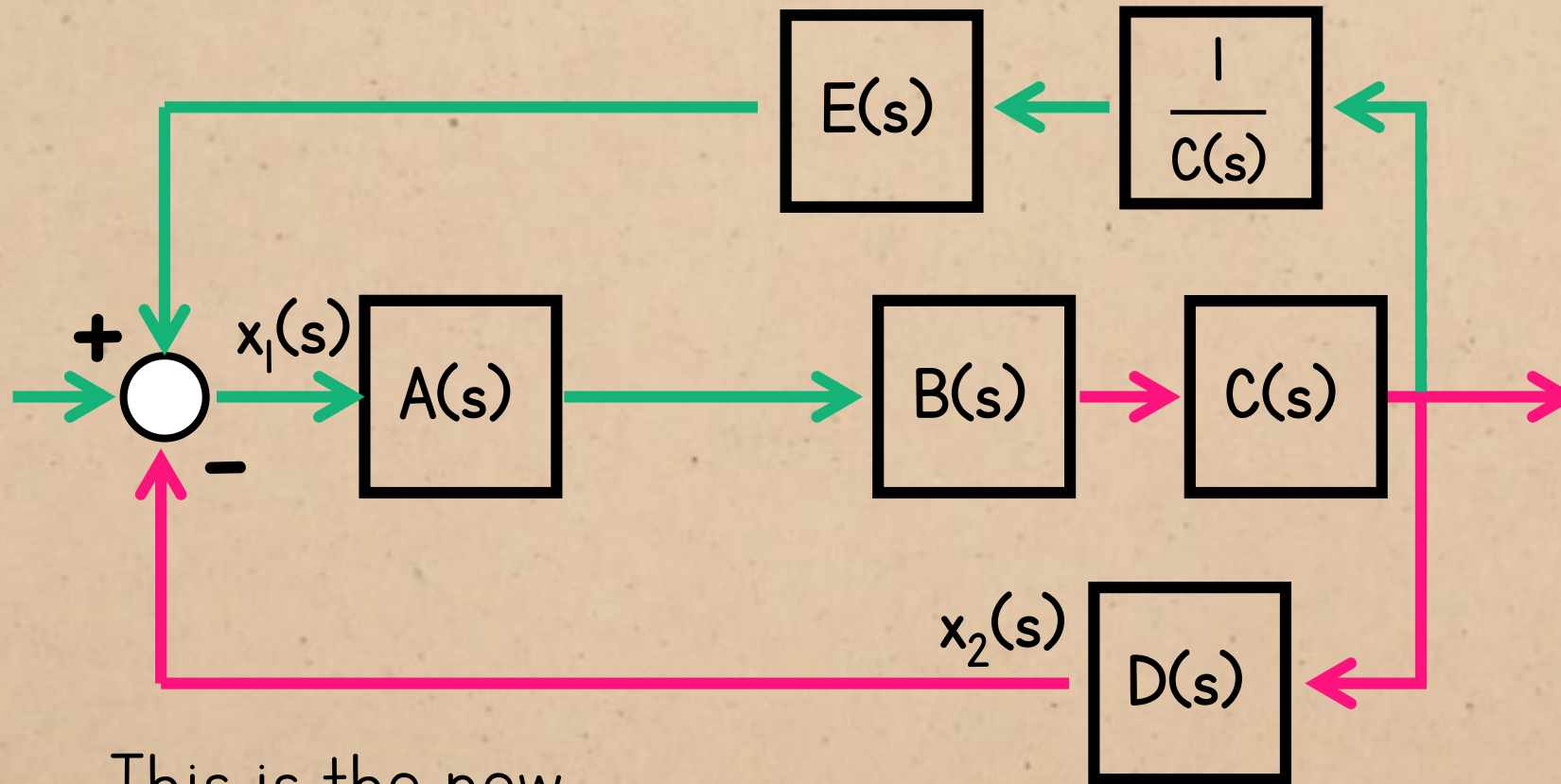


As before, we need the output to stay the same.

MOVING A SUMMING JUNCTION

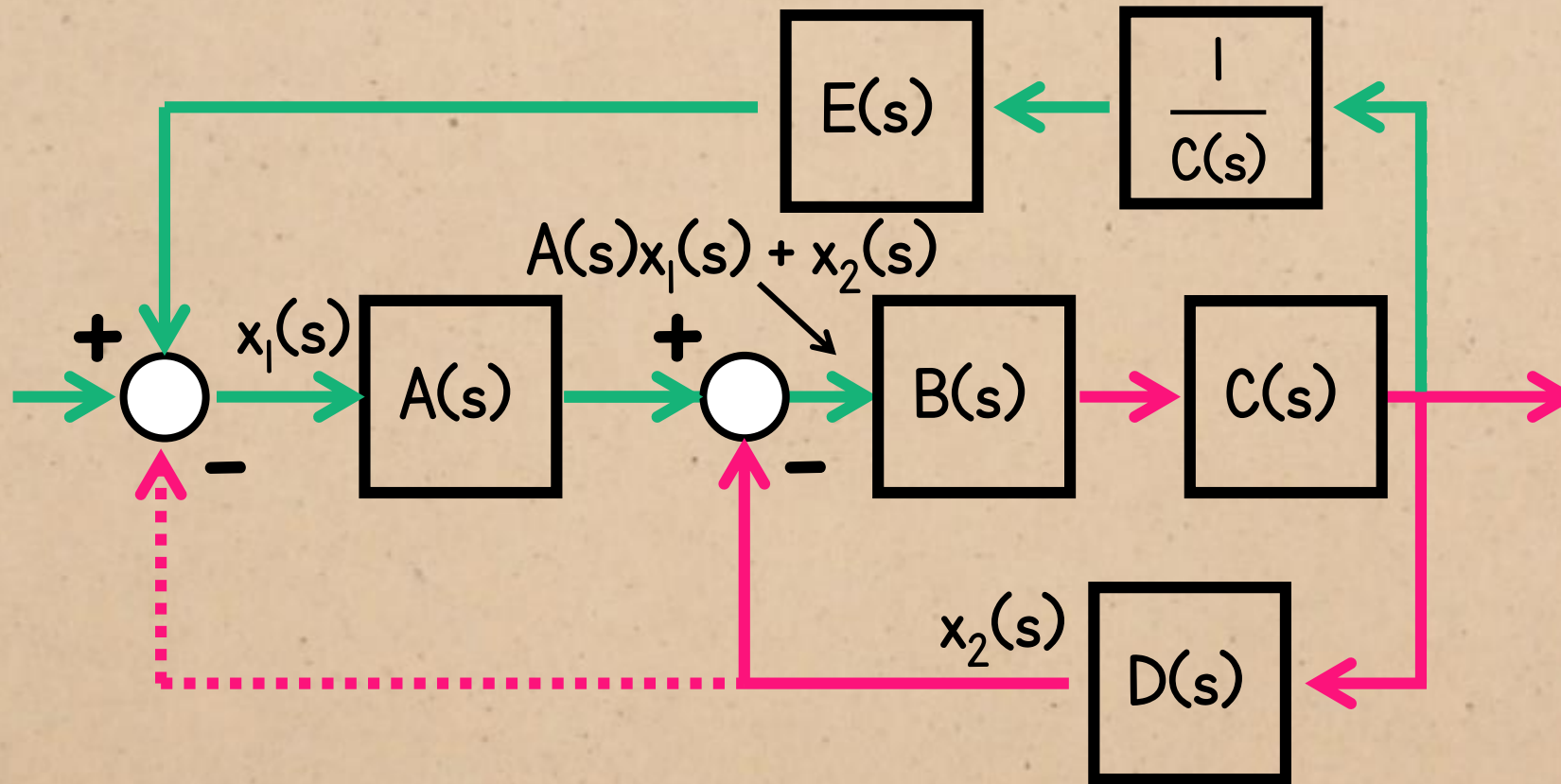


INTERLOCKING LOOPS



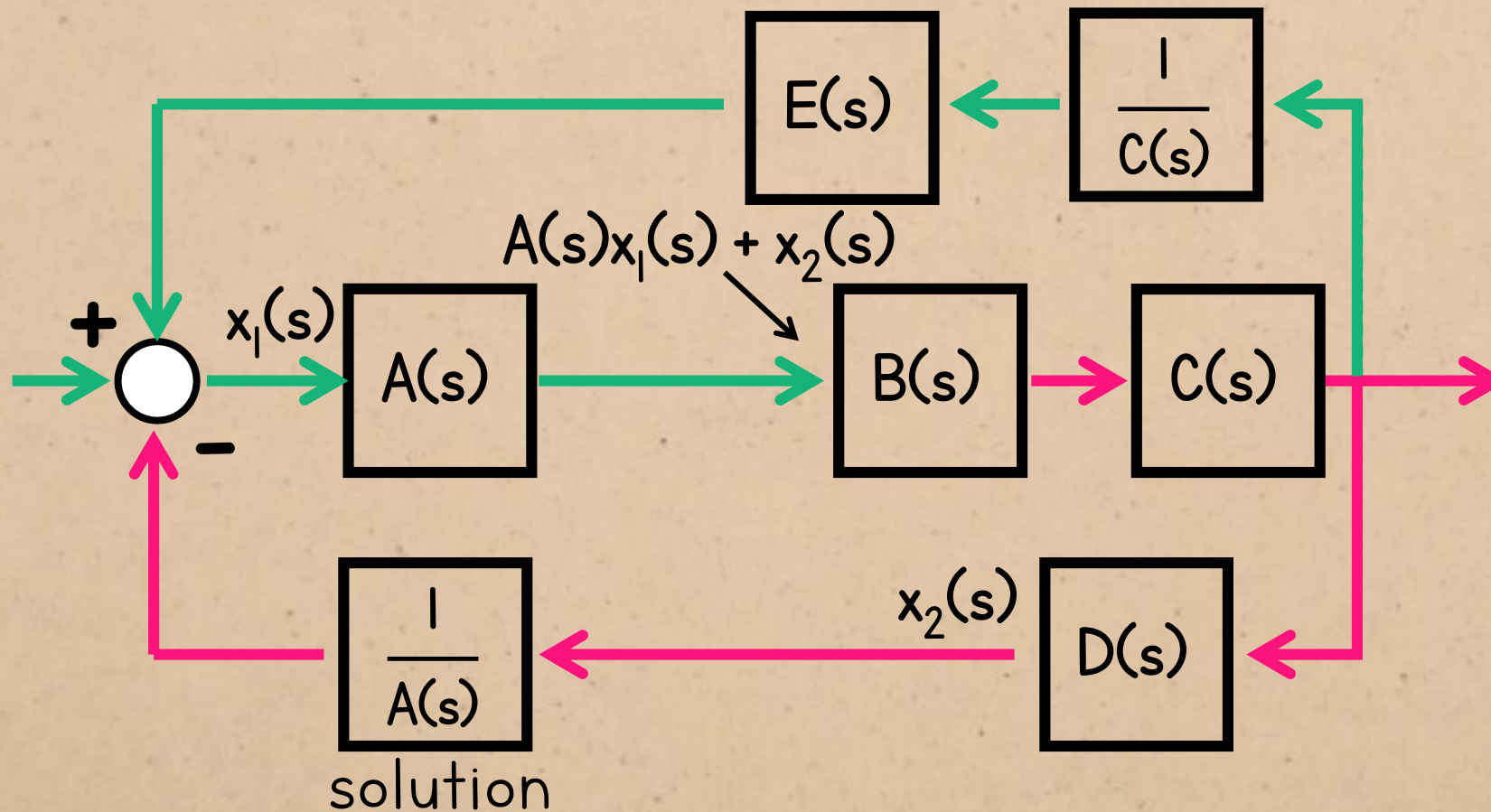
This is the new
configuration we want.

INTERLOCKING LOOPS

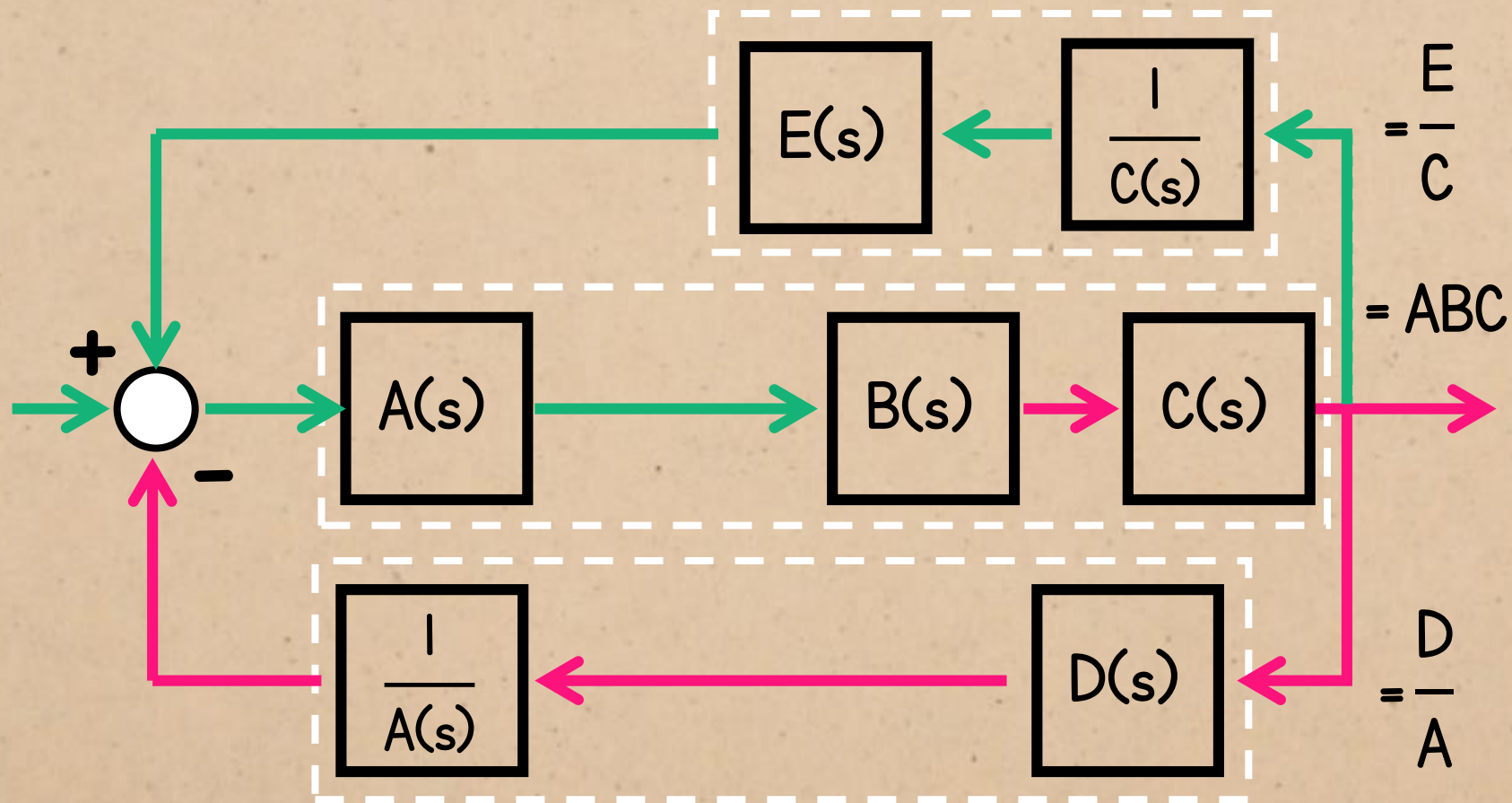


The input to B must be the same as in the original configuration.

INTERLOCKING LOOPS

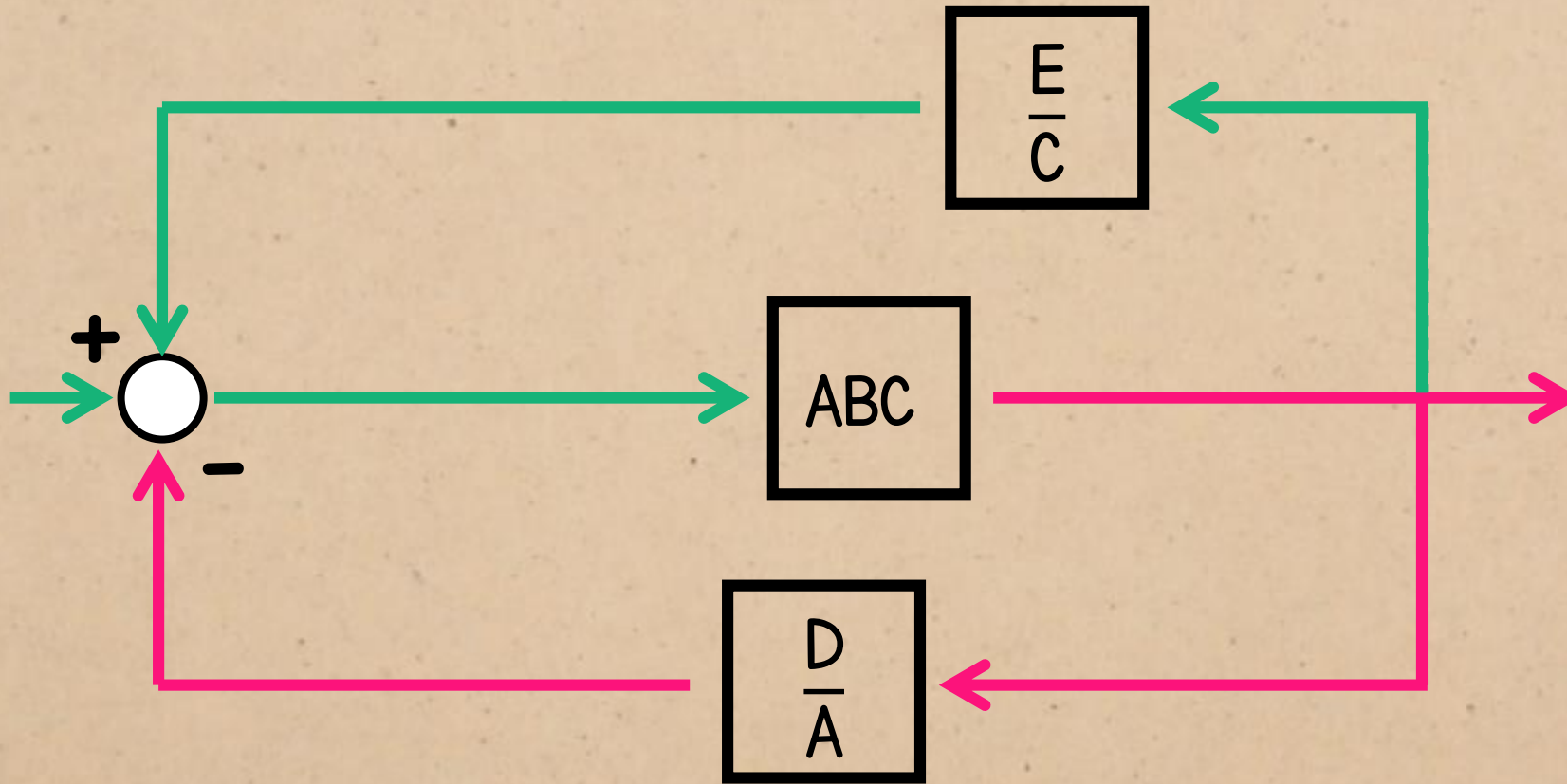


INTERLOCKING LOOPS

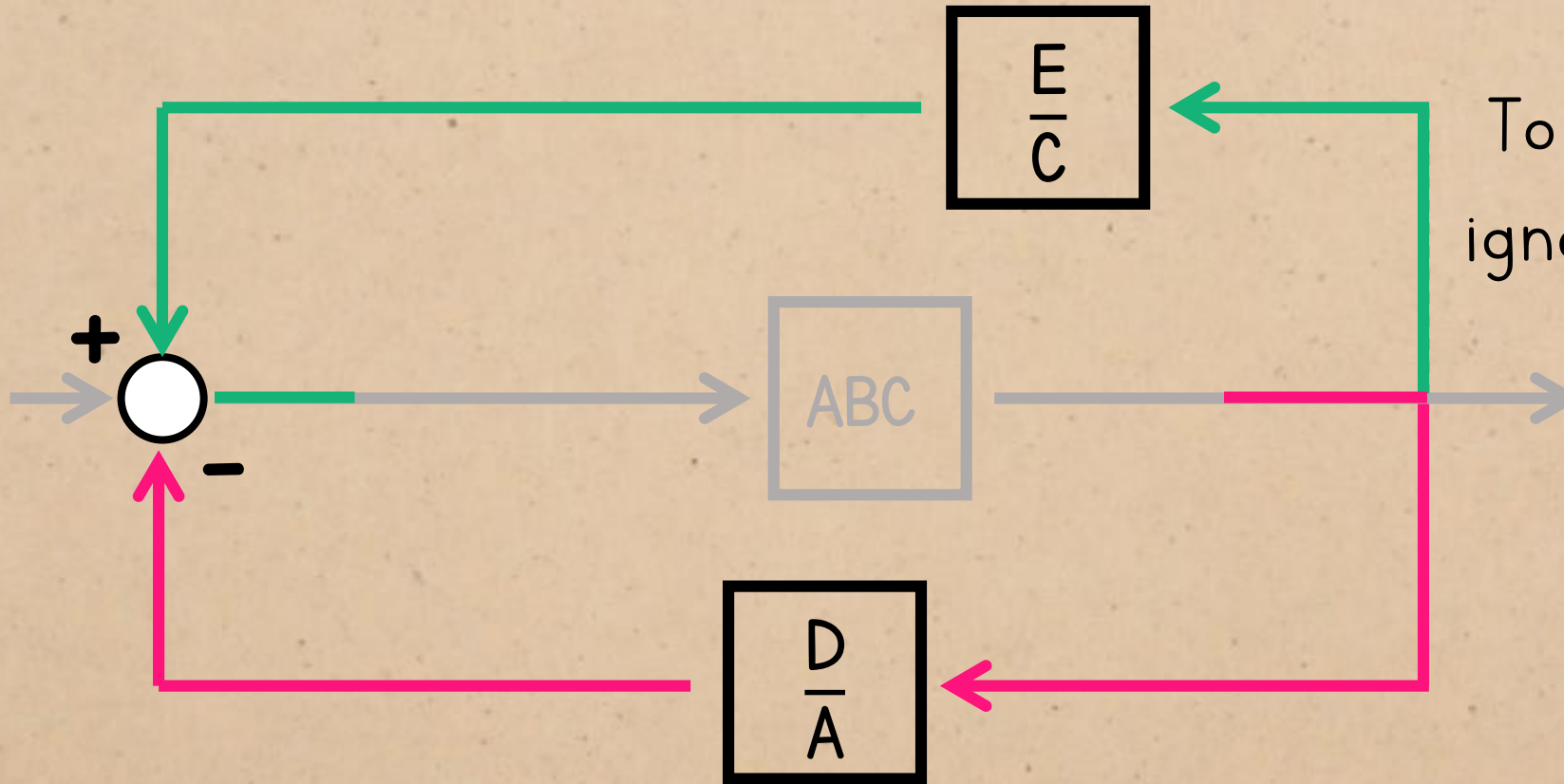


Now each of these cascade connections can be combined...

INTERLOCKING LOOPS

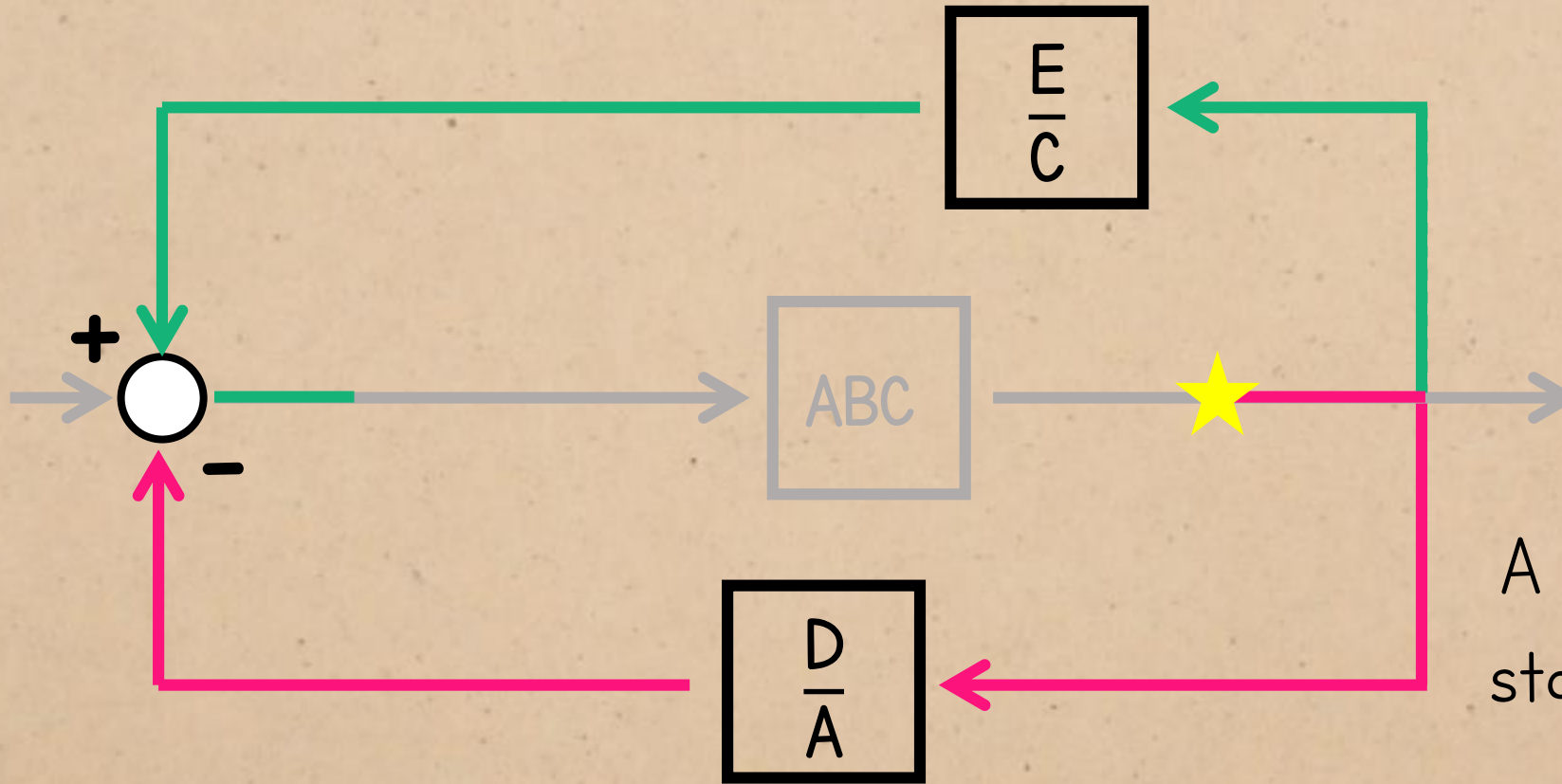


INTERLOCKING LOOPS



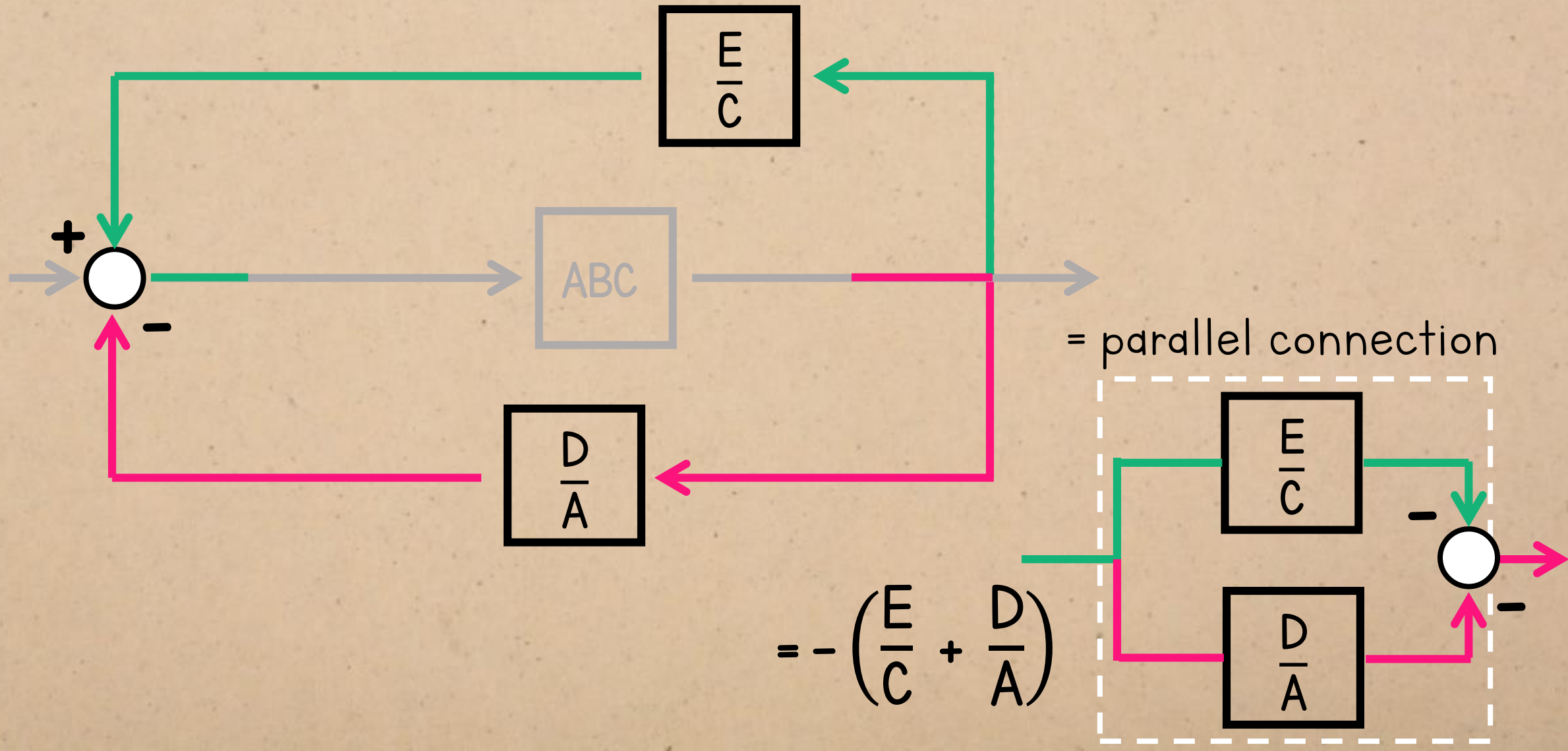
To see the next step, we ignore the middle branch.

INTERLOCKING LOOPS



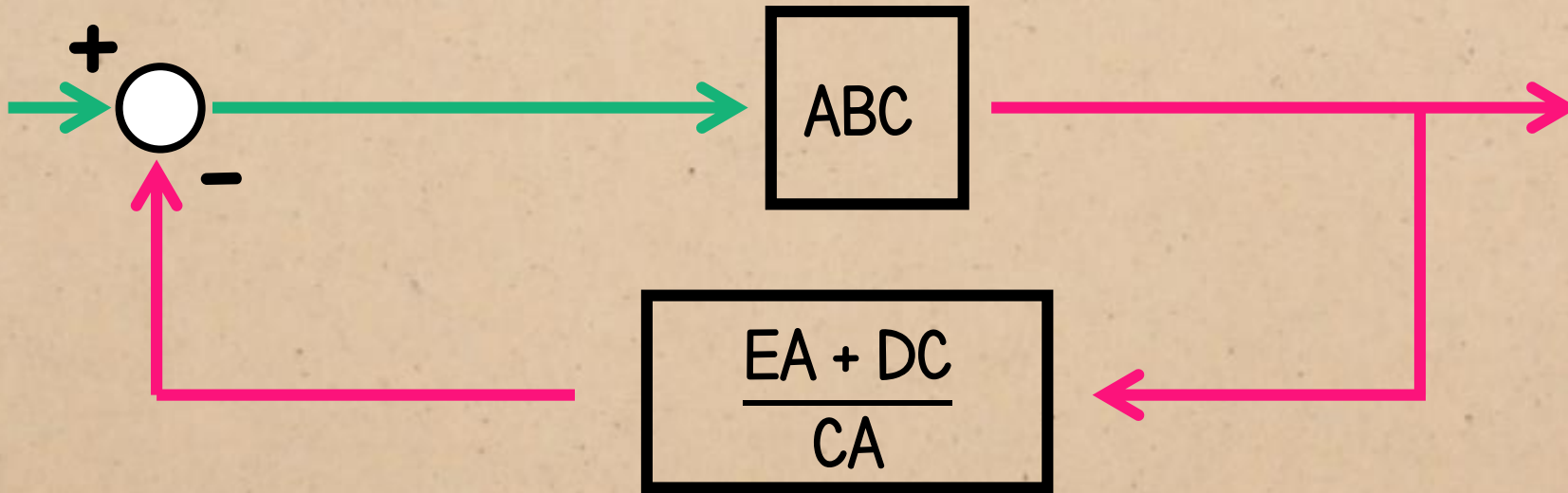
A signal starting at the star would flow through both branches and combine again at the sum.

INTERLOCKING LOOPS

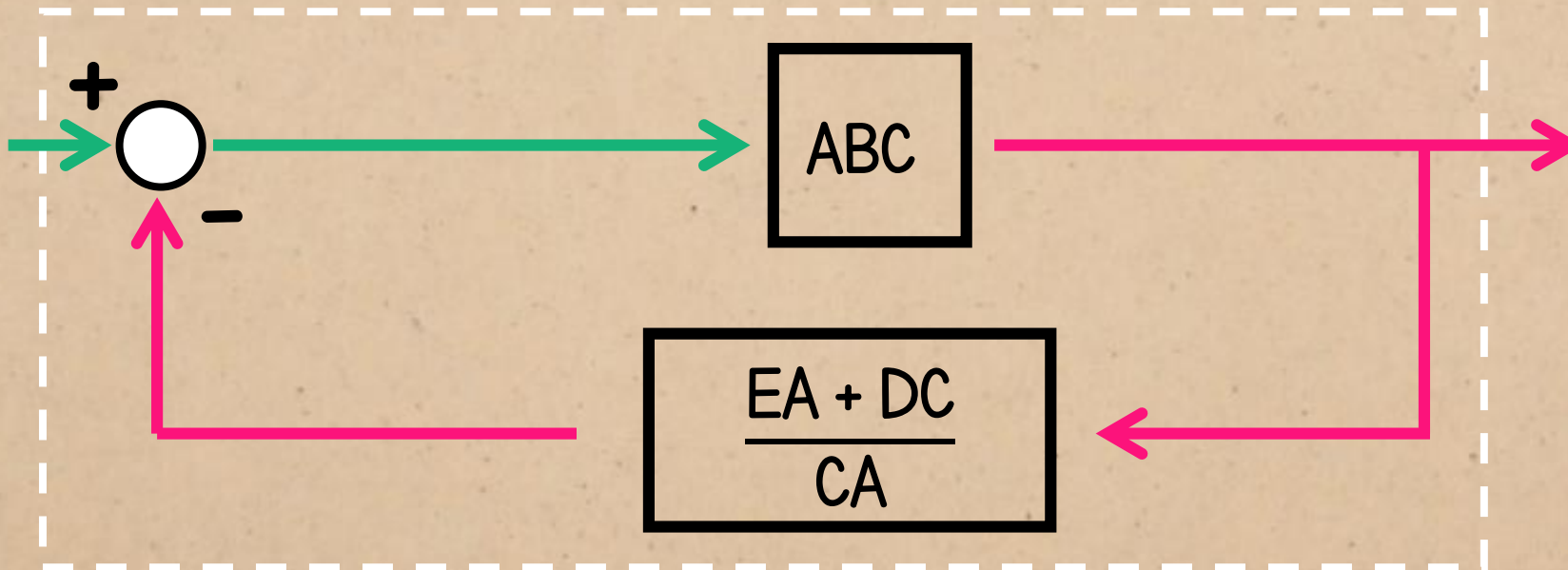


INTERLOCKING LOOPS

So now we have a simple loop again:



INTERLOCKING LOOPS



$$= \frac{ABC}{1 + ABC \left(\frac{EA + DC}{CA} \right)}$$

INTERLOCKING LOOPS

